

Chapter VI

Microscopic Coupled-Channels Study of Scattering and Breakup of Light Heavy-Ions

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The elastic and inelastic scattering and the breakup reactions of light heavy-ions, particularly ${}^6\text{Li}$, ${}^7\text{Li}$ and ${}^{12}\text{C}$, are investigated on the basis of a microscopic coupled-channels method. The bound states and resonant and non-resonant continuum states of the projectile nucleus are taken into account. The wave functions of those states are given by a microscopic cluster model assuming α -d, α -t and 3α for the three projectile nuclei, respectively. The unbound continuum states are approximately treated with the method of coupled discretized continuum channels (CDCC). The real parts of all the diagonal and coupling potentials are given by doubly folding the M3Y effective nucleon-nucleon interaction into the nucleon transition densities of the projectile nucleus and the ground-state density of the target nucleus. The microscopic CDCC method is successfully applied to the analyses of the scattering and the breakup of ${}^{6,7}\text{Li}$ and ${}^{12}\text{C}$ in a wide range of the target mass and the bombarding energy. The effect of the projectile virtual breakup on the elastic scattering is strikingly large and absolutely important to reproduce the observed cross sections. The dynamical polarization (DP) potentials induced by the breakup of the ${}^{6,7}\text{Li}$ and ${}^{12}\text{C}$ projectiles are found to be strongly repulsive and, for ${}^6\text{Li}$, very weakly absorptive, which is quite different from what has been known for DP potentials induced by the excitation of usual collective states. Unified understanding is presented for various types of possible DP potentials due to weak-coupling channels through strong-coupling channels. Further discussions are made on the roles of the coupling between the breakup channels and that between the elastic channel and the breakup channels in the scattering and breakup reactions of the light heavy-ions.

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§ 1. Introduction

In the study of nuclear direct reaction mechanisms, breakup processes of light heavy-ion projectiles are of current interest. Most of the direct reaction theories for heavy-ion reactions have been concerned with two-body systems in the intermediate and final stages of reactions. As for deuteron induced reactions, however, a lot of theoretical studies have been performed on the basis of the three-body model for proton and neutron and target nucleus (a review of these studies is given in Ref. 1); references are therein). Multi-step processes via the deuteron breakup states have been subjects of those studies both because of the simple structure of deuteron and because of a large number of available experimental data. Among various theoretical approaches the method of coupled discretized continuum channels (CDCC)^{2)~5)} with folded interactions has been recognized as a useful and practical tool for the theoretical studies of the deuteron breakup processes in deuteron induced nuclear reactions. Particularly, the method has successfully been applied by Kyushu group^{1),5)~9)} to the analyses of the experimental data for (d, d) , (d, d') , (d, p) , (p, d) and (d, pn) reactions. The essential features clarified in these studies are as follows: (i) The effect of the deuteron breakup states is universally important in deuteron induced elastic and inelastic scattering and transfer reactions. (ii) The coupling among the deuteron breakup continuum states is stronger and more essential

than the coupling between the ground state and the continuum states. (iii) The multi-step transitions among the breakup states also play an important role in the actual breakup of the deuteron projectile observed in the (d, pn) reactions. Similar results have been obtained for the breakup of ^3He projectile.^{9),10)}

It is, therefore, of great interest to apply the method of CDCC to the scattering of heavier projectiles and study the breakup processes of those projectiles; this is the main aim of the present paper. Extension of the method to heavier projectiles seems straightforward, but it is to be noted that there are some physical features which are characteristic of those projectiles. Namely, structure of excited states of the projectile nucleus becomes much more complicated as the projectile mass increases; there appear excited bound states and resonance states besides the non-resonant continuum states, and no such complex structure exists in the excited states of deuteron and ^3He . This structure of the heavier projectiles may bring about a variety of reaction mechanisms which are not seen in the light-ion induced reactions. Thus, it is essentially important to prepare good wave functions of the resonant and non-resonant continuum states of the projectile nucleus as well as its bound states before we enter theoretical investigations of the reaction mechanisms. As for the excited states of light nuclei ($A \lesssim 40$) in the region of $E_x \lesssim 15$ MeV, two kinds of states are known to coexist, namely, *shell-model-like* states and *cluster-like* states.¹¹⁾ Both kinds of the states have successfully been described with the use of the microscopic cluster models^{11),12)} in which the relative motion between the constituent clusters is solved dynamically with the total antisymmetrization taken into account correctly; the shell-model-like states and the cluster-like states can simultaneously be obtained from the same equation of motion as compactly coupling solutions and as loosely coupling solutions (both for the resonant and non-resonant continuum states), respectively. We therefore employ the microscopic cluster models to construct the internal wave functions of the projectile nuclei which are concerned in the present coupled-channel study. In heavy-ion reactions, direct-reaction processes are generally much suppressed due to the strong absorption compared with those in light-ion reactions, while projectile breakup reactions take place mainly at nuclear surface and the beyond as discussed later and do not suffer so much the suppression from the absorptive potential. Thus, it may be expected that the breakup process is more prominent among the direct reaction processes than in the cases of the deuteron and ^3He projectiles. Since the degree of the clusterization of light nuclei is seen mostly in the surface region, the analyses of the scattering and breakup of those light nuclei by the target nucleus may give a severe test of the adopted internal wave functions of the nuclei in the surface region and provide a good information on the clusterization.

Another reason to investigate the breakup processes of light heavy-ions, especially of ^6Li and ^7Li ions is closely related to the famous anomaly of the renormalization factor of the potential strength in the *double-folding model*¹³⁾ for the scattering of ^6Li , ^7Li and ^9Be projectiles;^{13)~22)} those projectiles have very low threshold energies of breakup into the constituent clusters. The double-folding model provides an attempt to understand the inter-nucleus potentials from a microscopic point of view. In this model, the real part of the potential is constructed by the folding of an effective nucleon-nucleon potential such as of M3Y type²³⁾ into nucleon densities of colliding nuclei in their ground states. The real potential derived is then multiplied by a renormalization factor, N_r , which is a free parameter to fit the elastic-scattering cross section. On the other hand, a flexible shape

such as the Woods-Saxon form is assumed for the imaginary part of the inter-nucleus potential. The optimum values of the imaginary-potential parameters as well as that of N_R are searched at the individual bombarding energies so that the observed elastic-scattering cross section is reproduced by the calculation with the potential. In spite of the simple assumption for the real potential, the double-folding model has been successful for elastic scattering of various heavy ions at low energies ($E/A \lesssim 20$ MeV);¹³⁾ “successful” means that the resultant optimum value of N_R is very close to 1.0.

Let us consider the implication of the success of the model. In general, the elastic scattering proceeds via a lot of non-elastic channels at the intermediate stage. Therefore, the *generalized optical potential*²⁴⁾ is, in general, complex and non-local and dependent on energy and angular momentum. Following Feshbach’s projection-operator formalism,²⁴⁾ the potential may be written as

$$U_{\text{opt}} = V_{00} + \sum_{\alpha\alpha'} V_{0\alpha} \left(\frac{1}{E - H - i\varepsilon} \right)_{\alpha\alpha'} V_{\alpha'0}, \quad (1.1)$$

where α and α' run over all the non-elastic channels, $\alpha=0$ denoting the elastic (incident) channel. For the moment we ignore the antisymmetrization between two nuclei for simplicity. The first term, V_{00} , is the double-folded potential which acts between the colliding nuclei in their ground states and is real and local. The remaining term, ΔU say, which we refer to as a *dynamical polarization (DP) potential*, arises from coupling to all the non-elastic channels. It is this part that makes the generalized optical potential complex. This term certainly gives rise to an imaginary potential and, in principle, also contribute to the real part of the optical potential. Thus, the success of the double-folding model implies that V_{00} alone is adequate for the real part of U and ΔU can be simulated well by a local imaginary potential only. In other words the real part of ΔU is negligibly small compared with the “zero-th order” term, V_{00} , at least in the spatial region which is important for the scattering. In fact, some approximate estimations^{25)~30)} have suggested that $|\text{Re}\Delta U| \ll |\text{Im}\Delta U|$, at least in the peripheral region, when calculated to the second order of $V_{0\alpha}$.

Exception to the success of the double-folding model is the elastic scattering of ${}^6\text{Li}$, ${}^7\text{Li}$ and ${}^9\text{Be}$.^{13)~22)} For the projectiles N_R must be reduced artificially by a factor of about two, namely, $N_R \approx 0.5$; as long as N_R is kept fixed at 1.0, one cannot satisfactorily reproduce the observed elastic-scattering cross section, irrespective of the incident energy and target nucleus, though the imaginary part is varied flexibly. The strong reduction of N_R was considered to imply that a strongly repulsive real potential is induced by some unknown effect. Since the breakup threshold energy is very small for the three nuclei, it was often stated that the reduction might be due to the effect of projectile breakup. In fact, a remarkable effect of ${}^6\text{Li}$ breakup on the elastic scattering was first found by Thompson and Nagarajan³¹⁾ on the basis of adiabatic three-body model³²⁾ in which the excitation energies of all the α -d breakup states were approximated by the ground-state energy, and the ${}^6\text{Li}$ -target interaction was assumed to be derived from the α -target and the d-target optical potentials. Though the interaction they used was not directly related to the double-folded ${}^6\text{Li}$ -target potential, their result which explained well the elastic-scattering cross sections suggests that the breakup effect can be a reasonable candidate for the origin of the reduction of N_R .

There remained, however, a puzzling problem noted by Satchler and Love,¹³⁾ if one generalizes the approximate estimation^{25)~30)} of the DP potential induced by the coupling of *collective* states, the coupling of the *breakup* states might similarly be expected to induce dominantly an imaginary potential accompanied by a very weakly attractive real potential. The observed effect is, however, strongly repulsive ($N_R \approx 0.5$) for the elastic scattering of ${}^6\text{Li}$, ${}^7\text{Li}$ and ${}^9\text{Be}$ projectiles. Therefore, a question to be answered was whether or not the explicit inclusion of breakup channels really induces such a large, repulsive real potential. It was this puzzle that motivated the present authors to study the breakup of light heavy-ions. To make clear the direct relation between $N_R \approx 0.5$ and the breakup effect, it is necessary to stand on the same ground of the double-folding model as for the interaction. To this end, we developed^{33),34)} a microscopic version of the CDCC method in which the real part of all the diagonal and the coupling potentials are calculated by the double folding of the M3Y nucleon-nucleon interaction into the target density and the transition densities of the projectile nucleus obtained by the microscopic cluster model. The double-folding model mentioned before is included in this CDCC framework as the single-channel case. In the present paper, we shall compare the result of the CDCC calculation and that of the single channel-calculation of the ${}^6\text{Li}$ scattering for a wide range of targets and bombarding energies,^{35),36)} and discuss the mechanism of the reduction of N_R due to the coupling of ${}^6\text{Li}$ breakup channels.

A similar investigation will be done for the ${}^7\text{Li}$ projectile. For ${}^7\text{Li}$, however, there is another candidate for the origin of the strong reduction of N_R . Some authors^{37),38)} claimed that the reduction can be explained by the effect of the re-orientation coupling of the deformed ground state of ${}^7\text{Li}$. We shall therefore examine the effects of both the breakup and the re-orientation coupling by the CDCC calculations³⁵⁾ in which they are simultaneously included.

As mentioned above, the double-folding model has been successful for the elastic scattering of various heavy-ions except for the ${}^{6,7}\text{Li}$ and ${}^9\text{Be}$ projectiles. For 300 MeV ${}^{12}\text{C} + {}^{12}\text{C}$ scattering, however, a similar anomaly of $N_R \approx 0.6$ was found by Bohlen et al.³⁹⁾ Since the double-folding-model analysis of ${}^{12}\text{C}$ scattering had been limited to the data at lower energies, $E_{\text{lab}}/12 \lesssim 10$ MeV,¹³⁾ no such anomaly of N_R had been recognized before the work of Ref. 39). Therefore, we shall make a double-folding-model analysis of recent data^{39)~44)} on the elastic scattering of ${}^{12}\text{C}$, ${}^{13}\text{C}$ and ${}^{16}\text{O}$ from ${}^{12}\text{C}$ at $E_{\text{lab}}/A_p = 10 \sim 25$ MeV, and investigate for what energies the double-folding model with $N_R = 1.0$ breaks down. We then discuss the result of the microscopic CDCC calculation^{35),45)} with the ${}^{12}\text{C} \rightarrow 3\alpha$ breakup states in order to make clear the relation between the reduction of N_R and the effect of the ${}^{12}\text{C}$ excitation and breakup.

On the basis of the microscopic CDCC analyses we next study a general property of the dynamical polarization (DP) potential which is induced by the coupling of the excited and the breakup states of ${}^6\text{Li}$ and ${}^{12}\text{C}$; we calculate it in the form of a wave-function-equivalent local potential. The DP potentials obtained^{34)~36)} are found to be of new type whose feature has not been expected from the conventional theories^{25)~30)} of the DP potential due to the excitation of collective states. This new type of DP potential is further investigated on the basis of a model calculation for the expression of Eq. (1.1). The systematic transition of the qualitative feature of the DP potentials from the well-known type to the newly-found type will unifiedly be understood.

The subjects of the present paper are summarized as follows:

- (i) Examination of the applicability of the microscopic CDCC method to the scattering and breakup of light heavy-ions.
- (ii) Investigation of the projectile-breakup effect on the elastic scattering and its relation to the strong reduction of N_R in the double-folding model.
- (iii) Analysis of the reaction mechanism of the inelastic scattering and breakup reactions in terms of multi-step processes via the excited and the breakup channels.
- (iv) Finding of new types of the dynamical polarization potential induced by the strong coupling of the breakup channels.
- (v) Unified understanding of dynamical polarization potentials due to weak coupling channels and those due to strong coupling channels.

In § 2, the present microscopic CDCC method is described. The wave functions of the bound and unbound states of ${}^6\text{Li}$, ${}^7\text{Li}$ and ${}^{12}\text{C}$ are given by microscopic α -d, α -t and 3α cluster models, respectively. In §§ 3~6, we analyze the experimental data of the scattering and breakup of ${}^6\text{Li}$, ${}^7\text{Li}$ and ${}^{12}\text{C}$ by means of the CDCC method. The anomaly of N_R in the double-folding model for these projectiles is discussed from the viewpoint of the virtual projectile breakup. In § 7, we calculate the DP potentials due to the excitation and the breakup of ${}^6\text{Li}$ and ${}^{12}\text{C}$ and discuss systematic features of the potentials. In § 8, a unified understanding is presented for the DP potentials due to various types of the coupling by excited channels. The final section is devoted to a summary.

§ 2. Microscopic coupled-channel method

We first expand the total wave function of the colliding system in terms of the eigenfunctions of the internal states of projectile, ϕ_i , and those of target, Φ_j , as

$$\Psi = \sum_{ij} \phi_i \Phi_j \chi_{ij}(\mathbf{R}), \quad (2.1)$$

where $\chi_{ij}(\mathbf{R})$ describes the relative motion between the projectile in state “ i ” and the target in state “ j ”. Here, we employ the cluster-model wave function for the ground and the bound-excited states of the projectile. Since we are interested in weakly-bound projectiles such as ${}^6\text{Li}$, we further include breakup states in Eq. (2.1). In actual CC calculations, the breakup continuum is approximately treated as a finite number of averaged-discrete states within the framework of the method of coupled discretized continuum channels (CDCC).^{2)~5), 34)~36)}

As for the target nucleus we include only the ground state in the present work, except in § 6.3 where the target excitation is considered. We assume that the effect of the target-excited channels is taken into account by modifying the imaginary part of the CC potentials. The assumption will be approximately valid for target nuclei which are not as strongly excited as the “soft-projectiles” such as ${}^6,7\text{Li}$ and ${}^{12}\text{C}$ (cf. the discussion in §§ 7 and 8).

Then, the wave functions of relative motion, $\chi_i(\mathbf{R})$, may be obtained as solutions of the following CC equations,

$$\left[-\frac{\hbar^2}{2\mu} \nabla^2 + U_{ii}(\mathbf{R}) - (E - \varepsilon_i) \right] \chi_i(\mathbf{R}) = \sum_{j \neq i} U_{ij}(\mathbf{R}) \chi_j(\mathbf{R}), \quad (2.2)$$

where $U_{ij}(\mathbf{R})$ denotes the diagonal ($i=j$) and the coupling ($i \neq j$) potentials. The potentials are calculated by folding an effective nucleon-nucleon interaction potential between the internal states of the colliding nuclei.

In the next subsection, we construct the wave functions of the internal states of ${}^6\text{Li}$, ${}^7\text{Li}$ and ${}^{12}\text{C}$ with microscopic cluster models and examine their reliability. The prescription for truncating and discretizing the breakup continuum is given in § 2.2. In § 2.3, the coupling potentials in Eq. (2.2) are calculated.

2.1. Microscopic cluster-model wave functions

In this subsection, we construct the wave functions of the internal states of ${}^6\text{Li}$, ${}^7\text{Li}$ and ${}^{12}\text{C}$ on the basis of microscopic α -d, α -t and 3α cluster models, respectively, with total antisymmetrization between constituent nucleons taken into account. The reliability of the calculated wave functions is closely examined as to whether they could reproduce the observed binding energies, electron-scattering form factors, root-mean-square (r.m.s.) charge radii, $B(E\lambda)$ values and the phase shifts of the scattering between the constituent clusters. For ${}^{12}\text{C}$, full detail of such studies has already been described in Refs. 46) and 47), so that only a brief review is given here. A detailed description is given for the cases of ${}^6\text{Li}$ and ${}^7\text{Li}$.

2.1.1. ${}^6\text{Li}$ wave functions

In the α -d cluster model of ${}^6\text{Li}$, the totally antisymmetrized wave functions are written as

$$\phi_{01+K}({}^6\text{Li}; \text{g.s.}) = \mathcal{A} \{ \varphi(\alpha) [\varphi_{1+}(\text{d}) \otimes u_{l=0}^{(1+)}(\mathbf{r})]_{1+K} \} \quad (2.3)$$

for the ground state, and

$$\phi_{uK}({}^6\text{Li}; k) = \mathcal{A} \{ \varphi(\alpha) [\varphi_{1+}(\text{d}) \otimes u_l^{(1)}(k, \mathbf{r})]_{IK} \} \quad (2.4)$$

for the α -d scattering state with the asymptotic momentum $\hbar k$. Here, $\varphi(\alpha)$ and $\varphi_{1+}(\text{d})$ are the internal wave functions of the alpha and deuteron clusters, and $u_l(\mathbf{r})$ represents the α -d relative motion with angular momentum $\hbar l$. The ground state is assumed to be a pure S state ($l=0$). $\phi_{01+K}({}^6\text{Li}; \text{g.s.})$ and $\phi_{uK}({}^6\text{Li}; k)$ are orthonormalized as

$$\langle \phi_{01+K}({}^6\text{Li}; \text{g.s.}) | \phi_{01+K'}({}^6\text{Li}; \text{g.s.}) \rangle = \delta_{K,K'}, \quad (2.5a)$$

$$\langle \phi_{01+K}({}^6\text{Li}; \text{g.s.}) | \phi_{uK}({}^6\text{Li}; k) \rangle = 0, \quad (2.5b)$$

$$\langle \phi_{uK}({}^6\text{Li}; k) | \phi_{u'K'}({}^6\text{Li}; k') \rangle = \delta_{u,u'} \delta_{K,K'} \delta(k-k'). \quad (2.5c)$$

The asymptotic form of $u_{lm}^{(1)}(k, \mathbf{r})$ is taken to be

$$u_{lm}^{(1)}(k, \mathbf{r}) \rightarrow \sqrt{2/\pi} \sin \left(kr - \eta \ln(2kr) - \frac{l\pi}{2} + \sigma_l^c + \delta_u \right) r^{-1} Y_{lm}(\hat{\mathbf{r}}), \quad (2.6)$$

where σ_l^c and δ_u are the Coulomb and the nuclear phase shifts respectively and η is the

Sommerfeld parameter. In the framework of the resonating group method (RGM),⁴⁸⁾ $u_l(\mathbf{r})$ is obtained as a solution of the following integro-differential equation,

$$\langle \varphi(\alpha) [\varphi_{1+}(\mathbf{d}) \otimes Y_l(\hat{\mathbf{r}})]_{IK} | H(^6\text{Li}) - \varepsilon_{\text{tot}} | \mathcal{A} \{ \varphi(\alpha) [\varphi_{1+}(\mathbf{d}) \otimes u_l^{(I)}(\mathbf{r})]_{IK} \} \rangle = 0, \quad (2.7)$$

where $H(^6\text{Li})$ is the six-nucleon Hamiltonian, and ε_{tot} the total energy of ^6Li . In this framework, there is no free adjustable parameter once a certain set of wave functions $\varphi(\alpha)$, $\varphi(\mathbf{d})$ and the effective nucleon-nucleon interaction are chosen. Therefore, it is not obvious whether the wave functions obtained reproduce *simultaneously* well the binding energy, the elastic and inelastic electron-scattering form factors, the $B(E\lambda)$ strengths and α -d scattering phase shifts.

Since our aim here is just to *prepare* reliable wave functions of ^6Li internal states for the CC study of ^6Li scattering, we take more flexible, though more approximate, method than RGM to determine $u_{lm}^{(I)}$. Namely, instead of Eq. (2.7), we assume that $u_{lm}^{(I)}$ satisfies the integro-differential equation,

$$\sqrt{1-\mathcal{K}} \left(-\frac{\hbar^2}{2\mu_r} \nabla^2 + V_{\text{eff}} + a_{l,l} \cdot V_{\text{eff}}^{(\text{SO})} - \varepsilon \right) \sqrt{1-\mathcal{K}} u_{lm}^{(I)} = 0. \quad (2.8)$$

Here ε is the α -d c.m. energy and μ_r is the α -d reduced mass, and $\mathcal{K}$ is the exchange-overlap kernel defined by

$$\langle \varphi(\alpha) \varphi_{1+\nu}(\mathbf{d}) \delta(\mathbf{r}_0 - \mathbf{r}) | \mathcal{A} \{ \varphi(\alpha) \varphi_{1+\nu}(\mathbf{d}) \delta(\mathbf{r}'_0 - \mathbf{r}) \} \rangle = \delta(\mathbf{r}_0 - \mathbf{r}'_0) - \mathcal{K}(\mathbf{r}_0 - \mathbf{r}'_0). \quad (2.9)$$

The operator $\sqrt{1-\mathcal{K}}$ eliminates the Pauli forbidden components out of $u_{lm}^{(I)}$. In the actual calculation of $\mathcal{K}$, the radial part of $\varphi(\alpha)$ and $\varphi_{1+}(\mathbf{d})$ are assumed to be $(0s)^4$ and $(0s)^2$ harmonic-oscillator shell-model wave functions with the oscillator constant $m\omega/\hbar = 0.50 \text{ fm}^{-2}$. The Pauli forbidden states of $u_{lm}^{(I)}$ are, therefore, the $0s$ and the $0p$ states.

The local potentials $V_{\text{eff}}(r)$ and $V_{\text{eff}}^{(\text{SO})}(r)$ are the effective central and spin-orbit potentials which are functions of the α -d relative distance r . They are parametrized by a two-range Gaussian form and by a two-range Gaussian-derivative form, respectively. $a_{l,l}$ stands for $\{I(I+1) - l(l+1) - 1(1+1)\}/2$. The potential parameters are chosen l -dependently so that the energies of the ground state, -1.47 MeV , and the 3^+ resonance at 0.71 MeV are reproduced exactly and the low-energy ($\varepsilon \lesssim 15 \text{ MeV}$) α -d scattering phase shifts are reproduced satisfactorily (Fig. 1). The potential parameters thus obtained are listed in Table I.

Equation (2.8) is a modified type of the

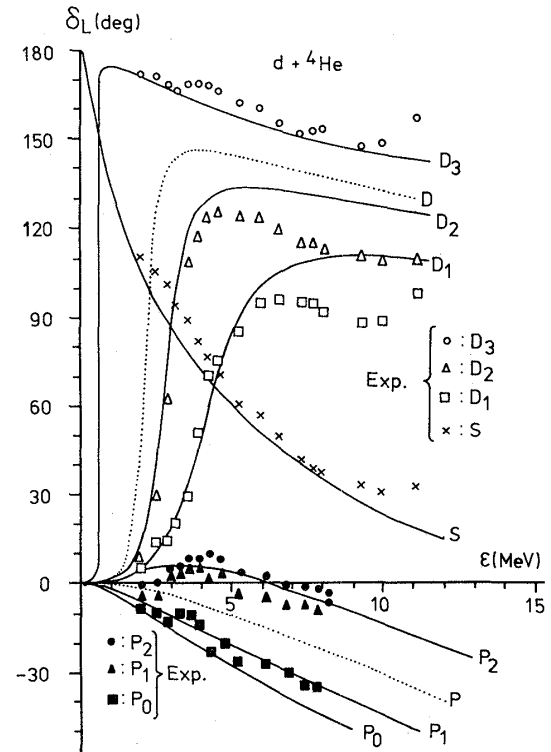


Fig. 1. α -d scattering phase shifts for $l=0, 1$ and 2 . The experimental data are taken from Ref. 53). The solid lines are given by the microscopic α -d cluster model. The dotted lines for $l=1$ and 2 are for the case of the α -d spin-orbit potentials switched off. ε is the α -d c.m. energy.

Table I. The parameters of the effective central and spin-orbit potentials between α and d in Eq. (2·8) for $l=0, 1$ and 2 .

parameters	$l=0$	$l=1$	$l=2$
$r_{1,l}(\text{fm})$	2.191	2.191	2.377
$r_{2,l}(\text{fm})$	1.607	1.607	1.852
$v_{1,l}(\text{MeV})$	-102.36	-49.41	-82.98
$v_{2,l}(\text{MeV})$	46.22	22.31	31.00
$v_{1,l}^{(SO)}(\text{MeV})$	—	11.76	2.31
$v_{2,l}^{(SO)}(\text{MeV})$	—	-9.87	-1.42
$R_{\text{CL}}(\text{fm})$	3.00	3.00	3.00

$$v_{\text{eff},l}(r) = v_{1,l}e^{-(r/r_{1,l})^2} + v_{2,l}e^{-(r/r_{2,l})^2} + V_{\text{CL}}(r).$$

$$V_{\text{eff},l}^{(SO)}(r) = v_{1,l}^{(SO)}re^{-(r/r_{1,l})^2} + v_{2,l}^{(SO)}re^{-(r/r_{2,l})^2}.$$

$$V_{\text{CL}}(r) = \begin{cases} Z_1Z_2e^2/r, & r \geq R_{\text{CL}} \\ (Z_1Z_2e^2/(2R_{\text{CL}}))(3 - r^2/R_{\text{CL}}^2), & r < R_{\text{CL}} \end{cases}$$

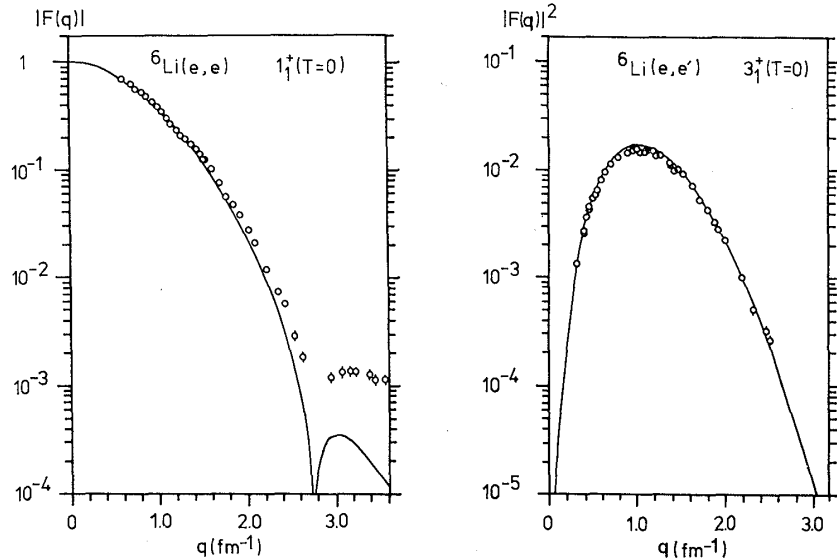


Fig. 2. Charge form factors of the elastic and inelastic electron scattering by ${}^6\text{Li}$. The experimental data are taken from Ref. 54). The solid lines represent the calculated results with the microscopic α -d cluster-model wave functions of the ground state and 3^+ resonance at $\varepsilon=0.71$ MeV. This figure is taken from Ref. 34).

equation of the orthogonality condition model (OCM)^{49),50)} and is considered to be a good approximation⁵¹⁾ of the RGM equation (2·7). This type of integro-differential equation can easily be solved for scattering states by the use of a variational method⁵²⁾ of the Kohn-Hulthen type.

The calculated microscopic wave functions of the ground and the 3^+ resonance states reproduce well the elastic and inelastic (g.s. $\rightarrow 3^+$) electron scattering form factors (Fig. 2). The calculated values of the r.m.s. charge radius of the ground state is 2.58 fm, and the $B(E2)$ of the transition from the ground state to the 3^+ state is $27.8 \text{ fm}^4 e^2$, which agree well with the observed values⁵⁵⁾ of $2.56 \pm 0.05 \text{ fm}$ and $21.8 \pm 4.8 \text{ fm}^4 e^2$, respectively.

In the CC calculations presented in the next section, we switch off $V_{\text{eff}}^{(SO)}$ in Eq. (2·8) after performing the examination of the wave functions described above and re-calculate

$u_{lm}^{(I)}$ using only V_{eff} in Table I, except when we analyze the experimental data for the ${}^6\text{Li}$ breakup reactions in §5. The ground-state energy is unchanged by this approximation, whereas the D -wave resonances (3^+ , 2^+ and 1^+), shown in Fig. 1, become degenerate at $\varepsilon \simeq 2$ MeV with the width of $\simeq 1$ MeV. The computational time for the CC calculation becomes much shorter than with $V_{\text{eff}}^{(S0)}$. The approximation is expected to be good since the ${}^6\text{Li}$ incident energies considered here are in most cases considerably higher ($E_{\text{lab}}/A_p \gtrsim 10$ MeV) than the energy-splitting between the three resonances (see Fig. 1). This has in fact been confirmed numerically in some of the cases presented here.

To simplify formulae in this approximation, we drop the deuteron spin and introduce the following new notations:

$$\psi_{00}({}^6\text{Li}; \text{g.s.}) = \mathcal{A}\{\varphi(\alpha)\varphi(d)u_{00}(\mathbf{r})\} \quad (2.10)$$

for the ground-state ($l=m=0$) wave function and

$$\psi_{lm}({}^6\text{Li}; k) = \mathcal{A}\{\varphi(\alpha)\varphi(d)u_{lm}(k, \mathbf{r})\} \quad (2.11)$$

for the scattering state wave function with the angular momentum $\hbar l$ and the momentum $\hbar k$.

2.1.2. ${}^7\text{Li}$ wave functions

We prepare the wave functions of the bound and α -t scattering states of ${}^7\text{Li}$ in the same manner as for ${}^6\text{Li}$. According to the microscopic α -t cluster model, the wave functions for the ground state ($3/2^-$) and the first excited state ($1/2^-$) are written as

$$\psi_{I=1,IK}({}^7\text{Li}) = \mathcal{A}\{\varphi(\alpha)[\varphi_{1/2^+}(t) \otimes u_{I=1}^{(I)}(\mathbf{r})]_{IK}\} \quad (2.12)$$

with $I=3/2$ and $I=1/2$, respectively, and that for the α -t scattering state with the asymptotic momentum $\hbar k$ and orbital angular momentum $\hbar l$ is written as

$$\psi_{lK}({}^7\text{Li}, k) = \mathcal{A}\{\varphi(\alpha)[\varphi_{1/2^+}(t) \otimes u_l^{(I)}(k, \mathbf{r})]_{IK}\}. \quad (2.13)$$

Here, $\varphi_{1/2^+}(t)$ stands for the internal wave function of the triton cluster, the radial part of which is assumed to be a harmonic-oscillator shell-model wave function with $(0s)^3$ configuration. The oscillator constant $m\omega/\hbar = 0.514 \text{ fm}^{-2}$ is used for both $\varphi(\alpha)$ and $\varphi(t)$.

In Eqs. (2.12) and (2.13) $u_{lm}^{(I)}$ is assumed to be given as a solution of the modified OCM equation (2.8) with the effective central and

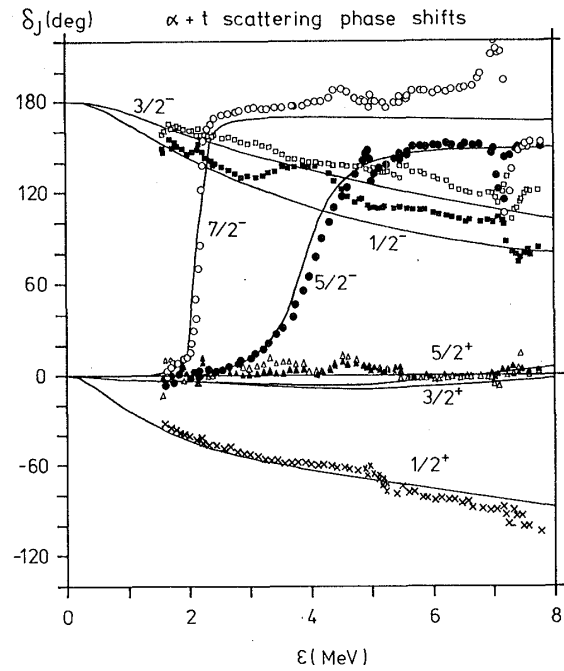


Fig. 3. α -t scattering phase shifts for $l=0$ ($I^\pi=1/2^+$), $l=1$ ($I^\pi=3/2^-, 1/2^-$), $l=2$ ($I^\pi=5/2^+, 3/2^+$) and $l=3$ ($I^\pi=7/2^-, 5/2^-$). The solid lines are given by the calculation with the microscopic α -t cluster model. The experimental data are taken from Ref. 56).

spin-orbit potentials. The potential parameters are, again, chosen l -dependently so as to reproduce well the energies of the ground state, -2.48 MeV, and the first excited state, -1.99 MeV, and the α -t scattering phase shifts⁵⁶⁾ (Fig. 3). It is well known that the effective α -t potential has a remarkable parity dependence.⁵⁷⁾ Keeping this in mind, we parametrize the effective central potential with a two-range Gaussian form for the even- l , with a one-range Gaussian form for the odd- l , and the spin-orbit potential with a one-range Gaussian form for all the l 's. For $l=1$, however, we adopt the Woods-Saxon derivative form for the spin-orbit potential in order to reproduce *simultaneously* the

Table II. The same as Table I for the potential between α and t for $l=0, 1, 2$ and 3. The form factor is taken to depend on parity.

parameters	$l=0$	$l=1$	$l=2$	$l=3$
$r_{1,l}(\text{fm})$	2.231	2.447	2.231	2.608
$r_{2,l}(\text{fm})$	3.310	—	3.310	—
$v_{1,l}(\text{MeV})$	-103.38	-86.29	-118.89	-75.38
$v_{2,l}(\text{MeV})$	13.78	—	15.85	—
$r_l^{(\text{SO})}(\text{fm})$	—	5.800	2.466	2.466
$v_l^{(\text{SO})}(\text{MeV})$	—	-6.000	-1.736	-2.736
$a_l^{(\text{SO})}(\text{fm})$	—	0.600	—	—

$$V_{\text{eff},l}(r) = \begin{cases} v_{1,l}e^{-(r/r_{1,l})^2} + v_{2,l}e^{-(r/r_{2,l})^2} & \text{for } l=\text{even}, \\ v_{1,l}e^{-(r/r_{1,l})^2} & \text{for } l=\text{odd}. \end{cases}$$

$$V_{\text{eff},l}^{(\text{SO})}(r) = \begin{cases} v_l^{(\text{SO})}e^{-(r/r_l^{(\text{SO})})^2} & \text{for } l \neq 1, \\ v_l^{(\text{SO})}e^x/(1+e^x), \quad x=(r-r_l^{(\text{SO})})/a_l^{(\text{SO})} & \text{for } l=1. \end{cases}$$

*) Coulomb potential form is the same as in Table I with $R_{\text{CL}}=2.0(\text{fm})$ for each l .

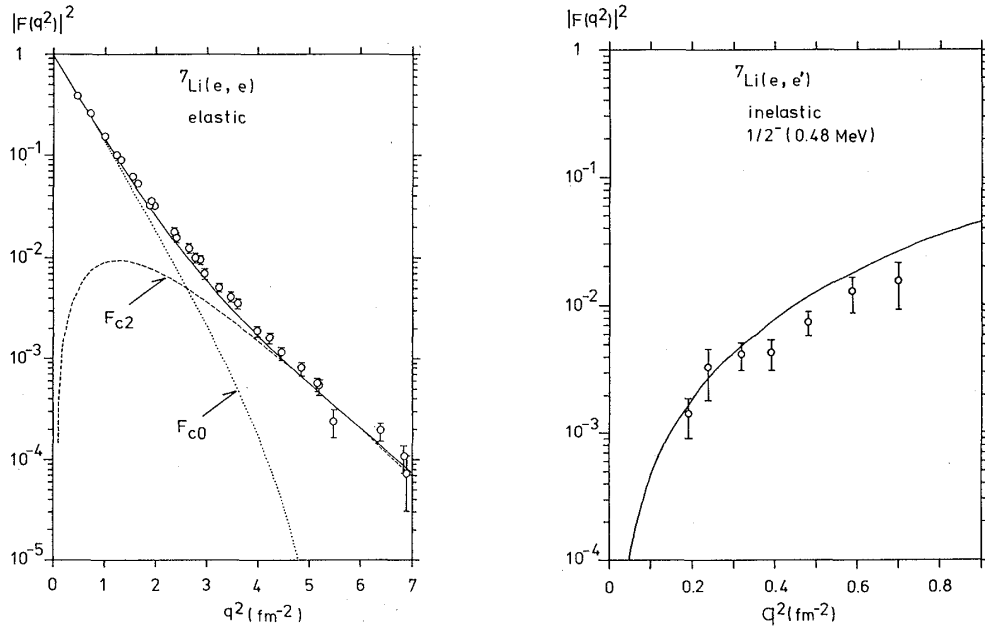


Fig. 4. Charge form factors of the elastic and inelastic electron scattering by ${}^7\text{Li}$. The solid lines represent the calculated results with the microscopic α -t cluster-model wave functions of the ground state ($3/2^-$) and the first excited state ($1/2^-$). The dotted and dashed lines represent respectively the monopole and the quadrupole contributions to the solid line. The experimental data are taken from Ref. 58) for the elastic scattering and from Ref. 59) for the inelastic one.

energy-splitting between the two bound states and the splitting between the phase shift for $I=3/2$ and that for $I=1/2$ (Fig. 3). The potential parameters obtained are listed in Table II.

The calculated wave functions of the ground state ($3/2^-$) and the first excited state ($1/2^-$) reproduce well the experimental data^{58,59} on the elastic and inelastic ($3/2^- \rightarrow 1/2^-$) electron-scattering form factors (Fig. 4). Since the ground state is a P -state ($l=1$), both monopole ($\lambda=0$) and quadrupole ($\lambda=2$) components of the wave function contribute to the elastic form factor. The calculated value of the r.m.s. charge radius is 2.44 fm and that of the quadrupole moment of the ground state is $-3.90 \text{ fm}^2 e^2$, which agree well with the observed values⁵⁸ of $2.39 \pm 0.03 \text{ fm}$ and $-4.20 \pm 0.25 \text{ fm}^2 e^2$, respectively. As for the $B(E2)$ value of the transition from the ground state to the first excited state, the calculated value $6.83 \text{ fm}^4 e^2$ agrees very well with the observed one,⁵⁹ $6.70 \pm 0.15 \text{ fm}^4 e^2$.

2.1.3. ^{12}C wave functions

The precise wave functions of the low-lying states of ^{12}C were given by Kamimura and Fukushima^{46,47} on the basis of the totally antisymmetrized 3α cluster model,

$$\psi_{IK}(^{12}\text{C}) = \mathcal{A}\{\varphi(\alpha)\varphi(\alpha)\varphi(\alpha)u_{IK}(\mathbf{r}, \mathbf{R})\}, \quad (2.14)$$

where $u_{IK}(\mathbf{r}, \mathbf{R})$ is the wave function of relative motion between 3α clusters with the total angular momentum $\hbar I$. They obtained it by solving the RGM equation for $\psi_{IK}(^{12}\text{C})$, diagonalizing the twelve-body total Hamiltonian in a subspace spanned by a set of L^2 -normalizable basis functions. In the RGM calculation, the Volkov potential of the second type⁶⁰ with the Majorana mixture $M=0.59$ was adopted as the effective nucleon-nucleon interaction, and the harmonic oscillator constant for the α clusters was taken to be $m\omega/\hbar=0.55 \text{ fm}^{-2}$. These values were determined so as to reproduce well the α - α scattering phase shifts in a wide range of energy as well as the binding energy and the size of the α particle itself.

The calculated wave functions reproduced well⁴⁷ the $B(E\lambda)$ strengths, the α -decay widths and the electron-scattering form factors of 0_1^+ , 2_1^+ , 4_1^+ , 0_2^+ , 2_2^+ , 1_1^- and 3_1^- states, as described in detail in Ref. 47).

In the present CC calculation, we employ 0_1^+ , 2_1^+ , 3_1^- , 0_2^+ and 2_2^+ states, the last two states being well-developed clustering states. In addition, we employ for $I^\pi=0^+$ and $I^\pi=2^+$ the lowest two states above the five states as the representative of the non-resonant 3α -breakup continuum. Other non-resonant states have been found to have negligible coupling to the ground state.

2.2. Discretization of breakup continua

As mentioned in the previous subsection, the 3α -breakup continuum of ^{12}C was described by a finite number of discrete states obtained by the diagonalization.

The breakup continuum of ^6Li (^7Li) was described as the α -d (α -t) scattering states (see Eqs. (2.4) and (2.13)). We employ the method of CDCC^{2)~5),34)~36)} to discretize it. In order to avoid complexity of notations, we describe the method only for the ^6Li case without $V_{\text{eff}}^{(\text{SO})}$, namely, with the notations of Eqs. (2.10) and (2.11).

The explicit form of Eq. (2.1) in this case is

$$\Psi_M = \left\{ \phi_{l=0}({}^6\text{Li}; \text{g.s.}) \chi_{l=0, L=J}^{(J)}(\hat{P}_0, \mathbf{R}) + \sum_{l=0}^{l_{\max}} \sum_L \int_0^{k_{\max}} [\phi_l({}^6\text{Li}, k) \otimes \chi_{lL}^{(J)}(P_k, \mathbf{R})]_{JM} dk \right\} \cdot \Phi_T, \quad (2.15)$$

where J and M are the total angular momentum and its z -component, L is the orbital angular momentum ($|J-l| \leq L \leq J+l$), and Φ_T stands for the ground-state wave function of the target nucleus. In Eq. (2.15), $l_{\max}$ and $k_{\max}$ are introduced to truncate the three-body model space. The wave functions $\chi_{0J}^{(J)}(\hat{P}_0, \mathbf{R})$ and $\chi_{lL}^{(J)}(P_k, \mathbf{R})$ describe the ${}^6\text{Li}$ -target relative motion in the incident channel and in the breakup channel, respectively. $\hbar\hat{P}_0$ and $\hbar P_k$ are the asymptotic momenta of the corresponding channels. They satisfy the energy-conservation relations as

$$\frac{\hbar^2}{2\mu} \hat{P}_0^2 + \hat{\varepsilon}_0 = \frac{\hbar^2}{2\mu} P_k^2 + \varepsilon_k = E. \quad \left(\varepsilon_k = \frac{\hbar^2}{2\mu_r} k^2 \right) \quad (2.16)$$

Here $\hat{\varepsilon}_0 (= -1.47 \text{ MeV})$ is the ${}^6\text{Li}$ ground-state energy with respect to the α -d breakup threshold and μ is the reduced mass of the ${}^6\text{Li}$ -target system.

Following the method of CDCC,^{2)~5), 34)~36)} the truncated k -continuum is divided into a finite number of narrow bins as $[k_{i-1}, k_i]$ ($i=1 \sim N$) with $k_0=0$ and $k_N=k_{\max}$. Though N could depend on l in practice, we drop the suffix l for simplicity. Here, we assume that, in each bin, any $\chi_{lL}^{(J)}(P_k, \mathbf{R})$ is approximately proportional to the function $\chi_{lL}^{(J)}(\hat{P}_i, \mathbf{R})$. Namely, we assume that

$$\chi_{lL}^{(J)}(P_k, \mathbf{R}) \simeq f_{il}(k) \chi_{lL}^{(J)}(\hat{P}_i, \mathbf{R}), \quad (k_{i-1} \leq k \leq k_i) \quad (2.17)$$

where $\hat{P}_i$ is related to the mean energy $\hat{\varepsilon}_i$ of the interval $[k_{i-1}, k_i]$ as

$$\frac{\hbar^2}{2\mu} \hat{P}_i^2 + \hat{\varepsilon}_i = E. \quad (2.18)$$

We take $f_{il}(k)$ and $\hat{\varepsilon}_i$ as follows:³⁴⁾

- (i) For a “non-resonance” bin where no α -d resonance exists

$$f_{il}(k) = 1 \quad (2.19)$$

and

$$\hat{\varepsilon}_i = \frac{\hbar^2}{2\mu} \left[\left(\frac{k_{i-1} + k_i}{2} \right)^2 + \frac{1}{12} (\Delta k_i)^2 \right], \quad (2.20)$$

where $\Delta k_i = k_i - k_{i-1}$.

- (ii) For the “resonance” bin in which an α -d resonance exists at $\varepsilon = \varepsilon_{\text{res}}$ with a width Γ , we take

$$f_{il}(k) = \left| \frac{i\Gamma/2}{\varepsilon_k - \hat{\varepsilon}_i + i\Gamma/2} \right| \quad (2.21)$$

and

$$\hat{\varepsilon}_i = \varepsilon_{\text{res}}. \quad (2.22)$$

Validity for (2·19) for the non-resonant continuum was extensively examined by Yahiro et al.^{4,5)} in the case of deuteron scattering. They checked the convergence of the elastic and breakup S-matrix elements with respect to narrowing the widths Δk_i . On the other hand, the reason for assuming the factor (2·21) for the resonance bin is as follows. The matrix element of the ${}^6\text{Li}$ -target coupling potential between the ${}^6\text{Li}$ ground state $\phi({}^6\text{Li}; \text{g.s.})$ and an α -d scattering state $\psi({}^6\text{Li}, k)$ in the resonance bin is found to be nearly proportional to the factor (2·21) times that between $\phi({}^6\text{Li}; \text{g.s.})$ and $\psi({}^6\text{Li}, k_{\text{res}})$ at $\varepsilon = \varepsilon_{\text{res}}$. This property holds well for a resonance with a narrow width.

Under the assumptions (2·17)~(2·22), the total wave function Ψ_{JM} is written as

$$\Psi_{JM} = \sum_{l=0}^{l_{\max}} \sum_L \sum_{i=0 \text{ or } 1}^N [\hat{\psi}_{il}({}^6\text{Li}) \otimes \hat{\chi}_{iL}^{(J)}(\mathbf{R})]_{JM} \cdot \Phi_T \quad (2\cdot23)$$

with

$$\hat{\psi}_{ilm}({}^6\text{Li}) = 1/\sqrt{N_{il}} \cdot \int_{k_{i-1}}^{k_i} \phi_{lm}({}^6\text{Li}; k) f_{il}(k) dk, \quad (2\cdot24)$$

$$\hat{\chi}_{iL}^{(J)}(\mathbf{R}) = \sqrt{N_{il}} \chi_{iL}^{(J)}(\hat{P}_i, \mathbf{R}), \quad (2\cdot25)$$

$$N_{il} = \int_{k_{i-1}}^{k_i} [f_{il}(k)]^2 dk, \quad (2\cdot26)$$

where new notations

$$\hat{\psi}_{i=0, l=0}({}^6\text{Li}) \equiv \psi_{l=0}({}^6\text{Li}; \text{g.s.}), \quad (2\cdot27)$$

$$\hat{\chi}_{i=0, l=0, L}^{(J)}(\mathbf{R}) \equiv \chi_{l=0, L}^{(J)}(\hat{P}_0, \mathbf{R}) \quad (2\cdot28)$$

are used for the incident channel. The summation over i in Eq. (2·23) includes $i=0$ only when $l=0$. By using Eqs. (2·5a)~(2·5c) and (2·24), one can easily find that $\hat{\psi}_{ilm}({}^6\text{Li})$ satisfy the orthonormal condition

$$\langle \hat{\psi}_{ilm}({}^6\text{Li}) | \hat{\psi}_{i'l'm'}({}^6\text{Li}) \rangle = \delta_{ii'} \delta_{ll'} \delta_{mm'}, \quad (2\cdot29)$$

which allows one to treat the discretized breakup channels as like usual inelastic channels.

Now we introduce the radial wave functions $\hat{\chi}_{iL}^{(J)}(R)$ by

$$\hat{\chi}_{iLM}^{(J)}(\mathbf{R}) = \hat{\chi}_{iL}^{(J)}(R) / R \cdot i^L Y_{LM}(\hat{\mathbf{R}}). \quad (2\cdot30)$$

The asymptotic form ($R \rightarrow \infty$) of $\hat{\chi}_{iL}^{(J)}(R)$ is taken to be

$$\hat{\chi}_{iL}^{(J)}(R) \rightarrow U_L^{(-)}(\hat{P}_0, R) \delta_{i0} \delta_{l0} \delta_{LJ} - \sqrt{\hat{P}_0 / \hat{P}_i} \hat{S}_{iL,00J}^{(J)} U_L^{(+)}(\hat{P}_i, R), \quad (2\cdot31)$$

where $U^{(-)}$ ($U^{(+)}$) is the incoming (outgoing) Coulomb wave function.

Now, it seems noteworthy that in the region which contributes most to the folded CC potentials the wave packet for the resonance bin is found to be very close to the approximate, “quasi-bound” wave function of the resonance. The latter is obtained by diagonalizing the norm and Hamiltonian kernels simultaneously in terms of appropriate basis functions. This fact may justify the approximation of 0_2^+ , 2_2^+ and 3_1^- resonance states of ${}^{12}\text{C}$

above the 3α threshold with corresponding quasi-bound states obtained by the diagonalization.

2.3. Coupled-channel equations with double-folding interactions

The diagonal and coupling potentials in the CC equations (2·2) are calculated by doubly-folding the effective nucleon-nucleon interaction with the nucleon densities of the target and ${}^6\text{Li}$:

$$\widehat{V}_{ilm,i'l'm'}(\mathbf{R}) = \int \rho_T(\mathbf{r}_T) \widehat{\rho}_{ilm,i'l'm'}(\mathbf{r}_P) v_{NN}(\mathbf{R} - \mathbf{r}_P + \mathbf{r}_T) d\mathbf{r}_P d\mathbf{r}_T. \quad (2\cdot32)$$

Here, the ground-state nucleon density of the target nucleus with mass number A is defined as

$$\rho_T(\mathbf{r}_T) = \langle \Phi_T | \sum_{j=1}^A \delta(\mathbf{r}_T - \mathbf{r}_j) | \Phi_T \rangle, \quad (2\cdot33)$$

and the projectile nucleon density between the states $\widehat{\phi}_{ilm}({}^6\text{Li})$ and $\widehat{\phi}_{i'l'm'}({}^6\text{Li})$ is defined^{(34)~(36)} as

$$\widehat{\rho}_{ilm,i'l'm'}(\mathbf{r}_P) = \langle \widehat{\phi}_{ilm}({}^6\text{Li}) | \sum_{j=1}^6 \delta(\mathbf{r}_P - \mathbf{r}_j) | \widehat{\phi}_{i'l'm'}({}^6\text{Li}) \rangle. \quad (2\cdot34)$$

In the calculation of $\widehat{\rho}_{ilm,i'l'm'}(\mathbf{r}_P)$, the total antisymmetrization between six nucleons is exactly treated by means of the generator-coordinate-method (GCM) technique.⁽⁶¹⁾

As for the effective nucleon-nucleon interaction v_{NN} , we use the spin- and isospin-singlet ($S=T=0$) component of the M3Y interaction.⁽²³⁾ The M3Y interaction used here is the same one as used in the double-folding model of Satchler and Love.⁽¹³⁾ The interaction is the *realistic* effective interaction which was derived by Bertsch et al.⁽²³⁾ from G -matrix elements. It was expressed as a sum of three Yukawa terms. One term was taken to be the OPEP which, however, does not contribute to the $S=T=0$ component. The ranges of the other terms were chosen to be 0.4 fm and 0.25 fm. The strengths of the terms for the even states were determined so as to reproduce the G -matrix elements calculated with the Reid soft-core potential, and those for the odd states were chosen to fit to the Sussex matrix elements.⁽²³⁾ It has the form⁽¹³⁾ as

$$v_{NN}(\mathbf{r}) = 7999 \frac{e^{-4r}}{4r} - 2134 \frac{e^{-2.5r}}{2.5r} - 262 \cdot \delta(\mathbf{r}), \quad (2\cdot35)$$

$$v_{NN}(\mathbf{r}) = 6315 \frac{e^{-4r}}{4r} - 1961 \frac{e^{-2.5r}}{2.5r} - 81 \cdot \delta(\mathbf{r}), \quad (2\cdot36)$$

where all energies are in MeV and lengths in fm. In Eqs. (2·35) and (2·36), the last terms were introduced to simulate the knock-on exchange contribution.⁽¹³⁾ It is well known⁽¹³⁾ that there is negligible difference between the double-folded potentials calculated with the two types of interactions. Thus we adopt the former in the actual calculations shown below.

Since the incident energies ($E/A \gtrsim 10$ MeV) considered in this work are considerably higher than the Coulomb-potential barrier, the coupling due to the Coulomb interaction is neglected. The usual Coulomb potential $V_{\text{coul}}(R)$ for the uniform-charge distribution is

assumed commonly for all the channels. The effect of the Coulomb interaction which induces the breakup of projectiles will be discussed briefly in § 5.

In general, the CC potentials U_{ij} in Eq.(2·2) are complex while the double-folded potentials in Eq. (2·32) are real because the M3Y interaction is real.²³⁾ We therefore assume U_{ij} to be given by

$$U_{iL,i'L'}^{(j)}(R) = (N_R + iN_I) \cdot \hat{V}_{iL,i'L'}^{(j)}(R) \quad (2\cdot37)$$

with

$$\hat{V}_{iL,i'L'}^{(j)}(R) = \sum_{mm'KK'} (lmLK|JM)(l'm'L'K'|JM) i^{L'-L} \int Y_{LK}^*(\hat{\mathbf{R}}) \hat{V}_{ilm,i'l'm'}(\mathbf{R}) Y_{L'K'}(\hat{\mathbf{R}}) d\hat{\mathbf{R}}, \quad (2\cdot38)$$

where N_R is the renormalization factor for the real potentials which is kept fixed at $N_R=1.0$ in all the CC calculations except in the cases of the inelastic scattering of ${}^6\text{Li}$ by ${}^{12}\text{C}$ and N_I is the strength factor of the imaginary part which is taken to be common to all the potentials in the CC equations. The value of N_I is the *only one* adjustable parameter in the present CC framework.

A reason why we assume the imaginary part to have the same geometry of the real part as in Eq. (2·37) is as follows: In Fig. 5, we illustrate the double-folding potential for the elastic channel which will be used in § 3.5 for the ${}^6\text{Li}$ scattering from ${}^{12}\text{C}$ and ${}^{208}\text{Pb}$ at 156 MeV; here $N_R=1.0$ and $N_I=0.47$ for ${}^{12}\text{C}$ and $N_R=1.0$ and $N_I=0.60$ for ${}^{208}\text{Pb}$. Also shown in the figure are the potentials for the same systems employed in the analysis³¹⁾ with the adiabatic-three-body model which was successful in describing the scattering as mentioned in § 1; the potential is obtained by folding the α -target and d-target optical

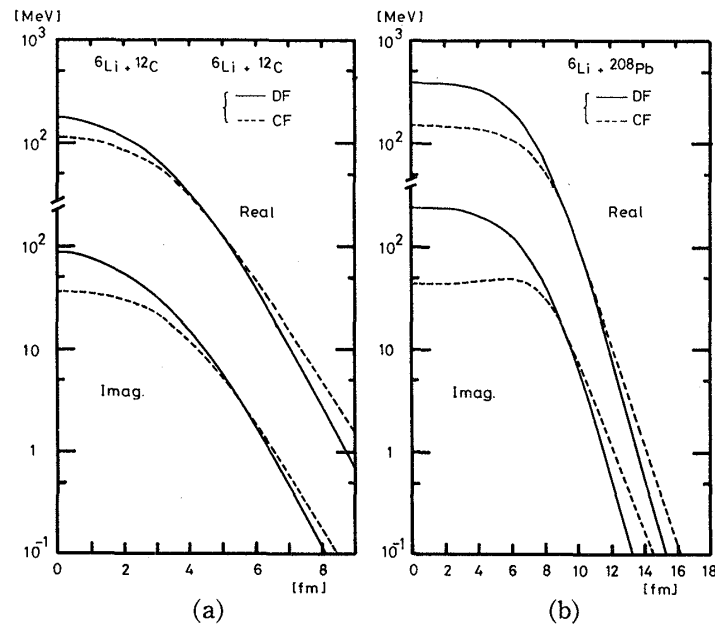


Fig. 5. Form factors of the double-folding-model potentials (solid curves) (a) for ${}^6\text{Li} + {}^{12}\text{C}$ system and (b) for ${}^6\text{Li} + {}^{208}\text{Pb}$ one compared with those of the cluster-folding model potentials³¹⁾ (dashed curves) for the systems evaluated at $E_{\text{lab}}=156$ MeV. For the double-folding-model potentials, the values of N_R and N_I are 1.0 and 0.47 for the ${}^{12}\text{C}$ target, respectively, and 1.0 and 0.60 for the ${}^{208}\text{Pb}$ target.

potentials into the α -d cluster-model wave functions of the ${}^6\text{Li}$ ground state. One sees that the assumption of Eq. (2·37) works well in the surface region which is most important for the scattering; also this is the case for the coupling potentials resulting an approximately common value of N_i .

The CC equations for $\hat{\chi}_{iL}^{(j)}(R)$'s are now written as

$$\begin{aligned} & \left[-\frac{\hbar^2}{2\mu} \frac{d^2}{dR^2} + \frac{\hbar^2 L(L+1)}{2\mu R^2} + V_{\text{coul}}(R) - (E - \varepsilon_i) \right] \hat{\chi}_{iL}^{(j)}(R) \\ & = - \sum_{i'l'} U_{iL, i'l'}^{(j)}(R) \hat{\chi}_{i'l'}^{(j)}(R). \end{aligned} \quad (2\cdot39)$$

($i=0 \sim N$, $l=0 \sim l_{\text{max}}$, $|J-l| \leq L \leq J+l$)

The right-hand side of Eq. (2·39) includes the diagonal potential. The single-channel case of the present CC framework corresponds to one of the versions of the double-folding model,¹³⁾ though more flexible shape of the imaginary part is allowed to be used in the model. Therefore, one can discuss directly, for example, the relation between the projectile-breakup effect of ${}^6\text{Li}$ and the anomalous value of $N_R (\simeq 0.5)$ mentioned in § 1.

§ 3. Elastic scattering of ${}^6\text{Li}$

In this section, we analyze the elastic scattering of ${}^6\text{Li}$ by ${}^{12}\text{C}$, ${}^{28}\text{Si}$, ${}^{40}\text{Ca}$, ${}^{48}\text{Ca}$, ${}^{58}\text{Ni}$ and ${}^{208}\text{Pb}$ at the incident energies $E_{\text{lab}} = 34 \sim 169$ MeV. The ground-state nucleon densities of ${}^{28}\text{Si}$, ${}^{40}\text{Ca}$ and ${}^{58}\text{Ni}$ are taken to be the same as those adopted in the double-folding-model study of Ref. 16), whereas the Hartree-Fock densities of Ref. 62) are employed for ${}^{48}\text{Ca}$ and ${}^{208}\text{Pb}$. As for the density of ${}^{12}\text{C}$, we adopt that calculated with the microscopic 3α -RGM wave functions⁴⁷⁾ discussed in § 2.1. All these densities reproduce well the electron-scattering form factors and the r.m.s. charge radii.

3.1. Truncation and discretization of breakup continua

Before performing the CC analyses of the elastic scattering of ${}^6\text{Li}$, we first set up the three-body model space for the CC calculation. For this purpose we have to choose appropriate values of l_{max} , k_{max} and N (or Δk) in Eqs. (2·15) and (2·23) to truncate and discretize the α -d breakup continua. To this end, we have examined the convergence of the S-matrix elements with respect to increasing l_{max} , k_{max} and narrowing Δk .^{34),35)} The examination is made in the case of ${}^6\text{Li}$ scattering from ${}^{28}\text{Si}$ at $E_{\text{lab}} = 99$ MeV. The potential factors, N_R and N_i , are taken to be 1.0 and 0.623, respectively, which are the realistic values to be used later.

For the check of convergence with increasing k_{max} and narrowing Δk , we took into account only the S-wave ($l=0$) continuum to save the computing time, which will be sufficient in the light of the experience^{4),5)} of similar test for deuteron scattering. First, we increased k_{max} according as $k_{\text{max}} = N\Delta k$ ($N=1 \sim 6$) with Δk fixed at $3/8 \text{ fm}^{-1}$, and next we diminished Δk according as $\Delta k = k_{\text{max}}/N$ ($N=1 \sim 10$) with k_{max} fixed at $3/2 \text{ fm}^{-1}$. Figure 6 shows an example of the change of elastic S-matrix elements for $J=30$ with respect to (a) increasing k_{max} and (b) narrowing Δk ; sufficient convergence is achieved at $k_{\text{max}} \simeq 1.0$ and $\Delta k \simeq 3/8$. Similar test has been done on the convergence of the S-matrix elements for other values of J and of the elastic-scattering cross sections, and we found that it is

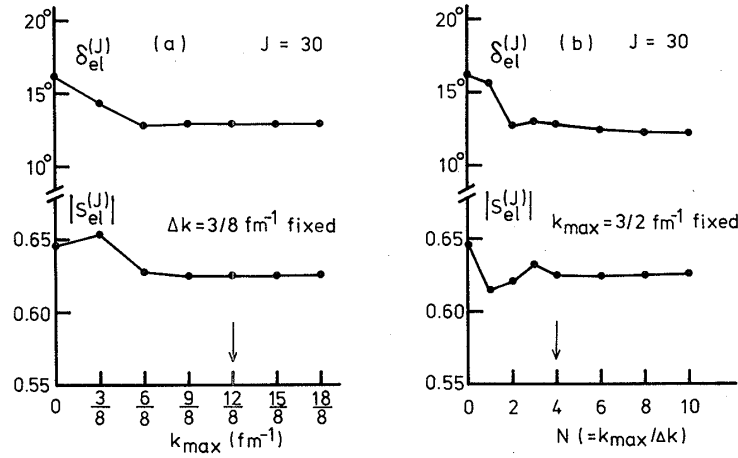


Fig. 6. Convergence of the elastic component of S -matrix elements $\hat{S}_{00,00J}^{(J)}$ for $J=30$, (a) with respect to increasing $k_{\max}$ and (b) with respect to narrowing the width Δk . $|S_{\text{el}}^{(J)}|$ and $\delta_{\text{el}}^{(J)}$ stand for the modulus and the nuclear-phase shift, respectively, which are defined as $\hat{S}_{00,00J}^{(J)} = |S_{\text{el}}^{(J)}| \exp(2i\delta_{\text{el}}^{(J)})$. This figure is taken from Ref. 34).

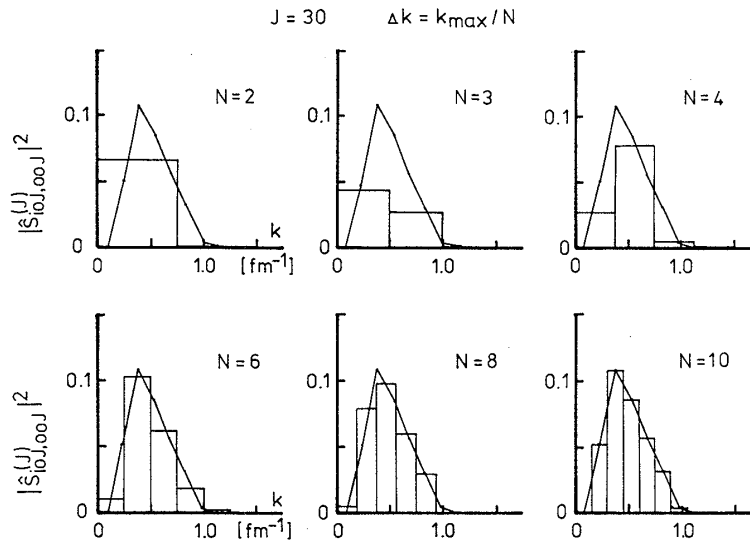


Fig. 7. Convergence of squared moduli of the S -wave ($l=0$) breakup component of the S -matrix elements, $\hat{S}_{i0,00J}^{(J)}$ ($i=1 \sim N$) for $J=30$ with respect to narrowing the width Δk .³⁵⁾ Here, $k_{\max}$ is fixed to 1.5 fm^{-1} . The shape of the $N=10$ case is illustrated by the folded line in each graph.

sufficient to take $k_{\max} = 1.5 \text{ fm}^{-1}$ and $\Delta k = 3/8 \text{ fm}^{-1}$ for the elastic scattering. On the other hand, Fig. 7 shows the change of the breakup components of the S -matrix elements with respect to narrowing Δk with $k_{\max}$ kept fixed at 1.5 fm^{-1} . From this test, it will be desirable to take $\Delta k \lesssim 0.25 \text{ fm}^{-1}$ when we calculate breakup cross sections.

Next, we examined $l_{\max}$. The elastic S -matrix elements have been calculated at several J values for $l_{\max} = 2$ and 4 , where $k_{\max}$ was fixed at 1.0 fm^{-1} and Δk at 0.5 fm^{-1} . We found that the S -matrix elements calculated with $l_{\max} = 2$ differ by less than 1 % from those calculated with $l_{\max} = 4$. Thus, the effect of the F -wave ($l=3$) and G -wave ($l=4$) is negligible in this case.

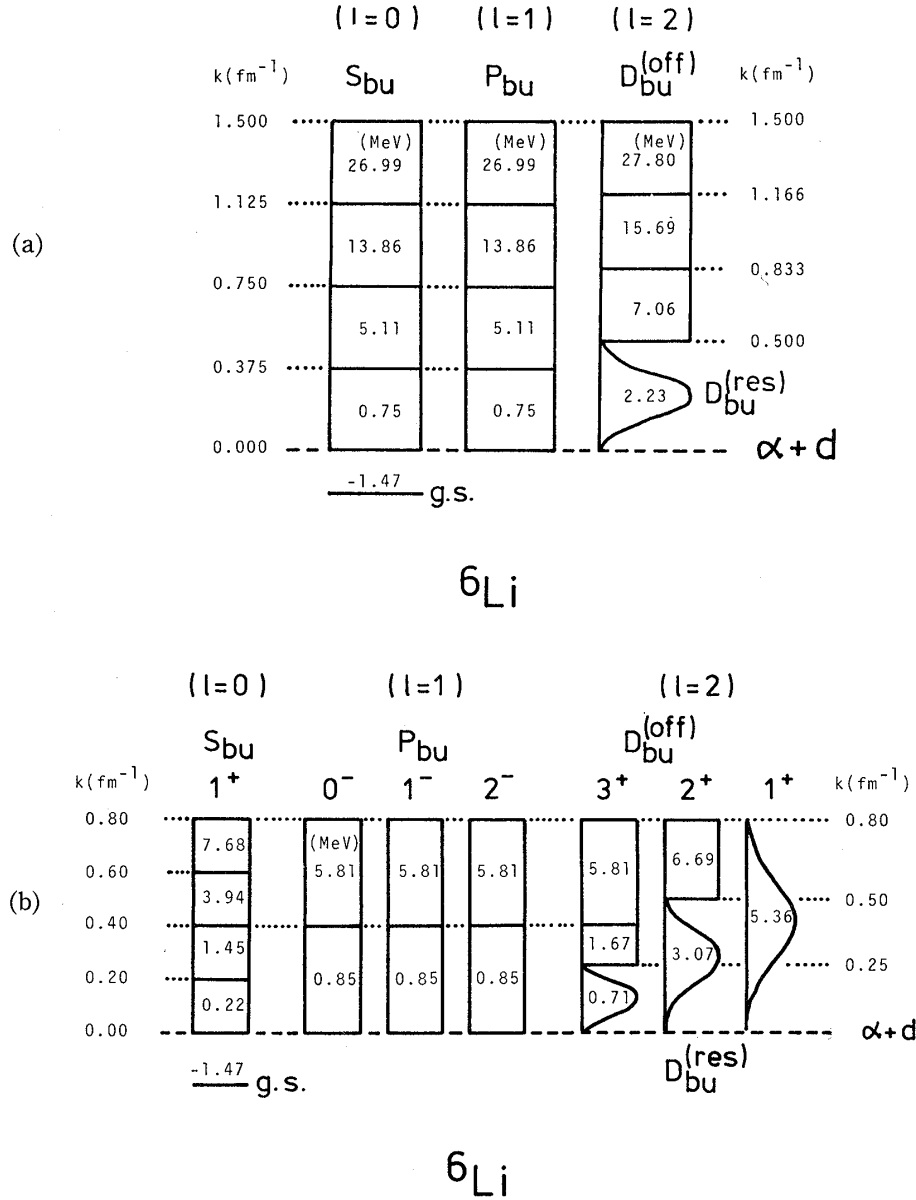


Fig. 8. (a) Truncation and discretization of the α -d breakup continuum of ${}^6\text{Li}$. Here, the spin-orbit potential $V_{\text{eff}}^{(\text{SO})}$ has been switched off. Each momentum continuum ($k \leq 1.5 \text{ fm}^{-1}$) of the α -d partial waves (S , P and D) is divided into four bins. The mean energies of each bin is presented in each “box”. The lowest bin of the D -wave stands for the D -wave resonance. (b) The same as (a) for more precise discretization of the α -d breakup continuum of ${}^6\text{Li}$ with $V_{\text{eff}}^{(\text{SO})}$ taken into account. Here k_{max} is taken to be 0.8 fm^{-1} . This is used in § 5 for the calculation of the ${}^6\text{Li}$ projectile-breakup cross sections.

Keeping those results in mind, we truncate and discretize the α -d breakup continua as follows :

- (i) When we investigate the virtual-breakup effects in the elastic scattering, we take $l_{\text{max}}=2$ and $k_{\text{max}}=1.5 \text{ fm}^{-1}$ and neglect the spin of the deuteron cluster, i.e., neglect the spin-orbit potential ($V_{\text{eff}}^{(\text{SO})}$) which acts between α and d , as discussed before. We then divide the momentum continuum of each partial wave into four bins ($N=4$) as illustrated in Fig. 8(a); each bin in the S -wave ($l=0$) and P -wave ($l=1$) has a common width of $\Delta k=3/8 \text{ fm}^{-1}$. The lowest bin of the D -wave ($l=2$) is a “reso-

nance bin" with $\Delta k = 1/2 \text{ fm}^{-1}$ which corresponds to the degenerated D -wave resonances mentioned in § 2.1.1, whereas the remaining "non-resonant" continuum in the D -wave is divided into three bins with a common width of $\Delta k = 1/3 \text{ fm}^{-1}$. The central energy and the width of the resonance, ε_{res} and Γ respectively, are taken to be the calculated values³⁴⁾ of 2.23 MeV and 0.75 MeV, respectively.

- (ii) When we analyze the experimental data on (${}^6\text{Li}$, αd) breakup reactions in § 5.1, we take $l_{\text{max}} = 2$ and $k_{\text{max}} = 0.80 \text{ fm}^{-1}$ and take into account the deuteron spin in the construction of the internal wave functions. As a result, the P -wave and D -wave continua are each split into three parts, 2^- , 1^- , 0^- and 3^+ , 2^+ , 1^+ respectively, while the S -wave one (1^+) remains unchanged. The truncated model space is discretized as shown in Fig. 8(b). The lowest bins of the D -wave in the figure correspond to the 3^+ , 2^+ and 1^+ resonances; the central energies and the widths are taken to be $\varepsilon_{\text{res}} = 0.71 \text{ MeV}$ and $\Gamma = 0.024 \text{ MeV}$ for the 3^+ resonance, 3.07 MeV and 0.710 MeV for the 2^+ one and 5.36 MeV and 1.80 MeV for the 1^+ one, respectively.

The adopted value of $k_{\text{max}} = 0.8 \text{ fm}^{-1}$ is somewhat smaller than the value ($k_{\text{max}} \gtrsim 1.0 \text{ fm}^{-1}$) recommended by the above examination and the widths (Δk) assigned to some of the non-resonance bins in the P - and D -waves are slightly larger than the recommended ($\Delta k \lesssim 0.25 \text{ fm}^{-1}$), as is shown in Fig. 8(b). These values are chosen to save the computing time for solving the coupled-channel equations. However, the model space and the discretization of the figure would be sufficient for an examination of the validity of the method of CDCC to be used in the analyses of ${}^6\text{Li}$ breakup reactions, particularly of the reactions in which the relative energy between the emitted α -particle and deuteron is relatively small, say $\varepsilon \lesssim$ several MeV. In fact, we shall analyze in § 5 such kind of breakup reactions and show that the calculated breakup cross sections reproduce well the measured ones.

For the sake of later convenience, we denote the total S -wave (P -wave) breakup channels by S_{bu} (P_{bu}), whereas the D -wave resonance channel by $D_{\text{bu}}^{(\text{res})}$ and the total D -wave off-resonance channels by $D_{\text{bu}}^{(\text{off})}$.

3.2. Single-channel calculations

First, we make a usual double-folding-model calculation, i.e., the single-channel case in the present CC framework. The optimum values of the potential strength parameters, N_R and N_I , are determined by fitting the calculated elastic-scattering cross

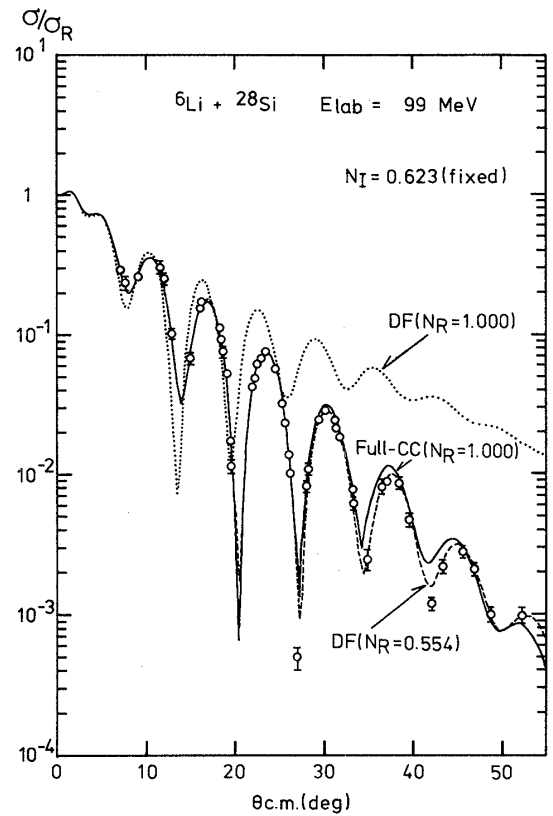


Fig. 9. The elastic scattering of ${}^6\text{Li}$ from ${}^{28}\text{Si}$ at $E_{\text{lab}} = 99 \text{ MeV}$. The experimental data are taken from Ref. 64). The solid curve represents the result of the full-CC calculation with $N_R = 1.0$. The dotted and dashed curves are the result of the double-folding-model (single-channel) calculation with $N_R = 1.0$ and 0.554, respectively. The latter is the best-fit result of the model calculation. In all the calculations, $N_I = 0.623$ is used. This figure is taken from Ref. 34).

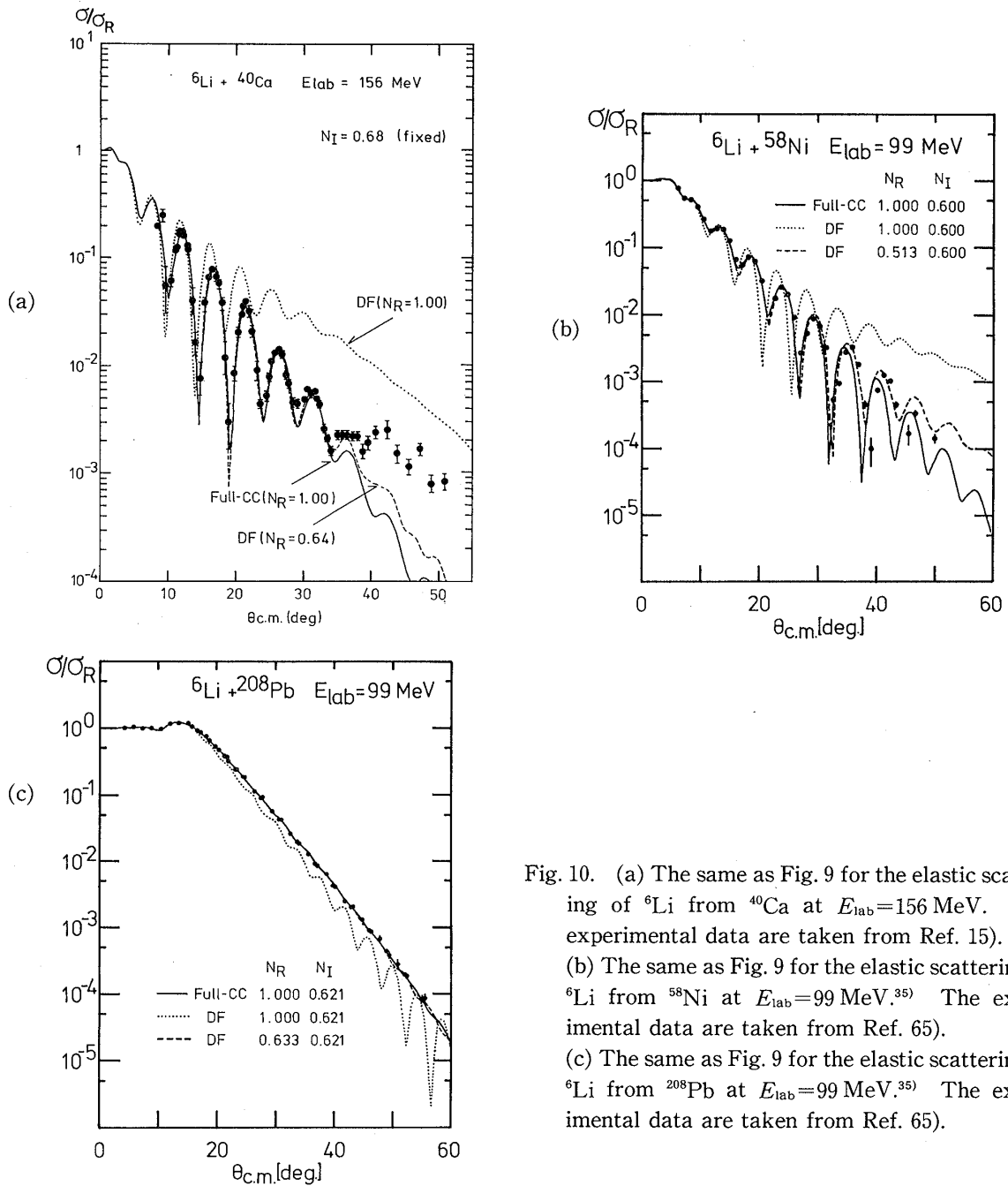


Fig. 10. (a) The same as Fig. 9 for the elastic scattering of ${}^6\text{Li}$ from ${}^{40}\text{Ca}$ at $E_{\text{lab}}=156$ MeV. The experimental data are taken from Ref. 15). (b) The same as Fig. 9 for the elastic scattering of ${}^6\text{Li}$ from ${}^{58}\text{Ni}$ at $E_{\text{lab}}=99$ MeV.³⁵⁾ The experimental data are taken from Ref. 65). (c) The same as Fig. 9 for the elastic scattering of ${}^6\text{Li}$ from ${}^{208}\text{Pb}$ at $E_{\text{lab}}=99$ MeV.³⁵⁾ The experimental data are taken from Ref. 65).

section to the measured one. An automatic local-potential search code (*ALPS*)⁶³⁾ is used in the search process. The dashed curve in Fig. 9 is the best-fit result for the ${}^6\text{Li}+{}^{28}\text{Si}$ scattering at $E_{\text{lab}}=99$ MeV. It reproduces well the observed data with $N_R=0.554$ and $N_I=0.623$. If we take $N_R=1.0$ with $N_I=0.623$, we get the dotted curve in Fig. 9 which deviates upward much from the data. As long as N_R is kept fixed at 1.0, one cannot obtain at all any satisfactory fit to the data even when we employ the Woods-Saxon shape for the imaginary potential. Thus, the large reduction of N_R from 1.0 to 0.554 seems inevitable to reproduce the data in the single-channel frame. This is a typical example of the famous anomaly of N_R seen in the double-folding-model analyses of ${}^6\text{Li}$ scattering.^{13)~18)}

A similar conclusion is reached for other targets and/or bombarding energies, another example being illustrated by the dashed curves in Figs. 10(a)~(c) for ${}^6\text{Li}+{}^{40}\text{Ca}$ scattering

at $E_{\text{lab}}=156 \text{ MeV}$, $^{15}\text{Li}+^{58}\text{Ni}$ at 99 MeV⁶⁵⁾ and $^6\text{Li}+^{208}\text{Pb}$ at 99 MeV.⁶⁵⁾

3.3. Virtual-breakup effects in the elastic scattering

Next, we perform the coupled-channel (CC) calculations including the breakup channels shown in Fig. 8(a). Throughout the CC calculations, we keep $N_R=1.0$. For N_I , however, we have no *a priori* choice of the value. In order to estimate the size of the CC effect, we first tried the same value as for the single-channel best-fit calculation mentioned above; for example, $N_I=0.623$ for the scattering of $^6\text{Li}+^{28}\text{Si}$

at 99 MeV. The solid curve in Fig. 9 is the result of the CC calculation with all the breakup channels ($S_{\text{bu}}+P_{\text{bu}}+D_{\text{bu}}^{(\text{res})}+D_{\text{bu}}^{(\text{off})}$) coupled, which we call the full-CC calculation. It reproduces the observed angular distribution very well up to backward angles. The large change from the dotted curve to the solid one represents a remarkable effect of the virtual breakup.

It is quite surprising that the experimental data are reproduced well by the full-CC calculation with using the same value of N_I as for the best-fit single-channel calculation (solid and dashed curves in Fig. 9). Furthermore, it is found³⁴⁾ that the two calculations give almost the same elastic S -matrix elements for the individual partial waves as shown in Fig. 11. One would usually expect that the coupling with non-elastic channels brings about considerable change in the imaginary part of the potential and affects little the real part. The situation in this case is completely opposite; the single-channel calculation with $N_R=0.554$ and $N_I=0.623$ gives almost the same result as that of the full-CC one with $N_R=1.0$ and $N_I=0.623$. These results show that the CC effect of the breakup channels is simulated well in the single-channel calculation by reducing the real-potential strength by a factor of two, little modification of the imaginary potential being necessary.

Much the same results^{34),35)} as above are also obtained in the case of the ^6Li scattering from ^{40}Ca at $E_{\text{lab}}=156 \text{ MeV}$, from ^{58}Ni at 99 MeV and from ^{208}Pb at 99 MeV (solid curves in Figs. 10(a)~(c)).

One can, therefore, consider that the coupling with the ^6Li -breakup channels is a dominant origin of the reduction of N_R for the double-folding-model potential for this scattering. This result suggests that the dynamical polarization (DP) potential due to the coupling must be strongly repulsive and have a very small imaginary part, at least in the potential-surface region to which the elastic scattering is sensitive. The strength of the repulsive real part must amount to about 40 % of the double-folded real potential. The DP potential will be calculated in § 7 where we shall confirm explicitly the property of the potential discussed here both qualitatively and quantitatively.

In order to understand the role of the individual breakup channels, we make four types of CC calculations in which each of S_{bu} , P_{bu} , $D_{\text{bu}}^{(\text{res})}$ and $D_{\text{bu}}^{(\text{off})}$ channels is individually

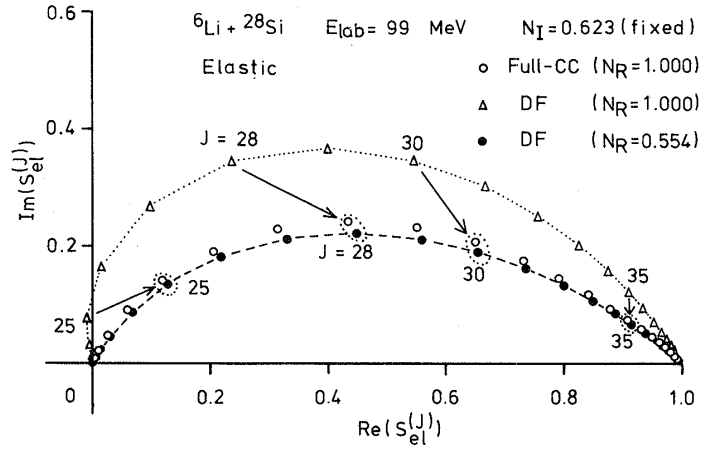


Fig. 11. The S -matrix elements for the elastic scattering of ^6Li from ^{28}Si at $E_{\text{lab}}=99 \text{ MeV}$ given by the three kinds of calculation shown in Fig. 8(a). This figure is taken from Ref. 34).

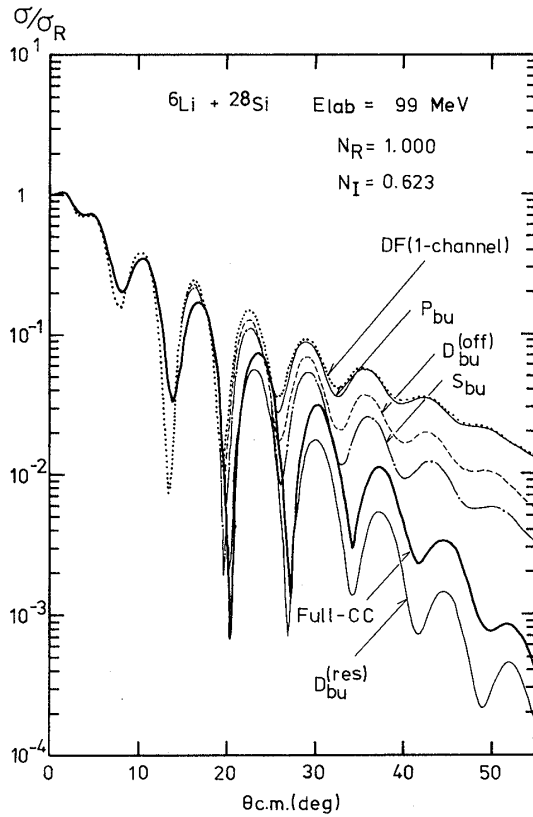


Fig. 12. Role of the individual breakup channels in the angular distribution of the elastic scattering of ^6Li from ^{28}Si at $E_{\text{lab}}=99$ MeV. The breakup channels taken are indicated for individual curves. This figure is taken from Ref. 34).

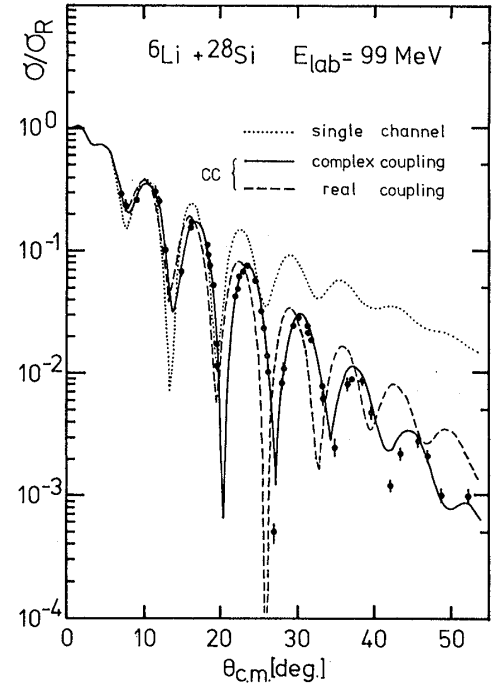


Fig. 13. Effects of the imaginary part of coupling potentials^{35),36)} on the angular distribution of the $^6\text{Li} + ^{28}\text{Si}$ elastic scattering at $E_{\text{lab}}=99$ MeV. The solid (dashed) curve represents the full-CC results with the complex (real) coupling potentials with $N_R=1.0$ and $N_I=0.623$ ($N_I=0.0$). The diagonal potentials of all the channels are fixed to be complex with $N_R=1.0$ and $N_I=0.623$. The dotted curve is given by the single-channel calculation.

coupled to the incident channel. Figure 12 shows the angular distribution given by the four types of CC calculation.³⁴⁾ We see that the D -wave resonance channel, $D_{\text{bu}}^{(\text{res})}$, gives the most important contribution among all the breakup channels, but non-resonant breakup channels also play a non-negligible role in reproducing the observed angular distribution (cf. the difference between “ $D_{\text{bu}}^{(\text{res})}$ ” and “Full-CC” in the figure).

3.4. Roles of the imaginary part of coupling potentials

All the coupling potentials as well as the diagonal ones are complex in the present CC framework (cf. Eq. (2.37)). It should be emphasized that the successful result of the CC calculation shown in Figs. 9 and 10 has been achieved with using complex coupling potentials with $N_R=1.0$ and $N_I \approx 0.6$. If we switch off the imaginary part of the coupling potentials in the CC calculation, we cannot reproduce the experimental data at all. In order to demonstrate the important role of the imaginary part, we compare in Fig. 13 two kinds of CC calculations for the $^6\text{Li} + ^{28}\text{Si}$ scattering at 99 MeV, one with real coupling potentials ($N_R=1.0$ and $N_I=0.0$) and the other with complex ones ($N_R=1.0$ and $N_I=0.62$). In both calculations, all the diagonal potentials are set to be complex with $N_R=1.0$ and $N_I=0.623$. It is obvious from the comparison that the complex coupling potentials with $N_I \approx 0.6$ are indispensable for the proper fit to the data; no matter how widely we changed the value of N_I for the diagonal potentials, no satisfying fit has been obtained as long as

we use real coupling potentials. Thus, one may understand that the imaginary part of the coupling potentials is very important in considering the effects of the ${}^6\text{Li}$ -breakup channels. Similar importance of the imaginary coupling potentials has been pointed out recently in Ref. 66) for the ${}^7\text{Li}$ scattering by ${}^{40}\text{Ca}$ at 89 MeV.

3.5. Energy and target dependence of breakup effects

Next, we investigate how the ${}^6\text{Li}$ -breakup effect depends on the incident energy and target mass number.³⁶⁾

The full-CC calculations including the breakup channels of Fig. 8(a) have been performed for the ${}^6\text{Li}$ scattering by ${}^{12}\text{C}$ at 50.6, 99, 123.5, 156 and 168.6 MeV, by ${}^{28}\text{Si}$ at 46, 99, 135 and 154 MeV, by ${}^{40}\text{Ca}$ at 50.6, 88, 99 and 156 MeV, by ${}^{48}\text{Ca}$ at 34 MeV, by ${}^{58}\text{Ni}$ at 50.6, 73.7 and 99 MeV and by ${}^{208}\text{Pb}$ at 36, 42, 50.6, 73.7, 88, 99 and 156 MeV. The calculated elastic cross sections³⁶⁾ are grouped according to incident energy and shown by the solid lines in Figs. 14~19. In general, the CC calculation with $N_R=1$ successfully reproduces the experimental data^{15)~18),67)~74)} for all the targets and the incident energies investigated here. The result implies that the projectile breakup effect well accounts for the reduction factor^{13)~18)} of $N_R=0.5\sim 0.6$ required in the DF-model (single-channel) studies of ${}^6\text{Li}$ scattering. Going into details, the fit to the data for ${}^{40}\text{Ca}$ at 88 MeV is exceptionally poor, which may, however, be due to an uncertainty of the experimental data since the data were known to have suffered from experimental difficulties.⁷⁵⁾ For higher energy cases in Figs. 18 and 19, large deviations from the data are seen at very large angles, which may be due

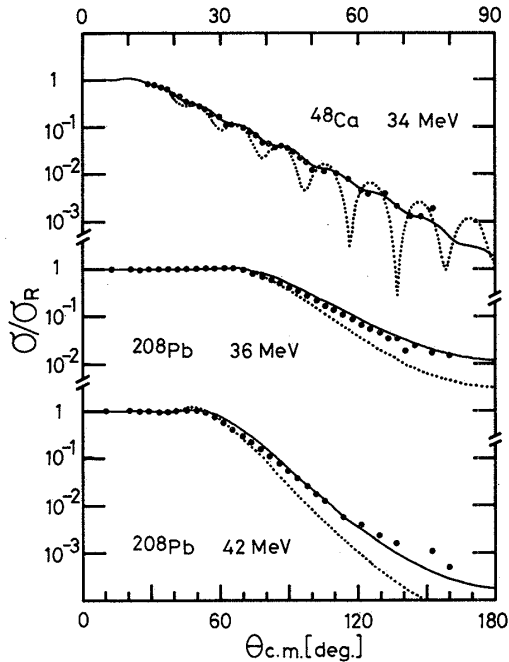


Fig. 14. The elastic scattering of ${}^6\text{Li}$ by ${}^{48}\text{Ca}$ at $E_{\text{lab}}=34$ MeV and by ${}^{208}\text{Pb}$ at 36 and 42 MeV. The experimental data are taken from Refs. 67) and 68). The upper angular scale is only for ${}^{48}\text{Ca}$ target and the lower scale for the others. The solid curve and the dotted one show the results of the CC calculation and the single-channel one, respectively, with N_R kept fixed at unity. This figure is taken from Ref. 36).

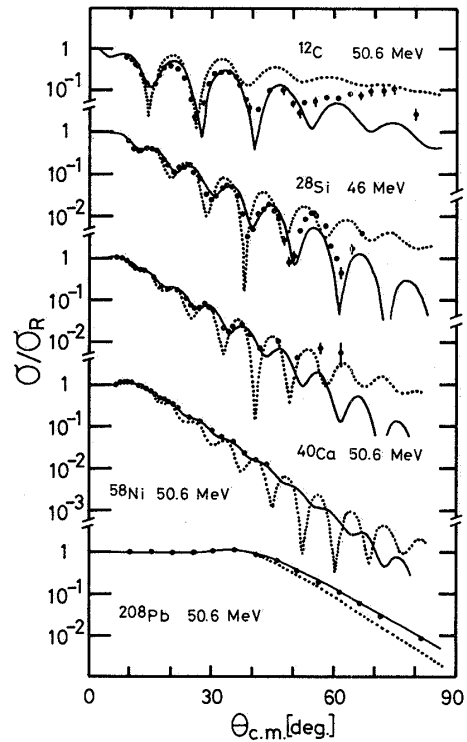


Fig. 15. The same as Fig. 14 for ${}^6\text{Li}$ scattering by ${}^{12}\text{C}$, ${}^{40}\text{Ca}$, ${}^{58}\text{Ni}$ and ${}^{208}\text{Pb}$ at 50.6 MeV and by ${}^{28}\text{Si}$ at 46 MeV. The data are taken from Refs. 18) and 69).

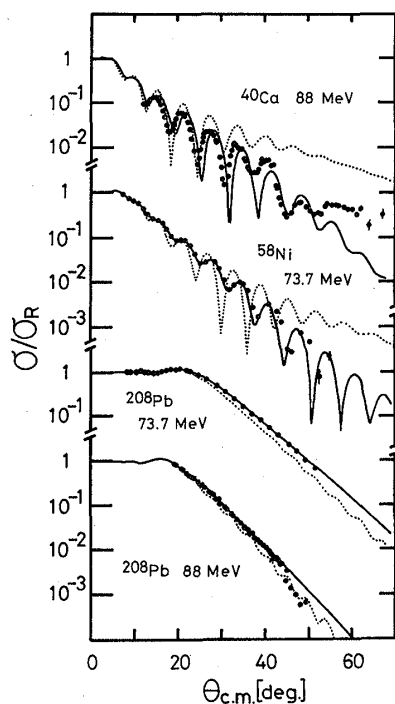


Fig. 16. The same as Fig. 14 for ^6Li scattering by ^{40}Ca and ^{208}Pb at 88 MeV and by ^{58}Ni and ^{208}Pb at 73.7 MeV. The data are taken from Refs. 17) and 70).

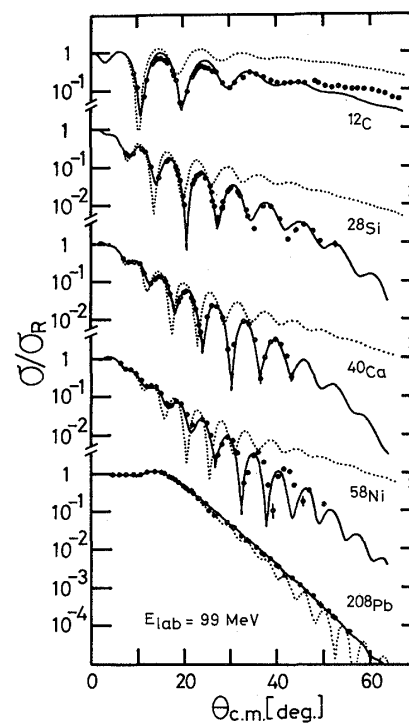


Fig. 17. The same as Fig. 14 for 99-MeV ^6Li scattering by ^{12}C , ^{28}Si , ^{40}Ca , ^{58}Ni and ^{208}Pb . The data are taken from Ref. 16).

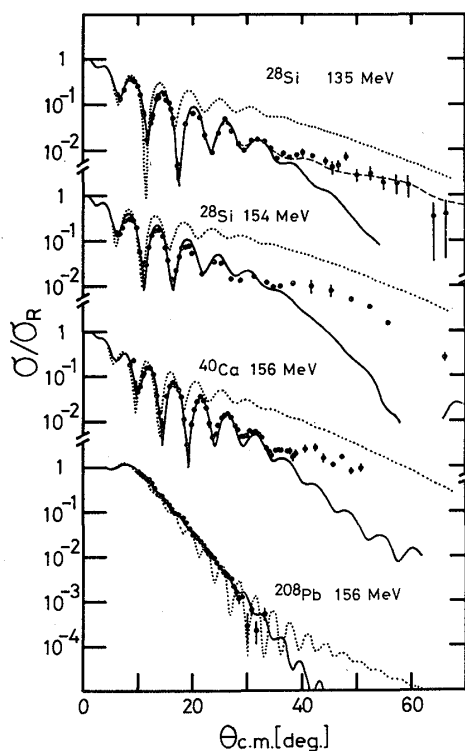


Fig. 18. The same as Fig. 14 for ^6Li scattering by ^{28}Si at 135 and 154 MeV and by ^{58}Ni and ^{208}Pb at 156 MeV. The data are taken from Refs. 15) and 71) ~ 73).

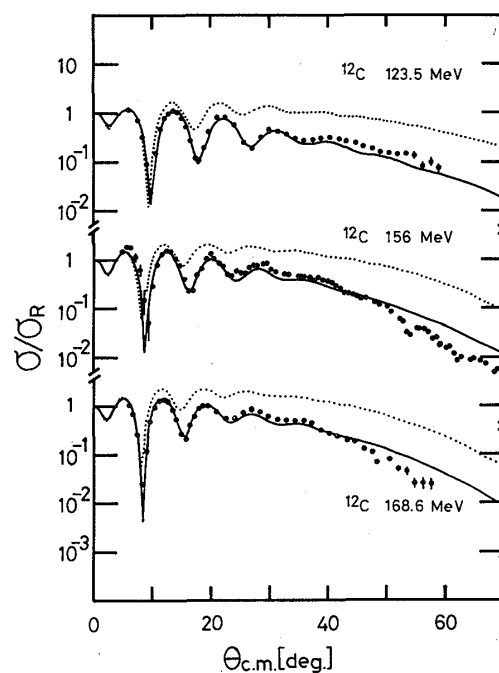


Fig. 19. The same as Fig. 14 but for ^6Li scattering by ^{12}C at 123.5, 156 and 168.6 MeV. The data are taken from Refs. 73) and 74).

to the simple assumption for the imaginary potential in Eq. (2·37), since, at higher energies, the cross section at large angles is sensitive to details of the potential at nuclear interior. An improvement of fit at those angles will be discussed later.

In Table III, listed are the optimum values of N_i deduced by the CC analyses of Figs. 14~19. No significant dependence on energy and target is seen, although smaller values ($N_i=0.4\sim0.5$) are required for ^{12}C and ^{48}Ca targets and for low energy scattering by ^{208}Pb target. For ^{208}Pb , one should not take seriously the decrease of N_i at low energies, since the fit to the data is not satisfactory around the grazing angle. Those incident energies are comparable with the Coulomb-barrier height for ^{208}Pb where considerable effects (e.g., long-range absorption) may be expected from the Coulomb excitations of the projectile and target which have been neglected in the present CC calculation.

The dotted lines in Figs. 14~19 show the single-channel calculation with the use of the same values of $N_R(=1)$ and N_i as those for the corresponding CC calculation. A significant effect of the breakup channels on the elastic scattering is seen by the difference between the dotted line and the solid one. As discussed above, a large fraction of the difference stems from the coupling with the D -wave resonance channel, although non-negligible contributions arise also from the non-resonant breakup channels. It should also be stressed that the couplings *among the breakup channels* are quite important for proper

Table III. Optimum values of N_i for the elastic scattering of ^6Li deduced by the full-CC calculations. The value of N_R is fixed at unity for all cases. J_{gr} denotes the grazing angular momentum.

target	$E_{\text{lab}}(\text{MeV})$	N_R	N_i	$J_{\text{gr}}(\hbar)$	Ref.
^{12}C	50.6	1.00	0.50	11	69)
	99	1.00	0.50	15	16)
	123.5	1.00	0.48	18	74)
	156	1.00	0.47	20	73)
	168.6	1.00	0.47	21	74)
^{28}Si	46	1.00	0.62	14	18)
	99	1.00	0.62	23	16)
	135	1.00	0.65	27	71)
	154	1.00	0.59	29	72)
^{40}Ca	50.6	1.00	0.60	17	69)
	88	1.00	0.60	24	17)
	99	1.00	0.64	26	16)
	156	1.00	0.68	33	15)
^{48}Ca	34	1.00	0.50	14	67)
^{58}Ni	50.6	1.00	0.60	19	69)
	74	1.00	0.60	25	70)
	99	1.00	0.60	30	16)
^{208}Pb	36	1.00	0.40	8	68)
	42	1.00	0.40	13	68)
	50.6	1.00	0.60	18	69)
	74	1.00	0.62	29	70)
	88	1.00	0.60	35	17)
	99	1.00	0.62	38	16)
	156	1.00	0.60	53	73)

estimation of the breakup effects.^{34),35)}

At first sight of Figs. 14~19, one would consider that the breakup effect on the elastic cross section varies much with the incident energy and target mass number. For light targets, for which the angular distribution shows the Fraunhofer diffraction pattern, the breakup effect reduces the cross section from the dotted line to the solid one, while, for heavy targets, the angular distribution being the Fresnel type, the cross section increases with the breakup effect. Both of them, however, reflect the repulsive nature of the breakup effect, which can be understood easily with the nearside/farside (N/F) decomposition of the scattering amplitude,^{76),77)} as shown below.

Following Ref. 76), we decompose the elastic scattering amplitude, $f(\theta)$, into the nearside and farside components as

$$f(\theta) = f_N(\theta) + f_F(\theta), \quad (3.1)$$

where each of $f_N(\theta)$ and $f_F(\theta)$ is the sum of the Coulomb and the nuclear part, respectively, as

$$f_N(\theta) = f_N^{(\text{Coul})}(\theta) + f_N^{(\text{nuc})}(\theta), \quad (3.2)$$

$$f_F(\theta) = f_F^{(\text{Coul})}(\theta) + f_F^{(\text{nuc})}(\theta). \quad (3.3)$$

Here, $f_N^{(\text{Coul})}(\theta)$ ($f_F^{(\text{Coul})}(\theta)$) is the nearside (farside) component of the Coulomb amplitude which is expressed with analytic functions.⁷⁶⁾ The nuclear part of the nearside and the farside component is expressed with the calculated S-matrix element for the elastic scattering, S_L , as

$$f_{N,F}^{(\text{nuc})}(\theta) = \frac{i}{2k} \sum_L (2L+1) \exp(2i\sigma_L^{(\text{C})}) (1 - S_L) \cdot Q_L^{(\pm)}(\cos \theta), \quad (3.4)$$

where $Q_L^{(+)}$ ($Q_L^{(-)}$) is the nearside (farside) component of the Legendre polynomial,⁷⁶⁾ and $\sigma_L^{(\text{C})}$ is the Coulomb phase shift.

Figures 20(a) and (b) show the N/F decomposition of the scattering of ${}^6\text{Li}$ by ${}^{28}\text{Si}$ and ${}^{208}\text{Pb}$ at 99 MeV. Both single-channel result and the CC one have been decomposed, where S_L 's in Eq. (3.4) are given by the single-channel and the CC calculation, respectively. The dotted and the dashed line show the individual contributions from the nearside and the farside component, respectively, and the solid line shows their coherent sum. In the single-channel case for ${}^{28}\text{Si}$ target, the Fraunhofer crossover⁷⁷⁾ between the nearside and the farside occurs around $\theta_{\text{cm}} \simeq 15^\circ$ and the farside dominates the scattering for angles larger than the crossover point, which is call the farside tail.⁷⁷⁾ The tail has reduced down in the CC calculation. Hence, the decrease of the cross section for those angles from the single-channel to the CC, as is seen in Fig. 17 for ${}^{28}\text{Si}$ target, is now understood as the result of the decrease of the farside tail. Conversely, the nearside component increases by the CC effect, which leads to that both components interfere hardly over the wide range of angle around the crossover point in the CC calculation, the coherent sum of them (solid line) reproducing the experimental data very well. For ${}^{208}\text{Pb}$ target, the breakup effect also causes an increase (decrease) in the nearside (farside) component. However, the scattering by the target is dominated by the nearside for most angles of interest here, which is due to the strong Coulomb repulsion.⁷⁷⁾ Thus, one may understand the reason for the

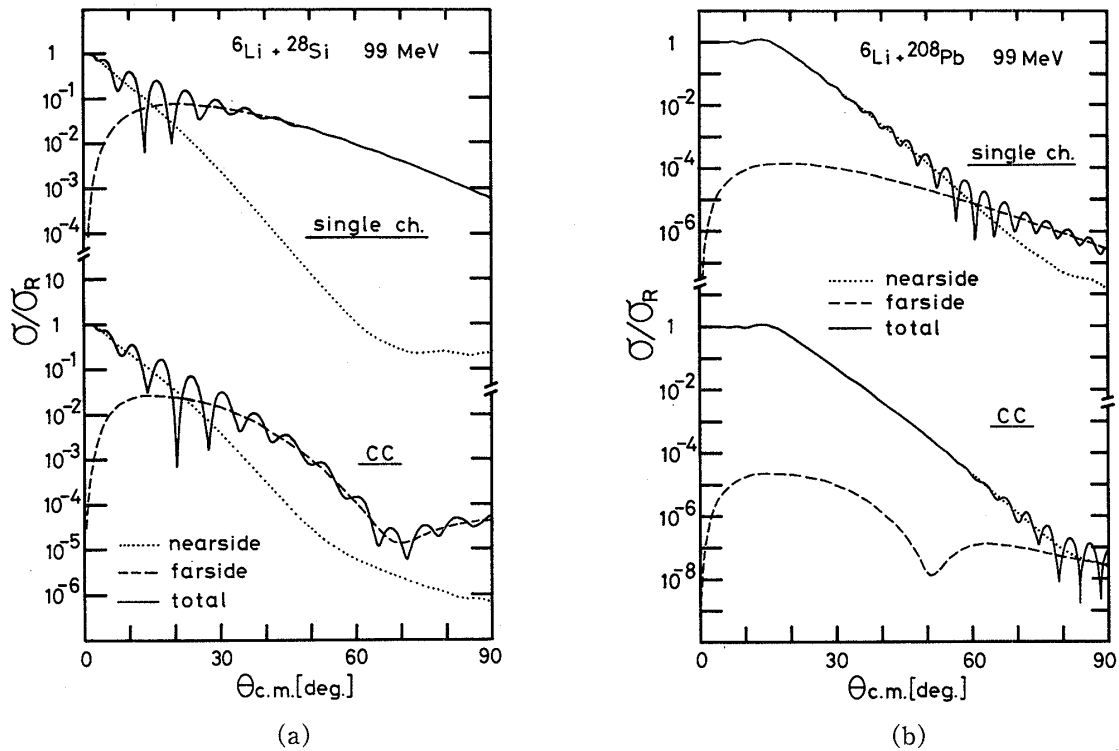


Fig. 20. The nearside/farside decomposition of the calculated ${}^6\text{Li}$ scattering (a) by ${}^{28}\text{Si}$ and (b) by ${}^{208}\text{Pb}$ at 99 MeV; both the single-channel (upper) and the CC (lower) results are decomposed into nearside (dotted curve) and farside (dashed curve) components; the coherent sum of them is shown by the solid line.

increase with damped oscillation of the cross section at large angles by the CC effect, as is seen in Fig. 17.

Since, in general, the attractive real potential plays a role of converging lens, the reduction of the real-potential strength causes a decrease of the farside amplitude while an increase of the nearside one.⁷⁷⁾ Therefore, both the decrease of the farside-dominant cross section for ${}^{28}\text{Si}$ target and the increase of the nearside-dominant one for ${}^{208}\text{Pb}$ target in Figs. 20(a) and (b) reflect the reduction in strength of the attractive (real) potential by the breakup effect. In fact, the results of the CC calculation can be simulated well by the single-channel calculation with the renormalization factor, $N_R=0.5\sim 0.6$, as is discussed already.

With the above consideration kept in mind, one may easily understand that, for other targets and/or energies in Figs. 14~19, the change in the cross section from dotted line to the solid one reflects the repulsive nature of the ${}^6\text{Li}$ -breakup effect.

3.6. An improvement of fits for higher-energy scattering

At higher incident energies, the fit to the data at large angles becomes rather poor (see Figs. 18 and 19) even by the CC calculation including the coupling to the breakup process, although large effects of the breakup are still evident. In the present CC framework, we have simply assumed the same geometrical form for the imaginary potentials as that of the corresponding real potentials (cf. Eq. (2·37)). As shown above, this assumption has been sufficient in order to discuss the essential features of the virtual breakup effects, which, however, may be somewhat too simplified to get an excellent fit to the large-angle

scattering at higher incident energies. In high energy scattering, cross sections at large angles are very sensitive to details of the potential at nuclear interior, which is indeed supported by the fact that the observed large-angle scattering in Fig.18 displays a nuclear-rainbow-like enhancement.¹⁸⁾

It turns out that much better fit to the data at those angles is obtained with a modified CC calculation^{35),36)} (the dashed line in Fig. 18), the only difference from the standard CC calculation (the solid line) being that the imaginary potential in the incident channel is replaced by a phenomenological one with the Woods-Saxon shape, while other potentials being unchanged. If the channel coupling induced an additional imaginary potential in the incident channel, it would be difficult (or at least tedious) to find out such a phenomenological bare potential⁷⁸⁾ with which the CC calculation fit the data, since the automatic potential search with the CC calculation requires a large amount of computing time. In the present case, however, the coupling of the breakup channels induces chiefly a real potential of repulsive nature and the contribution to the imaginary part is negligible, as will precisely be shown below. Since the repulsive effect of the coupling can be simulated well simply by the renormalization, $N_R \simeq 0.6$, to the real potential in the DF-model calculation (single channel), the bare imaginary potential for the modified CC calculation must be closed to the imaginary potential obtained by the DF-model fit to the elastic scattering. Thus, we adopt the latter as the bare imaginary potential.

The dashed line in Fig. 18 shows an example of a better fit to the data by such a modified CC calculation^{35),36)} for the scattering of ${}^6\text{Li}$ by ${}^{28}\text{Si}$ at $E_{\text{lab}}=135$ MeV. The parameter values for the bare imaginary potential are the following; $W_0=27.46$ MeV, $R_1=1.7 \cdot A_T^{1/3}$ fm where $A_T=28$, and $a_1=0.908$ fm, which have been obtained by the DF-model analysis of the same scattering.¹⁸⁾

§ 4. Elastic and projectile-inelastic scattering of ${}^7\text{Li}$

As mentioned briefly in § 1, the double-folding model with $N_R \simeq 1.0$ seems to fail for the elastic scattering of ${}^7\text{Li}$ in addition to those of ${}^6\text{Li}$ and ${}^9\text{Be}$.^{13)~22)} The double-folded real potential has to be reduced by a factor of about two ($N_R \simeq 0.5 \sim 0.6$) in order to fit the observed elastic cross sections. Some theoretical efforts^{10),37),38),74),79)~81)} have been devoted to explaining the anomaly of N_R . Since the $3/2^-$ ground state of ${}^7\text{Li}$ has a large quadrupole moment, the re-orientation coupling in the ground state may have some effect on the elastic scattering. Hnizdo et al. have performed a CC calculation³⁷⁾ including the re-orientation coupling. They have been able to fit the ${}^7\text{Li}$ elastic scattering from ${}^{40}\text{Ca}$ and ${}^{56}\text{Fe}$ at $E_{\text{lab}}=34$ and 48 MeV, respectively, by the CC calculation. At higher incident energies of $E_{\text{lab}} \simeq 60 \sim 90$ MeV, however, a renormalization of $N_R \simeq 0.6$ was required again to fit the data even by a CC calculation^{38),79)} including the ${}^7\text{Li}$ excitation to the $1/2^-$ state at 0.48 MeV as well as the re-orientation of the ground state. Cook et al. further investigated the target-excitation effect with a CC calculation⁷⁹⁾ including both the $1/2^-$ state of ${}^7\text{Li}$ and the first 2^+ state of the target nuclei (${}^{12}\text{C}$ and ${}^{24,26}\text{Mg}$) as well as the re-orientation effect of ${}^7\text{Li}$. The result was, however, negative, and they concluded that the coupling to the first excited states of target and projectile is little responsible for the renormalization, $N_R \simeq 0.6$, for the ${}^7\text{Li}$ scattering.

On the other hand, one can expect that the breakup of ${}^7\text{Li}$ may be responsible for the anomalous renormalization factor because the α -t breakup threshold energy of ${}^7\text{Li}$ is very

small (2.48 MeV), just as in the case of ${}^6\text{Li}$. In fact, a remarkable effect of the ${}^7\text{Li}$ breakup on the scattering was pointed out by Nagarajan et al.⁸⁰⁾ with an adiabatic three-body model,³²⁾ in which the ${}^7\text{Li}$ -target interaction was composed of the α -target and the t-target optical potentials. They have taken into account the α -t P -wave and F -wave breakup processes in the adiabatic approximation. Though the interaction they used is not directly related to the double-folded ${}^7\text{Li}$ -target potential, their result suggests that the breakup effect can be a promising candidate for the origin of the anomalous renormalization factor.

In this section, we present a microscopic CC calculation^{10),74),81)} of the elastic scattering by including all the possible effects mentioned above, and discuss directly the relation between the renormalization factor of the double-folded potential and the effects mentioned above. Since we employ the CDCC method for taking into account the breakup processes, it becomes possible to distinguish the contribution of the resonant and the non-resonant parts of the breakup continua, which could not be resolved in the previous work of Ref. 80). We analyze the elastic scattering of ${}^7\text{Li}$ from ${}^{12}\text{C}$, ${}^{48}\text{Ca}$, ${}^{58}\text{Ni}$ and ${}^{208}\text{Pb}$ at the higher incident energies of $E_{\text{lab}}=63\sim 148$ MeV. As for the ground-state nucleon densities of ${}^{48}\text{Ca}$ and ${}^{208}\text{Pb}$, we adopt the Hartree-Fock densities given by Negele⁶²⁾ with a correction for the center-of-mass motion. The density of ${}^{58}\text{Ni}$ is taken to be the same as that used in the double-folding-model calculation of Ref. 16), while the ${}^{12}\text{C}$ density is given by the microscopic cluster-model wave function⁴⁷⁾ mentioned before. All these densities reproduce the observed electron-scattering form factors and r.m.s. charge radii.

In the rest of this section, we compare the calculated cross sections for the projectile-inelastic scattering leading to the $1/2^-$ excited state with measured ones. Since the only parameter of the calculation, i.e., N_i , is determined by fitting the calculated elastic-scattering cross section to the measured one, no additional free parameter is left in the prediction of the inelastic-scattering cross sections, and of the breakup ones, too. Therefore, the analyses of the inelastic scattering will impose an additional examination of the present CDCC framework and the reaction mechanism concerned.

4.1. Truncation and discretization of breakup continua

We first set up the three-body model space for the CC calculation. Here, we do not examine the convergence of the S -matrix elements with respect to increasing l_{max} , k_{max} and narrowing Δk . Following the examination of the ${}^6\text{Li}$ case discussed in the previous section, we truncate and discretize the α -t breakup continuum as follows.

First, in restricting the α -t relative angular momentum, it seems helpful to note the following property of the ${}^7\text{Li}$ -target coupling potential which may be inferred if one assumes for the moment the three-body model of α , t and target nucleus.⁸²⁾ In this model, the coupling potential between the ground state and a breakup state of ${}^7\text{Li}$ consists of a matrix element of the α -target potential and that of the t-target potential. Since the ground state is a P -state ($l=1$), the two matrix elements are added constructively for a breakup state with an odd- l but destructively for an even- l breakup state. The resultant coupling potential is much smaller for the even- l case than for the odd- l case, if the α -target and t-target optical potentials are similar. This is analogous to the cases of deuteron⁵⁾ and ${}^6\text{Li}$ in which the coupling effect of the P -wave breakup channels is much smaller than those of the S -wave and D -wave ones since the ground state is an S -state ($l=0$). This property of the coupling potential must hold also for the double-folded-

interaction model adopted here. We assume, therefore, that in the ${}^7\text{Li}$ scattering the coupling effect of the α -t S -wave and D -wave breakup channels is negligible compared with those of the P -wave and F -wave ones. Thus, we restrict the α -t relative angular momentum to $l=1$ (P wave) and $l=3$ (F wave), assuming that the contribution of the breakup states with l larger than 4 is negligible⁸⁰⁾ as far as the breakup effect on the elastic and the inelastic scattering of ${}^7\text{Li}$ is concerned.

Now, we switch on the spin-orbit potential between α and t, $V_{\text{eff}}^{(\text{SO})}$. The P -wave and the F -wave continua then split into two parts, $3/2^-$, $1/2^-$ and $7/2^-$, $5/2^-$, respectively; it is noted that there exists a bound state in each of the $3/2^-$ and $1/2^-$ states, the former being the ground state of ${}^7\text{Li}$ and the latter being the first excited state at $E_x=0.48$ MeV. We take $k_{\text{max}}=1.0$ fm $^{-1}$ for each continuum. As illustrated in Fig. 21, the P -wave continuum ($3/2^-$ and $1/2^-$) is divided into four bins with an equal width $\Delta k=0.25$ fm $^{-1}$. There is no resonance in the P wave. In the F wave, there is a resonance in each of the $7/2^-$ and the $5/2^-$ continua (Fig. 3). We divide the $7/2^-$ and the $5/2^-$ continua into three bins and two bins, respectively, as shown in the figure. The lowest bin for $7/2^-$ and $5/2^-$ corresponds to the resonance bin. The central energies for the $7/2^-$ and the $5/2^-$ resonances are taken to be the calculated values of 2.19 MeV and 4.04 MeV, while their widths, Γ , to be 0.11 MeV and 1.10 MeV. The remaining three bins in the F wave are the non-resonance bins with an equal width $\Delta k=0.25$ fm $^{-1}$. The mean energy $\bar{\epsilon}_i$ of each bin is presented in Fig. 21. For simplicity, we denote the total P -wave breakup channels by P_{bu} , whereas the F -wave resonance channels by $F_{\text{bu}}^{(\text{res})}$ and the total F -wave off-resonance channels by $F_{\text{bu}}^{(\text{off})}$.

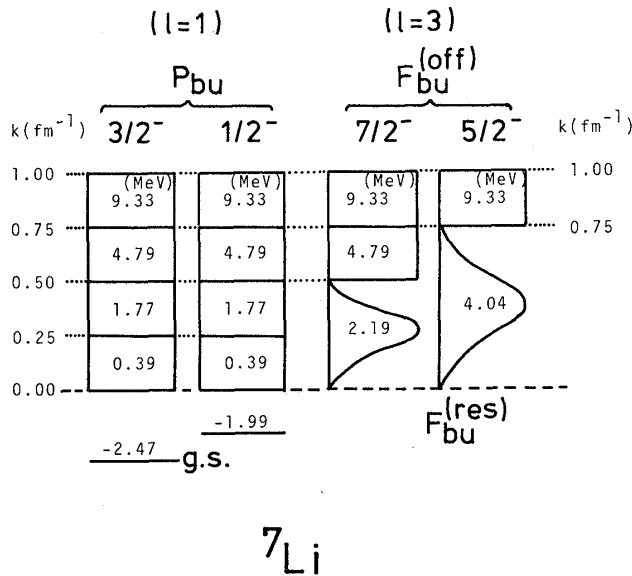


Fig. 21. The truncation and the discretization of the α -t breakup continuum of ${}^7\text{Li}$ for the CC calculation. The P -wave continuum ($3/2^-$ and $1/2^-$) is divided into four bins with an equal width $\Delta k=0.25$ fm $^{-1}$. The lowest bin for $7/2^-$ and $5/2^-$ is the F -wave resonance bin and the remaining three bins in the F -wave are the non-resonance bins with $\Delta k=0.25$ fm $^{-1}$. The mean energy, $\bar{\epsilon}_i$, of each bin is presented in each "box" in the figure.

4.2. Elastic scattering of ${}^7\text{Li}$

First, we make the single-channel calculation of the ${}^7\text{Li}$ scattering from several target nuclei. The optimum values of N_R and N_I are searched by using the automatic-search code *ALPS*.⁶³⁾ In Figs. 22(a)~(d), some examples of the best-fit results are shown for the ${}^7\text{Li}$ scattering from ${}^{12}\text{C}$ at $E_{\text{lab}}=132$ MeV,⁷⁴⁾ from ${}^{48}\text{Ca}$ at 89 MeV,²⁰⁾ from ${}^{58}\text{Ni}$ at 99 MeV,⁸³⁾ and from ${}^{208}\text{Pb}$ at 148 MeV.⁸⁴⁾ The observed data are reproduced well, although the oscillation minima in the calculated cross section seems to be somewhat too deep compared with the observed ones. The smearing out of the minima in the observed angular distribution may be due to the second-rank tensor interaction⁸²⁾ arising from the large deformation of the ${}^7\text{Li}$ ground state. The optimum values of N_R and N_I obtained are 0.68 and 0.58, respectively, for ${}^{12}\text{C}$ target, 0.487 and 0.410 for ${}^{48}\text{Ca}$, 0.447 and 0.450 for ${}^{58}\text{Ni}$, and

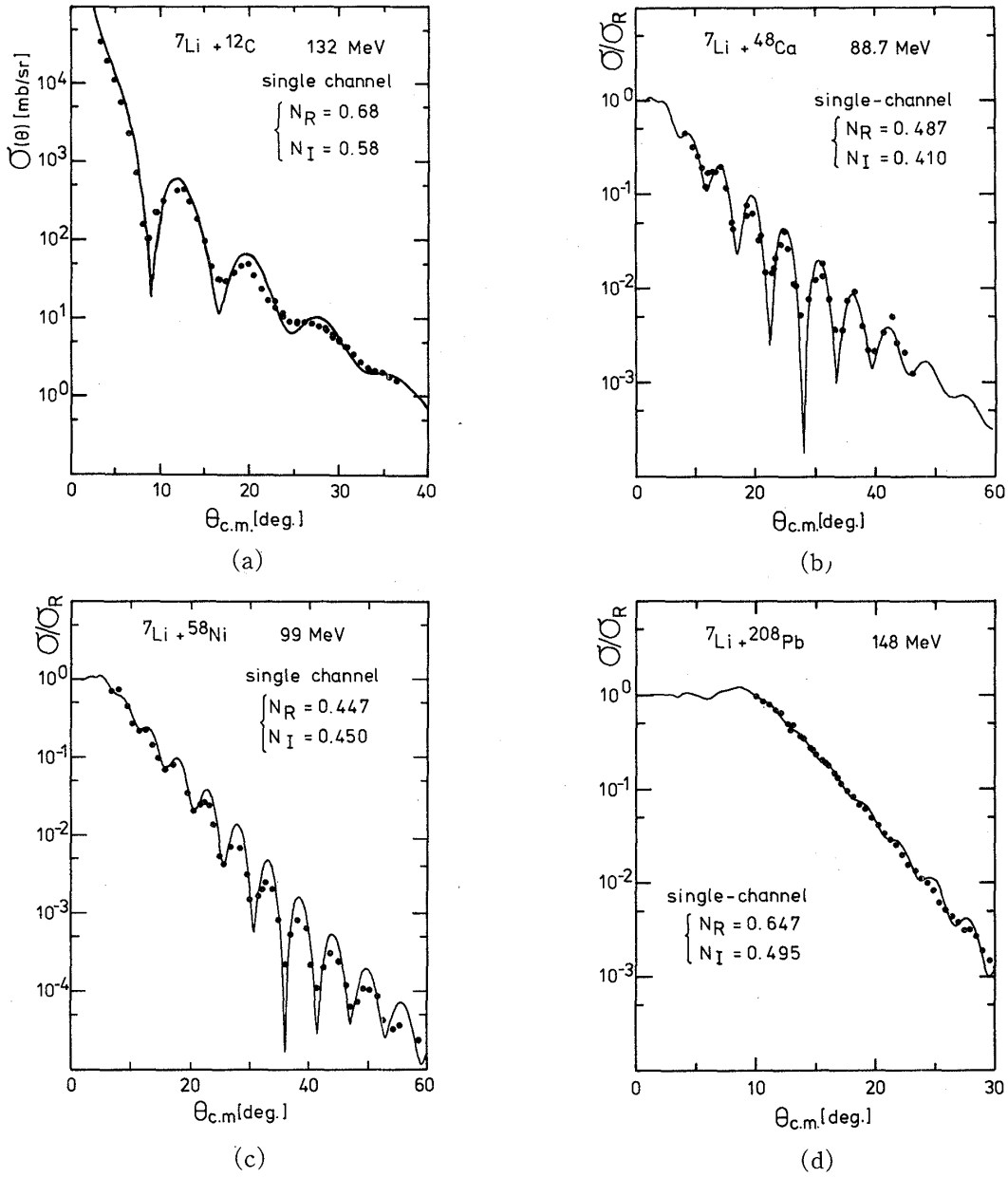


Fig. 22. The elastic scattering of ${}^7\text{Li}$ (a) from ${}^{12}\text{C}$ at $E_{\text{lab}}=132$ MeV, (b) from ${}^{48}\text{Ca}$ at 88.7 MeV, (c) from ${}^{58}\text{Ni}$ at 99 MeV and (d) from ${}^{208}\text{Pb}$ at 148 MeV. The solid curve in each figure represents the single-channel best-fit result.³⁵⁾ The optimum values of N_R and N_I are given in each figure. The experimental data are taken from Ref. 74) for (a), from Ref. 20) for (b), from Ref. 84) for (c) and from Ref. 83) for (d).

0.647 and 0.495 for ${}^{208}\text{Pb}$. The large reduction of N_R from 1.0 to 0.5~0.6 is inevitable for fitting the data.

Next, we perform a full-CC calculation including all the breakup channels shown in Fig. 21 as well as the $1/2^-$ state at 0.48 MeV. The re-orientation coupling of the ground state is also included. The solid curves in Figs. 23(a) and (b) show the results of the calculation (a) for ${}^{48}\text{Ca}$ target at $E_{\text{lab}}=89$ MeV and (b) for ${}^{12}\text{C}$ target at $E_{\text{lab}}=63$ and 78.7 MeV, respectively. The observed cross sections are reproduced well by the calculation with $N_R=1.0$. The optimum value of N_I is 0.54 for ${}^{48}\text{Ca}$ and 0.40 and 0.50 for ${}^{12}\text{C}$ at 63 and

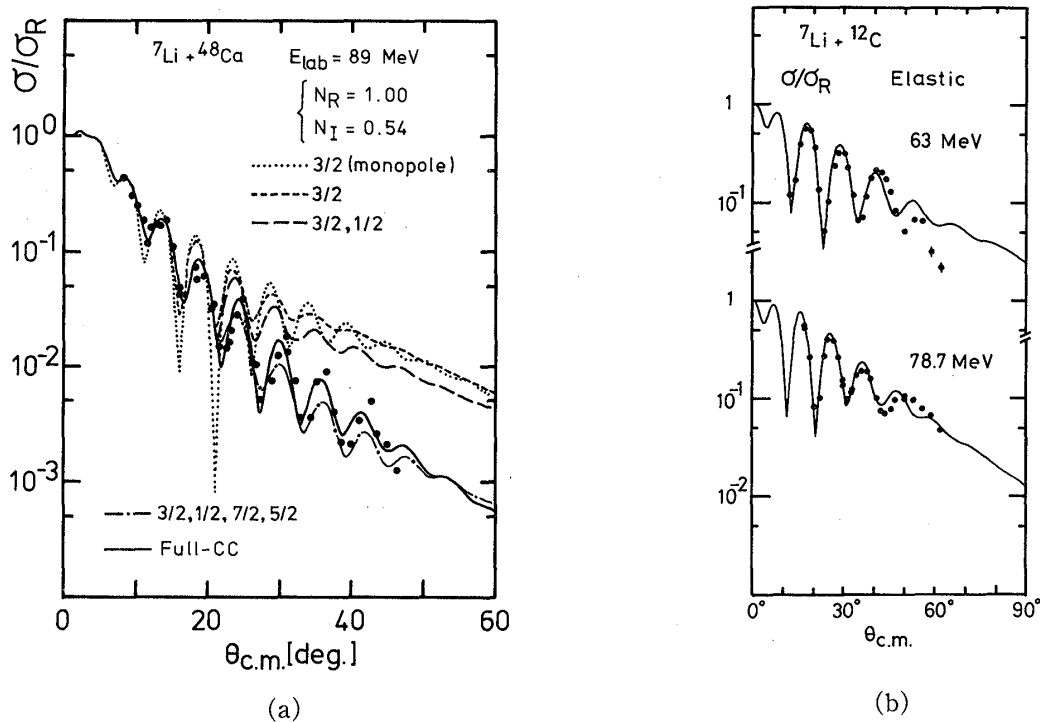


Fig. 23. (a) The elastic scattering of ${}^7\text{Li}$ from ${}^{48}\text{Ca}$ at $E_{\text{lab}} = 89$ MeV. The solid curves are the result of the full-CC calculation³⁵⁾ including all the channels shown in Fig. 21. The dotted (short-dashed) curve shows the result of the single-state calculation without (with) the re-orientation coupling. In the long-dashed and the dot-dashed curves further included are the $1/2^-$ state only and the $1/2^-$, $7/2^-$ and $5/2^-$ states, respectively. The re-orientation coupling is also included. In all the calculation, the values of $N_R = 1.0$ and $N_I = 0.54$ have been used. The experimental data are taken from Ref. 20).

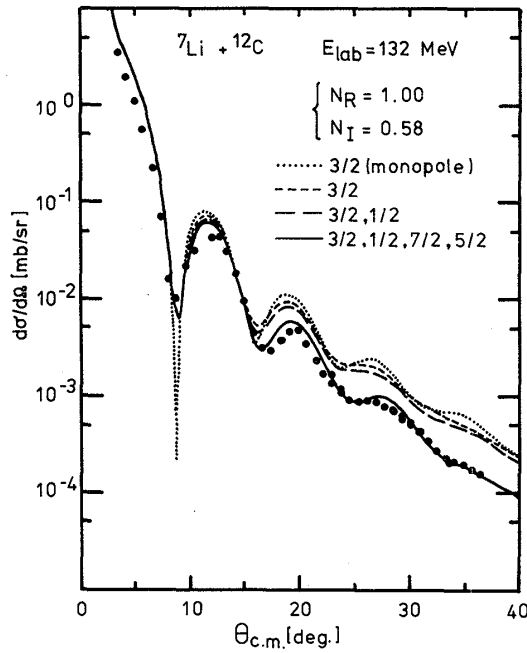
(b) The elastic scattering of ${}^7\text{Li}$ from ${}^{12}\text{C}$ at $E_{\text{lab}} = 63$ and 78.7 MeV. The solid curves are the result of the full-CC calculation including all the channels shown in Fig. 21. The experimental data are taken from Ref. 85).

78.7 MeV, respectively.

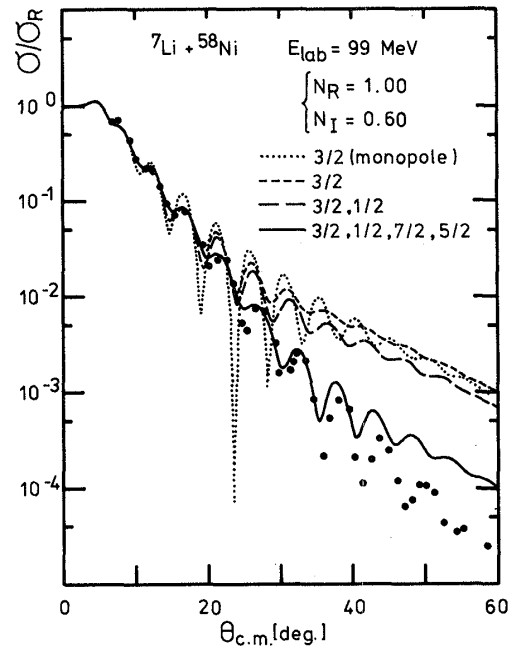
In order to investigate the relative importance of the several effects included in the full-CC calculation, we make four other kinds of CC calculations including the single-channel one for the case ${}^{48}\text{Ca}$ target. The same value of N_R and N_I as for the full-CC case is used here. The dotted curve in Fig. 23(a) shows the result of the single-channel calculation, which deviates upward much from the data and has deep minima in the diffraction pattern. The inclusion of the re-orientation coupling in the ground state does not recover the deviation at all although it smears out the deep minima, as shown by the short-dashed curve. As long as N_R is kept fixed at 1.0, the data cannot be reproduced by the calculation including the re-orientation effect only, even if the imaginary potential is allowed to vary. The difference between the short-dashed and the long-dashed curves represents the effect of the inclusion of the $1/2^-$ state. The latter curve still overestimates the data, although the fit at forward angles is somewhat improved. These results are consistent with those of Refs. 79) and 80). The most important contribution comes from the coupling of the F -wave resonance channels ($7/2^-$ and $5/2^-$), as shown by the difference between the long-dashed and the dot-dashed curves. The coupling improves the fit very much except at large angles. The effect of the $7/2^-$ resonance channel is found to dominate over that of the $5/2^-$ one, though not shown in the figure. Finally, the difference

between the dot-dashed and the solid curves represents the coupling effect of the non-resonant breakup channels, P_{bu} and $F_{bu}^{(off)}$. The coupling improves the fit up to backward angles.

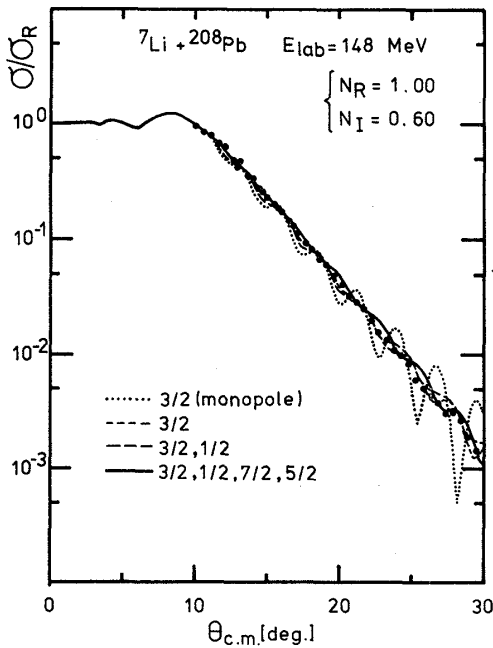
A similar results have been obtained for the ${}^7\text{Li}$ scattering from ${}^{12}\text{C}$ at $E_{lab}=132$ MeV (Fig. 24(a)), from ${}^{58}\text{Ni}$ at 99 MeV (Fig. 24(b)), and from ${}^{208}\text{Pb}$ at 148 MeV (Fig. 24(c)). Again, the optimum value of N_I is increased slightly from that in the single-channel best-fit calculation, except for the case of ${}^{12}\text{C}$ target. In these calculations, the non-resonant breakup channels have been neglected to save the computing time. The fit at large angles



(a)



(b)



(c)

Fig. 24. (a) The angular distribution of the elastic scattering of ${}^7\text{Li}$ from ${}^{12}\text{C}$ at $E_{lab}=132$ MeV. The meaning of the dotted curve, the short-dashed one and the long-dashed one is the same as that in Fig. 23(a), while the solid curve represents the result of the four-channel CC calculation including the $3/2^-$, $1/2^-$, $7/2^-$ and $5/2^-$ states. In all the calculation, the values of $N_R=1.0$ and $N_I=0.58$ have been used.

(b) The same as Fig. 24(a) for the elastic scattering of ${}^7\text{Li}$ from ${}^{58}\text{Ni}$ at $E_{lab}=99$ MeV.

(c) The same as Fig. 24(a) for the elastic scattering of ${}^7\text{Li}$ from ${}^{208}\text{Pb}$ at $E_{lab}=148$ MeV.

in the figures will be somewhat improved if we include the non-resonant breakup channels.

From these results, we conclude that the coupling of the ${}^7\text{Li}$ -breakup channels, especially that of the $7/2^-$ resonance channel, is the main origin of the strong reduction of N_R in the double-folding model for the ${}^7\text{Li}$ scattering, at least for the cases discussed here. The dynamical polarization potential due to the breakup channels must be strong enough to explain the reduction of N_R mentioned above, just as in the case of ${}^6\text{Li}$. However, it may have some imaginary part, since the optimum value of N_I for the CC calculation is slightly different from that of the single-channel best-fit one.

4.3. Inelastic scattering of ${}^7\text{Li}$ leading to the $1/2^-$ excited state

Next, we compare the calculation and the experimental data on the inelastic scattering of ${}^7\text{Li}$ leading to the first-excited state at $E_x=0.48$ MeV. As mentioned already, no additional free parameter is left in the predicted inelastic-scattering cross section after the only parameter (N_I) is determined by fitting the calculated elastic-scattering cross section to the measured; the cross sections for all the states of ${}^7\text{Li}$ coupled are simultaneously derived by the CDCC calculation.

In Figs. 25(a) and (b), we compare the calculated cross section with the observed one of the inelastic scattering of ${}^7\text{Li}$ by ${}^{48}\text{Ca}$ target at $E_{\text{lab}}=89$ MeV⁸⁰⁾ and by ${}^{12}\text{C}$ target at $E_{\text{lab}}=63$ and 79 MeV.⁸⁵⁾ The value of N_R and N_I is the same as in Figs. 23(a) and (b). The solid curve shows the full-CC calculation. Both the absolute values and the angular distribu-

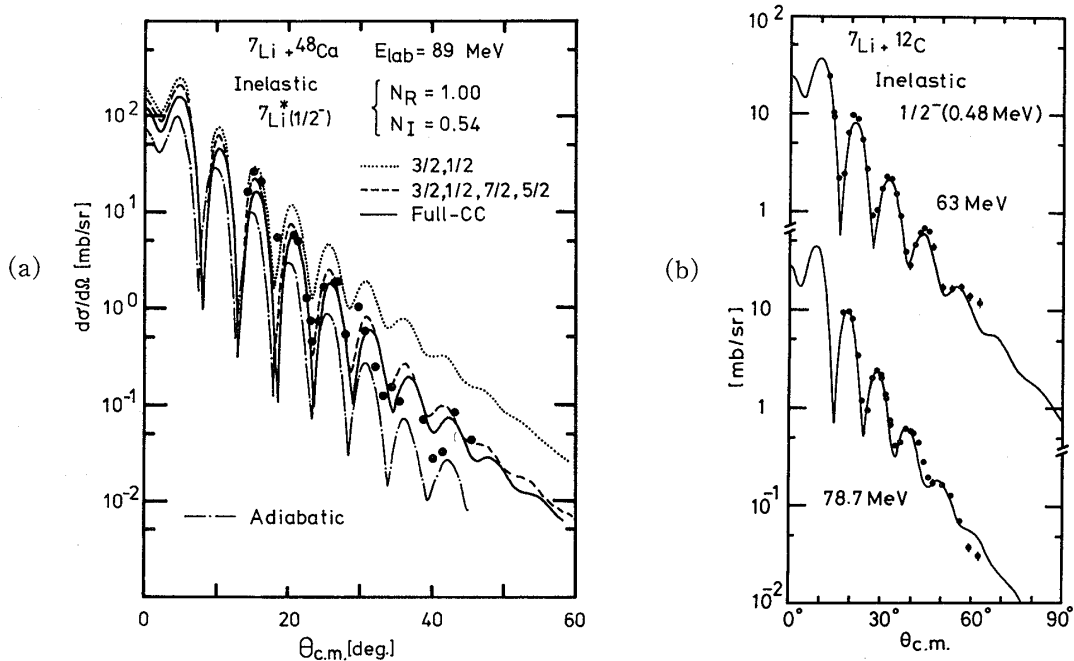


Fig. 25. (a) The inelastic scattering of ${}^7\text{Li}$ by ${}^{48}\text{Ca}$ at $E_{\text{lab}}=89$ MeV leading to the $1/2^-$ excited state of ${}^7\text{Li}$ at $E_x=0.48$ MeV. The experimental data are taken from Ref. 80). The solid curve represents the result of the same full-CC calculation as in Fig. 23(a). In the dotted and the dashed curves included are the $1/2^-$ states only, and the $1/2^-$ state as well as the $7/2^-$ and the $5/2^-$ resonance states, respectively. The dot-dashed curve shows the result of the adiabatic three-body-model calculation of Ref. 80).

(b) The inelastic scattering of ${}^7\text{Li}$ by ${}^{12}\text{C}$ at $E_{\text{lab}}=63$ and 78.7 MeV leading to the $1/2^-$ excited state of ${}^7\text{Li}$ at $E_x=0.48$ MeV. The experimental data are taken from Ref. 85). The solid curves represent the result of the full-CC calculation.

tions of the measured cross sections are reproduced well by the calculation. The dotted and the dashed curves in Fig. 25(a) represent the CC calculation including the $1/2^-$ state only and that including the $F_{bu}^{(res)}$ as well as the $1/2^-$ state, respectively. The re-orientation coupling in the ground state is taken into account in both calculations. The difference among these three curves indicates the importance of the breakup channels. For comparison, we also include in the figure the result of the adiabatic three-body-model calculation of Nagarajan et al.⁸⁰⁾ (dot-dashed curve), in which the effect of the re-orientation, the $1/2^-$ state, and the P -wave and F -wave breakup continua are included in the adiabatic approximation. The prediction of the present full-CC calculation seems to be much better than that of the adiabatic-three-body model. The difference between these two curves may stem chiefly from the difference in the interaction employed (i.e., the DF interactions for the former and the α -target and t -target optical potentials for the latter) rather than from the defect of the adiabatic approximation since the approximation seems good at the present bombarding energy regions.³⁴⁾

§ 5. Projectile breakup reactions of ${}^6\text{Li}$ and ${}^7\text{Li}$

In the CDCC framework, the S -matrix elements for breakup channels are obtained simultaneously together with those for the elastic channel. Since the only parameter of this framework, i.e., N_i , is determined by fitting the calculated elastic-scattering cross section to the measured, we have no additional free parameter for calculating the breakup cross sections. Therefore, it will be a stringent test of this theoretical framework and the reaction mechanisms concerned to examine to what extent the calculation reproduces the measured breakup cross sections. In this section, we investigate the validity of the method of CDCC to be used in the analyses of the projectile breakup reactions of ${}^6\text{Li}$ and ${}^7\text{Li}$, say ${}^6\text{Li} \rightarrow \alpha + d$ and ${}^7\text{Li} \rightarrow \alpha + t$ reactions. Since the mutual excitations of the projectile and target nuclei are not explicitly included in the present CDCC calculations, the breakup processes discussed here are the so-called elastic-breakup ones in which the target nucleus is left in the ground state at the final stage of the reaction.

Recently, exclusive measurements have become available^{86)~88)} of the cross sections for $({}^6\text{Li}, \alpha d)$ and $({}^7\text{Li}, \alpha t)$ elastic-breakup reactions with considerable accuracy. In the following, we show preliminary results of the CDCC analyses of some experimental data on the reactions. In particular, we concentrate our interest upon such processes in which the kinetic energy of the relative motion between the emitted breakup fragments (α and d for ${}^6\text{Li}$ and α and t for ${}^7\text{Li}$) is relatively small, say less than several MeV. This is because the model space of the present CDCC calculation is not so large as is suitable for the analyses of the breakup processes in which the projectiles are excited to much higher-energy states in the continuum. An extension of the method so as to be used for the breakup reactions to such highly-excited states will be a future subject.

One of the interesting points in the analyses of the breakup reactions to the low-energy continua of ${}^6\text{Li}$ and ${}^7\text{Li}$ is the relative magnitude between the breakup cross section from the low-lying resonance states and that from the non-resonant continuum states, as well as the competition of these processes through the coupling among the breakup channels. The CDCC frame is quite suitable for such kind of analyses of the reaction mechanism. Another point we shall discuss in this section is the variation of the total elastic-breakup cross section with the bombarding energy and with the mass number of

the target nucleus. Such investigation will be valuable for our knowledge on the gross features of the projectile-breakup reaction mechanisms.

5.1. Elastic breakup reactions of ${}^6\text{Li}$

First we analyze the exclusive cross sections for $({}^6\text{Li}, \alpha d)$ elastic-breakup reactions by ${}^{12}\text{C}$ and ${}^{208}\text{Pb}$ targets measured at a bombarding energy of 178 MeV. The measured cross sections⁸⁶⁾ have been given for the breakup from the continuum at $0.0 < \varepsilon < 1.0$ MeV (which we call as region I) and for that from the continuum at $1.0 < \varepsilon < 4.0$ MeV (region II), ε being the relative-motion energy between α and d measured with respect to the α - d threshold ($E_x = 1.47$ MeV) of ${}^6\text{Li}$; the 3^+ resonance at $E_x = 2.19$ MeV and the 2^+ one at $E_x = 4.31$ MeV are located in regions I and II, respectively. The double-differential cross section, $d^2\sigma/d\Omega_{\text{Li-T}}d\Omega_{\alpha-d}$, for each of regions I and II has been constructed by integrating the measured triple-differential cross section, $d^3\sigma/d\Omega_{\text{Li-T}}d\Omega_{\alpha-d}d\varepsilon$, over the corresponding region in ε . Therefore, the cross section in each of the regions contains contributions from both the resonance part and the non-resonance part of the region.⁸⁶⁾ On the other hand, the calculation presented here gives the single-differential cross sections of the breakup reaction. Therefore, in order to compare the calculated breakup cross section with the

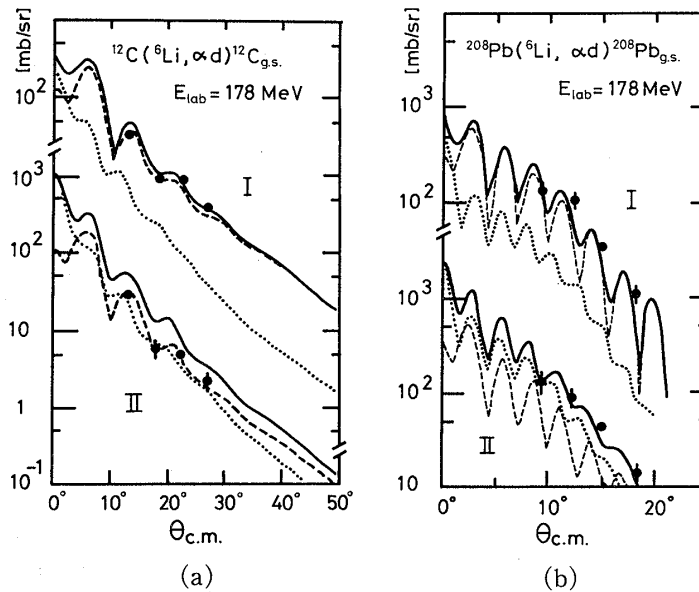


Fig. 26. (a) Observed⁸⁶⁾ and calculated⁹¹⁾ breakup cross sections for the ${}^{12}\text{C}({}^6\text{Li}, \alpha d){}^{12}\text{C}_{\text{gs}}$ reaction at $E_{\text{lab}} = 178$ MeV, shown with respect to the scattering angle of the c.m. of the breakup fragments. The upper part is for the breakup from the α - d continuum at $0.0 < \varepsilon < 1.0$ MeV (region I) and the lower is for the breakup from the continuum at $1.0 < \varepsilon < 4.0$ MeV (region II), ε being the relative energy between α and d . The dashed curve is the contribution of the resonance states (3^+ for region I and 2^+ for region II) and the dotted curve is that of the non-resonant continuum states, the solid curve being the sum of them.

(b) The same as Fig. 26(b) for the ${}^{208}\text{Pb}({}^6\text{Li}, \alpha d){}^{208}\text{Pb}_{\text{gs}}$ reaction at $E_{\text{lab}} = 178$ MeV.

measured, we further integrate the measured double-differential cross section over the α - d relative solid angles, $d\Omega_{\alpha-d}$, under the assumption that the decay of the breakup fragments, α and d , is isotropic in the center-of-mass (c.m.) frame of α and d . The integration is then replaced simply by a multiplication of $d^2\sigma/d\Omega_{\text{Li-T}}d\Omega_{\alpha-d}$ by 4π . We simply denote the resultant single-differential cross section as $d\sigma/d\Omega_{\text{cm}}$. The measured cross sections thus obtained are shown in Figs. 26(a) and (b) for ${}^{12}\text{C}$ target and ${}^{208}\text{Pb}$ one, respectively; θ_{cm} denotes the scattering angle of the c.m. of the breakup fragments, α and d , with respect to the beam direction.

In order to analyze the $({}^6\text{Li}, \alpha d)$ breakup reactions, we perform the CDCC calculation in the model space of Fig. 8(b) where the spin of deuteron is taken into account and the discretization is somewhat improved from that of Fig. 8(a). In the CDCC framework, what we obtain directly as the solutions of the coupled-channel equation (2.39) are

the S -matrix elements for individual bins defined in Eq. (2.31), with which we can calculate the breakup cross sections for the bins. The calculated cross section for the bin which contains the $3^+(2^+)$ resonance may be a good representative of the resonance component of the cross section in region I (II), since the width of the resonance is narrow enough. On the other hand, for the non-resonance part of region I (II), we make an interpolation through the cross-section values divided by the momentum width, $d^2 \bar{\sigma}_{iu}/d\Omega_{\text{cm}} dk_i$, for bins in individual partial wave such as 1^+ ($l=0$), 3^+ ($l=2$), etc., yielding $d^2 \bar{\sigma}_u/d\Omega_{\text{cm}} dk$. We then integrate them over k within the momentum interval corresponding to region I (II) and sum up the integrated cross sections over the partial waves:

$$d\sigma_{\text{I(II)}}/d\Omega_{\text{cm}} = \sum_u \int_{\text{I(II)}} (d^2 \bar{\sigma}_u/d\Omega_{\text{cm}} dk) \cdot dk. \quad (5.1)$$

The results of the calculation are compared with the experimental data⁸⁶⁾ in Figs. 26 (a) and (b). In these calculations, the values of N_l used are 0.47 and 0.60 for ^{12}C target and ^{208}Pb one, respectively, which have been determined by fitting the calculated elastic-scattering cross sections to the measured ones at $E_{\text{lab}}=169$ MeV for ^{12}C and 156 MeV for ^{208}Pb (cf. Table III in § 3.5). Although the present incident energy of 178 MeV is slightly larger than those of the above values, it will be justified to adopt the values of N_l deduced at slightly lower energies since the optimum value of N_l to fit the elastic scattering is almost independent of the energy, as is shown in § 3.5.

For each of region I (upper) and II (lower) in the Figs. 26(a) and (b), the dashed curve and the dotted one show the contribution of the resonance part and that of the non-resonance part, respectively, and the sum of them is given by the solid curve. In region I, the breakup occurs mainly from the 3^+ resonance for both targets, while, in region II, the contribution of the breakup process from the non-resonant continuum is as significant as that of the breakup process from the 2^+ resonance. When we compare the result for ^{12}C target with that for ^{208}Pb target, we notice that the contribution of the non-resonant breakup in region II is more prominent for the latter target than the former. For both targets, the agreement between the calculation and the experimental data⁸⁶⁾ is quite satisfactory both for the magnitudes and for the angular distributions. These results may guarantee the validity of the present theoretical framework to be used in the analyses of the breakup reactions as well as of the elastic scattering.

In the above, we have discussed the cross sections for the ^6Li breakup from the resonance states and that from the non-resonant continuum states. We further note that the breakup cross sections are much affected by the coupling between the resonance states and the non-resonance ones. For this purpose, we compare in Fig. 27 three kinds of calculations of the breakup cross sections from the 3^+ and 2^+ resonances in the case of $^6\text{Li} + ^{12}\text{C}$ at 178 MeV: (a) The same full-CC calculation including all the states in Fig. 8(b) as introduced above (solid curves which are the same as the dashed curves in Fig. 26(a)), (b) a CC calculation (dashed curves) in which the D - and P -wave non-resonant states are omitted from (a), (c) a CC calculation (dotted curves) in which the D -, P - and S -wave non-resonant states are omitted from (a), i.e., only the three resonances being included. The significant difference between the curves shows the importance of the multi-step transitions between the resonance states and the non-resonant continuum states through the coupling between them; the cross sections for the resonance states are reduced by a

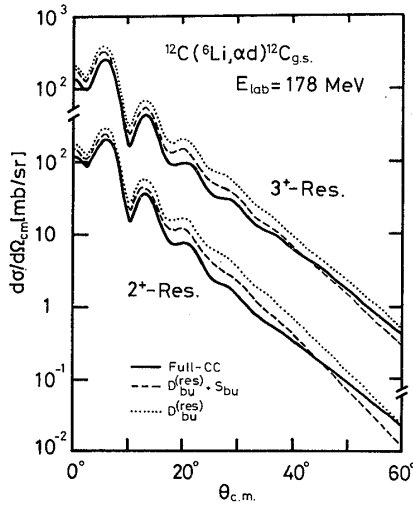


Fig. 27. The angular distributions of the ${}^6\text{Li} \rightarrow \alpha + d$ breakup reaction from the 3^+ and 2^+ resonances of ${}^6\text{Li}$ for the ${}^{12}\text{C}$ target at $E_{\text{lab}} = 178$ MeV. The solid curves represent the result of the full-CC calculation⁸⁵⁾ including all the states shown in Fig. 8(b). The dotted and the dashed curves show the CC calculation including $D_{\text{bu}}^{(\text{res})}$ only, and including S_{bu} and $D_{\text{bu}}^{(\text{res})}$, respectively. In all the calculation, the values of $N_R = 1.0$ and $N_I = 0.466$ have been used.

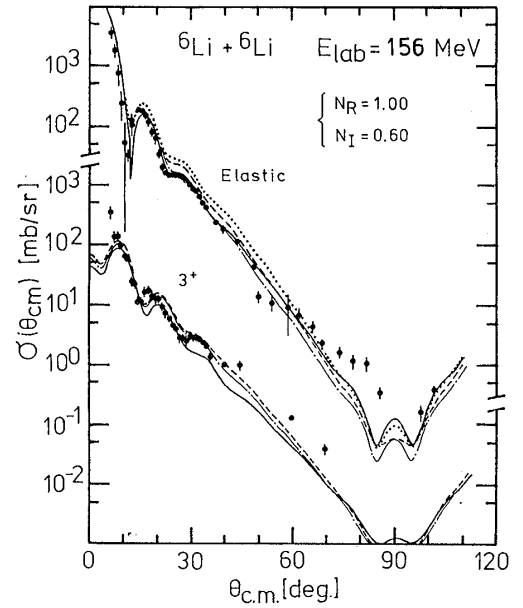


Fig. 28. The elastic scattering of ${}^6\text{Li} + {}^6\text{Li}$ at $E_{\text{lab}} = 156$ MeV and the inelastic one leading to the 3^+ resonance state of the target. The calculation and the experimental data are taken from Ref. 90). The solid curves represent the result of the full-CC calculation. The dotted curve shows the single-channel calculation. The dashed and the dot-dashed curves represent the result of the CC calculation in which the 1^+ ground state is coupled with 3^+ resonance channel only and with 3^+ , 2^+ and 1^+ resonance channels, respectively. In all the calculation, the values of $N_R = 1.0$ and $N_I = 0.6$ have been used.

factor of two to three by the inclusion of the coupling with little change of the angular dependence. Also, it is found that the breakup from the non-resonant states is much affected by the multi-step transition to the states via the resonance states.

Let us consider an inelastic scattering of a projectile from unbound excited states of ${}^6\text{Li}$ target. If we consider the interchange of the projectile and target, the observed target-inelastic cross section for the unbound states can be identified with the cross section for the projectile breakup leaving the target nucleus in the ground state. In many cases, such a measurement⁸⁹⁾ of the inelastic scattering of a projectile from the ${}^6\text{Li}$ target seems much simpler and easier than the coincident detection of the breakup fragments. Thus, a good quality of the angular distribution data can be secured, which may provide a good test of the CDCC calculation for the projectile breakup. This type of measurement for ${}^6\text{Li} + {}^6\text{Li}$ scattering has been done recently with using 156-MeV ${}^6\text{Li}$ beam at Karlsruhe.⁹⁰⁾ Figure 28 shows the observed differential cross sections for the elastic scattering and the inelastic one to the 3^+ resonance state of the target ($E_x = 2.19$ MeV). In the figure, three kinds of CC calculations⁹⁰⁾ are shown by the solid curve (full-CC calculation), the dot-dashed one (four-channel one) and the dashed one (two-channel one), respectively, together with the single-channel calculation (dotted curve). In the calculation, N_R is kept

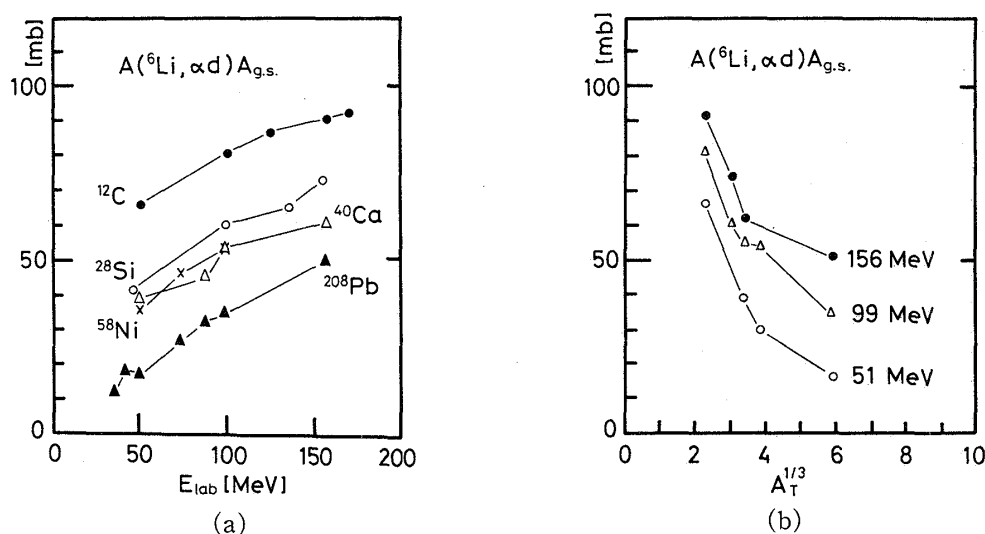


Fig. 29. Angle-integrated cross sections for the elastic breakup of ${}^6\text{Li}$ projectile predicted by the CDCC calculations of Figs. 14~19; (a) the incident-energy dependence and (b) the target-mass dependence.

fixed at 1.0, whereas $N_i=0.6$ is determined so that the full-CC calculation (solid curves) reproduce the elastic scattering. The mutual-excitation channels are not taken into account here for simplicity. Both the observed elastic and the inelastic cross sections are reproduced well by the full-CC calculation, except at extremely forward angles and at large angles. The difference between those curves shows the effect of multi-step transitions discussed above which is still important for the present system.

Finally, we discuss briefly the dependence of the total elastic-breakup cross section on the bombarding energy and the target mass number based on the CDCC calculations shown in § 3.5. By considering the results of Figs. 26~28, such discussions based on the CDCC calculations will make sense, because the absolute magnitude of the measured breakup cross sections is reproduced well by the theory. Figures 29(a) and (b) show the CDCC prediction⁹¹⁾ of the variation of the total elastic-breakup cross section with the bombarding energy and the target mass number, respectively. For all the targets examined here, the cross section increases almost linearly with the bombarding energy within the energy region investigated here. Since it is found that, within the energy region, the calculated total reaction cross section is almost saturated at the complete absorption limit, the result of Fig. 29(a) indicates that the fraction of the elastic-breakup cross section in the reaction cross section increases as the energy increases. On the other hand, the elastic-breakup cross section is much larger for light targets than for

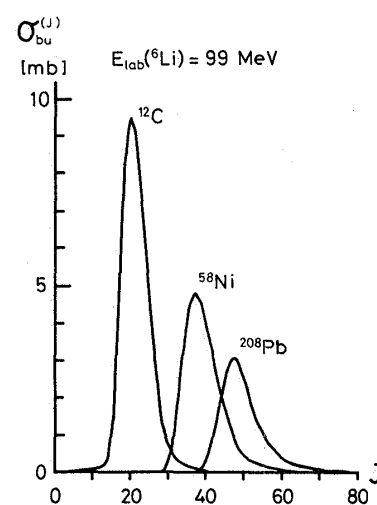


Fig. 30. Partial cross sections for the elastic breakup of ${}^6\text{Li}$ projectile by ${}^{12}\text{C}$, ${}^{58}\text{Ni}$ and ${}^{208}\text{Pb}$ targets at $E_{\text{lab}}=99$ MeV. Abscissa J is the incident angular momentum.

heavy targets as clearly seen in Fig. 29(b). Since it is known³⁴⁾ that the elastic-breakup occurs at the periphery of the target nucleus, one would expect that the cross section increases almost proportionally to $A_T^{1/3}$, where A_T is the target mass number. However, the result of the calculation is completely opposite to the above conjecture. This unexpected A_T -dependence of the cross section can be seen also in Fig. 30 which shows the partial cross sections of the elastic-breakup by ^{12}C , ^{58}Ni and ^{208}Pb targets at $E_{\text{lab}}=99$ MeV. As seen in the figure, the elastic-breakup occurs at a narrow region of the periphery of targets with a common width of about 2 fm as mentioned already, but the magnitude of the partial cross section at the peak position decreases rapidly with the increase of the target mass number, which results in the decrease of the total elastic-breakup cross section with A_T shown in Fig. 29(b). These results suggest that the flux is removed out from the elastic-breakup channels to other reaction channels, such as inelastic-breakup channels and fusion ones, more heavily for heavy targets than for light targets. Therefore, further investigation of the energy and the target dependence of the elastic-breakup cross sections both theoretically and experimentally together with those of the inelastic-breakup ones will be an interesting subject in the future.

5.2. Elastic breakup reactions of ^7Li

Next, we analyze the cross sections for (^7Li , αt) elastic-scattering reactions by ^{12}C , ^{120}Sn and ^{208}Pb targets measured at a bombarding energy of 70 MeV. The coincident measurements^{87),88)} were done of the emitted α -particles and tritons from the $7/2^-$ resonance state at $E_x=4.63$ MeV ($\varepsilon=2.16$ MeV with respect to the α -t breakup threshold) and those from the non-resonant continuum at $0.29 < \varepsilon < 3.0$ MeV. Since the measured cross section for the resonance state is separated from that for the non-resonant continuum states, the analyses of the experimental data will provide another stringent test of the CDCC method and the reaction mechanisms concerned. Particularly, it is quite interesting to investigate whether the relative magnitude of the breakup cross section from the resonance and that from the non-resonance are reproduced by the calculation. The experimental data^{87),88)} were given in the similar form of the single-differential cross section to that for the ^6Li case mentioned in § 5.1, i.e., the differential cross section for the

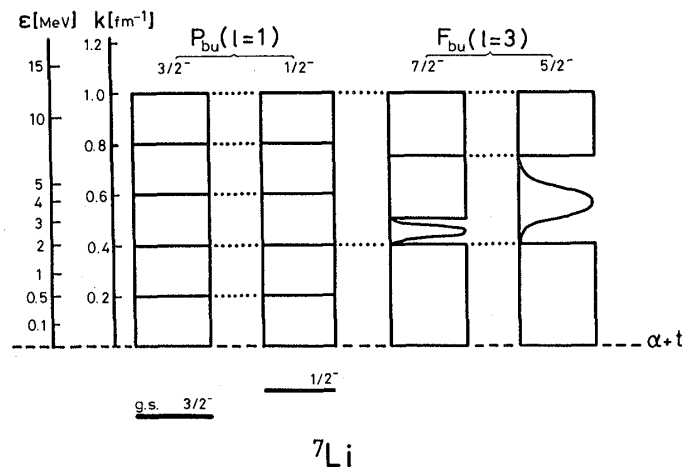


Fig. 31. The same as Fig. 21 for more precise discretization of the α -t breakup continuum of ^7Li . This is used for the CDCC calculation of the ^7Li projectile-breakup cross sections.

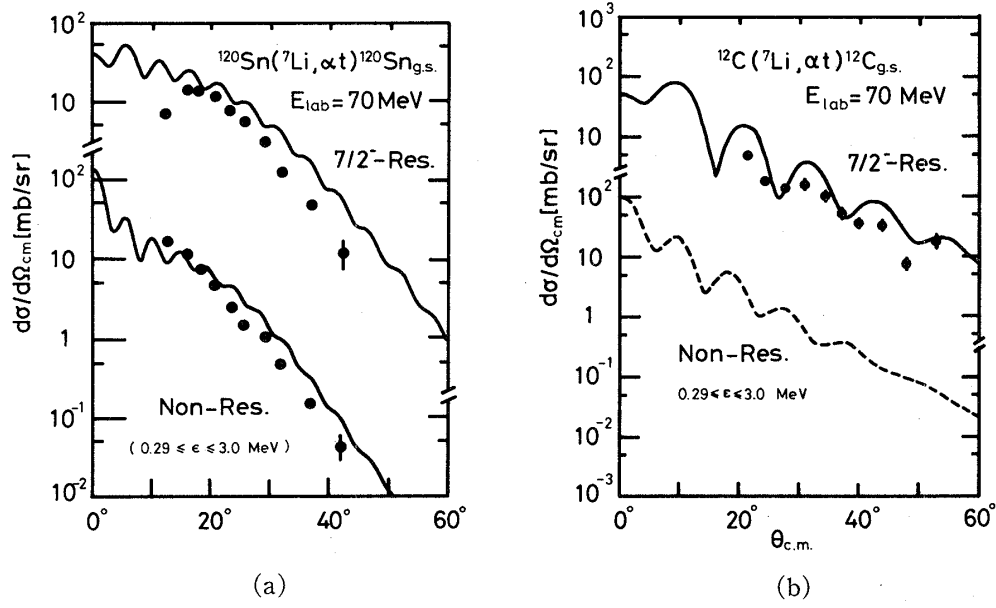


Fig. 32. (a) Observed⁸⁸⁾ and calculated⁹¹⁾ breakup cross sections for the $^{120}\text{Sn}(^7\text{Li}, \alpha t)^{120}\text{Sn}_{g.s.}$ reaction at $E_{lab} = 70$ MeV. The values of N_R and N_I used in the calculation are 1.0 and 0.65, respectively. The upper part is for the breakup from the $7/2^-$ resonance state and the lower is for the breakup from the non-resonant continuum at $0.29 < \epsilon < 3.0$ MeV, ϵ being the relative energy between α and t . (b) Observed⁸⁷⁾ and calculated⁹¹⁾ breakup cross section for the $^{12}\text{C}(^7\text{Li}, \alpha t)^{12}\text{C}_{g.s.}$ reaction $E_{lab} = 70$ MeV. The solid curve is the predicted cross section for breakup from the $7/2^-$ resonance and the dashed curve is for breakup from the non-resonant continuum at $0.29 < \epsilon < 3.0$ MeV. Breakup from the non-resonant continuum is not seen in the experiment.⁸⁷⁾

scattering angle of the c.m. of the emitted α -particle and triton with respect to the beam direction.

As like the ^6Li -breakup reactions, we regard the calculated cross section for the $7/2^-$ -resonance bin as a representative of the cross section for the resonance state, while we make an interpolation for the non-resonance bins and integrate the interpolated cross section similar to Eq. (5.1) in order to compare with the experimental data. To improve the accuracy of the interpolation for the non-resonant part, the model space for the ^7Li -breakup process has been somewhat improved from that employed in the previous section for the analyses of the elastic scattering (Fig. 21); the model space employed here is illustrated in Fig. 31. Figure 32(a) shows the measured⁸⁸⁾ and calculated⁹¹⁾ cross sections for the ^7Li elastic-breakup by ^{120}Sn target from the $7/2^-$ resonance (upper) and those from the non-resonant continuum (lower). The agreement between the calculated cross section and the measured is satisfactory for the $7/2^-$ resonance and very good for the non-resonant continuum both in the magnitude and in the angular distribution, except at forward angles for the former. In the calculation, the value of $N_I = 0.65$ has been employed; since there exist no experimental data on the elastic scattering of ^7Li by ^{120}Sn at 70 MeV, we have adopted the value of N_I by considering the fact that the elastic scattering of ^7Li by heavy targets are reproduced well by the CDCC calculations with $N_I \approx 0.6$, as shown in § 4. Therefore, we have not tried to adjust the value of N_I so as to fit the experimental data on the breakup reactions.

The calculation has been performed for the case of ^{12}C target and compared with the experimental data⁸⁷⁾ in Fig. 32(b). Since there exist no data for the elastic scattering at

$E_{\text{lab}}=70$ MeV, we have determined the value of N_i by averaging the values determined in §4 (Figs. 23(b) and 25(b)) at $E_{\text{lab}}=63$ MeV ($N_i=0.40$) and at 78.7 MeV ($N_i=0.50$), which gives $N_i=0.45$ for 70 MeV. The agreement between the calculation (the solid curve) and the experimental data⁸⁷⁾ is satisfactory for the cross section from the $7/2^-$ resonance though the calculated cross section is more oscillatory than the measured. On the other hand, for this target, no experimental evidence was found⁸⁷⁾ of the breakup from the non-resonant continuum at $0.29 < \varepsilon < 3.0$ MeV, whereas the CDCC calculation predicts non-negligible magnitude of this breakup (the dashed curve) compared with the breakup from the $7/2^-$ resonance. By considering the success of the present theory for ^{120}Sn target, particularly on the non-resonant breakup cross section, it is highly desired to confirm experimentally whether the cross section is really negligible for this target.

In the calculation shown above, we have ignored the projectile breakup by the Coulomb interaction. In order to investigate the effects of the Coulomb interaction, especially in the case of heavy targets, we perform a preliminary calculation including both the nuclear and Coulomb interactions which induce the projectile breakup. The Coulomb interactions are derived by folding the nucleon-target Coulomb potential with the transition densities between the states of ^7Li . The CDCC calculation is performed in the four-channel frame including the ground state ($3/2^-$), the first excited state ($1/2^-$) and the two resonance states ($7/2^-$ and $5/2^-$); the coupling with the non-resonant breakup channels is neglected for simplicity. Therefore, the discussion is made of the effect of the

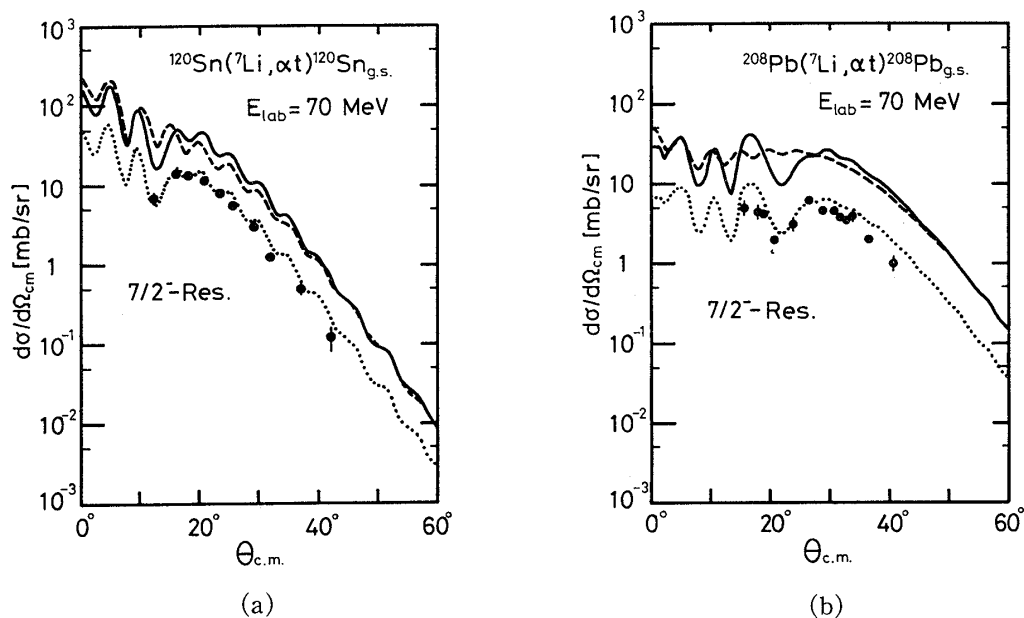


Fig. 33. (a) Contribution of the Coulomb interaction to the ^7Li projectile breakup from the $7/2^-$ resonance in the $^{120}\text{Sn}(^7\text{Li}, \alpha t)^{120}\text{Sn}_{\text{g.s.}}$ reaction at $E_{\text{lab}}=70$ MeV. The data are taken from Ref. 88). The calculation is made only with the four-channel scheme ($3/2^-$ and $1/2^-$ bound states and $7/2^-$ and $5/2^-$ resonance states). The values of N_R and N_i are the same as those used in Fig. 32(a). The dashed curve is the calculated cross section for the breakup induced by the nuclear interaction only and the solid curve is for the breakup induced by the nuclear and Coulomb interactions. The dotted curve is simply given by reducing the solid curve by a factor of three (see text). (b) The same as Fig. 33(a) for the $^{208}\text{Pb}(^7\text{Li}, \alpha t)^{208}\text{Pb}_{\text{g.s.}}$ reaction at $E_{\text{lab}}=70$ MeV. The values of N_R and N_i used are 1.0 and 0.60, respectively. The dotted curve is given simply by reducing the solid curve by a factor of four (see text).

Coulomb interaction on the breakup cross section for the $7/2^-$ resonance state. In Figs. 33 (a) and (b), the calculated cross section (the solid curve) is compared with the experimental data^{87),88)} and with the cross section given by the same calculation but neglecting the Coulomb interaction (the dashed curve) (a) for the ^{120}Sn target and (b) for the ^{208}Pb one, respectively. The effect of the Coulomb interaction is significant, particularly at forward angles, and the angular distributions of the calculated cross sections well reproduce the measured ones for both targets as shown by the dotted curves, although the absolute magnitude of the calculated is larger than that of the measured by a factor of three for the ^{120}Sn target and by a factor of four for the ^{208}Pb one, respectively (cf. the dotted curves to see the factor). The discrepancy of the magnitude is, however, interpreted as the effect of the non-resonant breakup channels which have been ignored in the calculation. Namely, as we have seen in the previous subsection for the ^6Li breakup, the resonance-breakup cross section suffered much reduction from the multi-step process via the non-resonance channels through the coupling with the channels (cf. the reduction from the dotted curve to the solid one in Fig. 27). This reaction mechanism may hold also for the ^7Li -breakup reactions. Therefore, much improvement of the fit with the experimental data will be expected for the calculated cross section for the $7/2^-$ state if we perform a calculation in the full space of Fig. 31 with including the Coulomb interaction. Such a calculation is in progress and will be reported elsewhere.

In conclusion, as far as the elastic-breakup reactions presented here are concerned, the projectile breakup reactions of ^6Li and ^7Li are induced mainly by the nuclear interaction from the target nucleus and the CDCC calculation including the breakup by the nuclear interaction well reproduces the absolute magnitudes and the angular distributions of the measured cross sections for the breakup both from the low-lying resonance states and from the surrounding non-resonant continuum states. The projectile breakup induced by the Coulomb interaction from the target nucleus is also important for the breakup reactions by heavy target nucleus at low bombarding energies and the Coulomb-breakup amplitude interferes much with the nuclear-breakup one at forward angles, but the effect of the Coulomb interaction is not so significant at large angles or for the reactions at higher bombarding energies where the nuclear interaction is dominant.

§ 6. Scattering and breakup of ^{12}C

As mentioned before, the double-folding model with $N_R=1.0$ has been successful in describing the lower-energy ($E_{\text{lab}}/A_p \lesssim 20$ MeV) scattering of various heavy ions except $^6,7\text{Li}$ and ^9Be .¹³⁾ However, Bohlen et al. pointed out strong reduction of N_R (≈ 0.6) in the $^{12}\text{C} + ^{12}\text{C}$ scattering at $E_{\text{lab}}=300$ MeV.³⁹⁾ They have measured the elastic scattering and the inelastic scattering leading to the 2^+ state at 4.44 MeV. They performed a CC calculation including the 2^+ state of one of the ^{12}C nuclei, neglecting the mutual-excitation channel. In the calculation, the real part of the diagonal and the coupling potentials was constructed by the double-folding procedure just as in the present work and was multiplied by a renormalization factor, N_R , as an adjustable parameter, whereas the imaginary part was parametrized with a deformed Woods-Saxon shape. Then, the optimum value of the five parameters, N_R , the strength, the geometry and the deformation parameters, W_0 , r_1 and a_1 , and β_2 , respectively, of the imaginary part was searched so that the observed elastic and inelastic cross sections are reproduced simultaneously well by the CC calculation. It was

found that the best fit requires a renormalization of $N_R=0.50$. If N_R is kept fixed at 1.0, one cannot reproduce the correct phase of the diffraction of the observed cross sections.

The double-folding-model analyses of ^{12}C scattering were limited to data at lower energies, $E_{\text{lab}}/12 \lesssim 10$ MeV.¹³⁾ Thus, in order to confirm the reduction of N_R mentioned above and investigate their incident-energy dependence, we first analyze the recent data^{39)~44)} on the elastic scattering of ^{12}C , ^{13}C and ^{16}O from ^{12}C at $E_{\text{lab}}/A_p=10\sim 25$ MeV with the double-folding model. Next, we perform a microscopic CC calculation including the $^{12}\text{C} \rightarrow 3\alpha$ breakup channels, and discuss the origin and the incident-energy dependence of the anomaly in N_R . We also analyze the inelastic scattering of $^6,7\text{Li}$, ^{12}C and ^{16}O from ^{12}C leading to some low-lying states of ^{12}C .

6.1. Double-folding model for higher-energy ^{12}C scattering

First, we make the double-folding-model (single-channel) calculation for the elastic scattering of $^{12}\text{C}+^{12}\text{C}$ at $E_{\text{lab}}/12=10\sim 25$ MeV. The ground-state nucleon density of ^{12}C is given by the 3α -RGM wave function discussed in § 2.1. Since the projectile and the target nuclei are identical particles, the symmetry of the scattering amplitude is taken into account. We make two types of calculations. In case (a), the shape of the imaginary potential is taken to be the same as that of the double-folded potential and the strength factor of the real and the imaginary part, N_R and N_I respectively, are searched. In case (b), the Woods-Saxon shape is assumed for the imaginary potential, and the optimum values of N_R and the three parameters of the imaginary part, W_0 , r_1 and a_1 , are searched. An automatic potential-parameter search code *ALPS*⁶³⁾ is used in the analysis.

The solid curves in Fig. 34 show the best-fit results^{35),92)} of case (b). There is no significant difference between the results of cases (a) and (b), except at very large scattering angles. The optimum values of the potential parameters obtained are given in Table IV. The value of N_R tends to decrease as the incident energy increases, though there is a small fluctuation. We plot the value of N_R versus the incident energy per nucleon, E_{lab}/A_p , by the solid circles in Fig. 35(a). One could understand that the double-folding model with $N_R=1.0$ breaks down for $E_{\text{lab}}/A_p \gtrsim 10$ MeV. If N_R is kept fixed at 1.0, we cannot obtain a good fit to the data, as shown by the dashed curve in Fig. 34 for the case $E_{\text{lab}}=300$ MeV. As mentioned above, the reduction of N_R cannot be removed even by taking into account the channel-coupling effect of the 2^+ state at 4.44 MeV.³⁹⁾ Therefore, it is interesting to investigate whether the reduction of N_R and its incident-energy dependence are explained as the virtual-

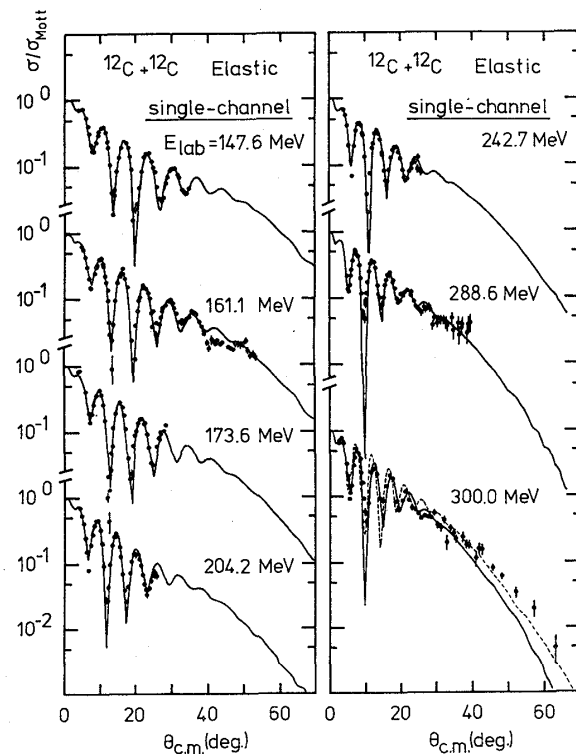


Fig. 34. The elastic scattering of $^{12}\text{C}+^{12}\text{C}$ at $E_{\text{lab}}=147.6\sim 300$ MeV. The solid curves show the best-fit result of the single-channel calculation of case (b). The optimum values of the potential parameters are given in Table IV. The experimental data are taken from Refs. 39)~41).

Table IV. The double-folding-model potential parameters for the $^{12}\text{C}+^{12}\text{C}$ scattering at $E_{\text{lab}}=147.6 \sim 300$ MeV. In case (a), the shape of the imaginary potential is taken to be the same as that of the real potential, while the Woods-Saxon form is used for the imaginary potential in case (b). χ^2/N denotes the chi-square value per data point. In the lowest line, N_R is kept fixed at unity.

E_{lab} (MeV)	(a)			(b)				
	N_R	N_I	χ^2/N	N_R	W_0 (MeV)	r_1 (fm)	a_1 (fm)	χ^2/N
147.6	0.918	0.692	15.5	0.916	216.0	1.366	0.763	9.4
161.1	0.834	0.672	19.3	0.834	241.5	1.384	0.736	14.5
173.6	0.875	0.725	29.8	0.876	275.0	1.335	0.752	22.8
204.2	0.776	0.701	17.5	0.776	252.0	1.215	0.814	10.6
242.7	0.751	0.638	17.1	0.752	253.0	1.241	0.790	11.7
288.6	0.827	0.769	12.1	0.830	271.1	1.243	0.817	7.5
300.0	0.657	0.633	24.2	0.658	298.8	1.094	0.844	12.1
300.0				1.000	442.0	1.285	0.747	97.0

(a) $V(R)=(N_R+iN_I) \cdot V_{\text{DF}}(R)$.

(b) $V(R)=N_R \cdot V_{\text{DF}}(R)-iW_0/(1+\exp((R-R_I)/a_I))$, $R_I=r_1 A^{1/3}$, $A=12$.

*) Coulomb radius is fixed to $R_c=2.4 \cdot A^{1/3}$ for both (a) and (b) cases.

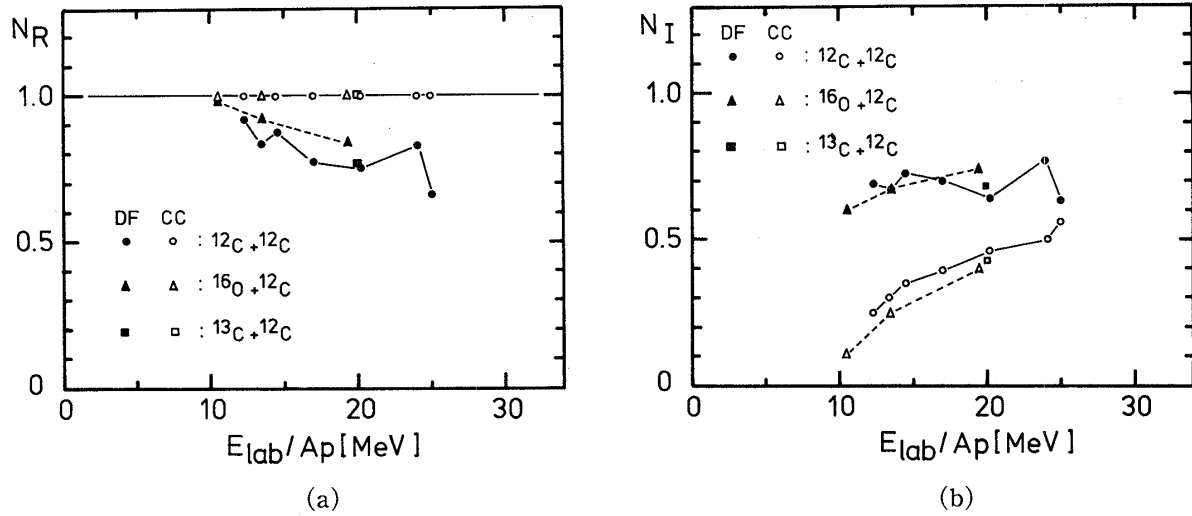


Fig. 35. (a) The incident-energy dependence of the optimum value of N_R for the full-CC calculation (open symbols) and for the single-channel (double-folding-model) one (solid symbols) of the scattering of $^{12}\text{C}+^{12}\text{C}$, $^{16}\text{O}+^{12}\text{C}$ and $^{13}\text{C}+^{12}\text{C}$.

(b) The same as Fig. 35(a) for the value of N_I .

breakup effect of ^{12}C above the 3α -breakup threshold at 7.27 MeV.

Similar results³⁵⁾ have been obtained for the $^{16}\text{O}+^{12}\text{C}$ scattering^{43),44)} at $E_{\text{lab}}(^{16}\text{O})=168$, 216 and 311 MeV and for the $^{13}\text{C}+^{12}\text{C}$ scattering⁴²⁾ at $E_{\text{lab}}(^{13}\text{C})=260$ MeV. In the calculation, the ground-state nucleon density of ^{16}O (^{13}C) is taken from Ref. 62) (from Ref. 93)). The optimum values of N_R are plotted by the solid symbols in Fig. 35(a).

For the sake of later discussions, we also plot the optimum values of N_I for case (a) of Table IV by the solid symbols in Fig. 35(b) for the three systems.

6.2. $^{12}\text{C} \rightarrow 3\alpha$ virtual-breakup effects in the elastic scattering

Next, we perform the CC calculation including the $^{12}\text{C} \rightarrow 3\alpha$ breakup states as well as the 2^+ state at 4.44 MeV. As discussed before, the resonant and the non-resonant breakup continua are approximated by a finite number of discrete states obtained by the 3α -RGM calculation.^{46),47)} The model space for the present CC calculation is illustrated in Fig. 36. The three states of 0_2^+ , 2_2^+ and 3_1^- in the figure are the resonance states above the 3α -breakup threshold, while the other four states above the threshold are the discretized non-resonant breakup states. The 0_1^+ , 2_1^+ and 3_1^- states are of shell-model-like character and are represented as spatially compact wave functions of the relative motion between the totally antisymmetrized 3α clusters. On the other hand, the 0_2^+ , 2_2^+ and the non-resonant breakup states are well-developed 3α -clustering states.

First, we analyze the $^{12}\text{C} + ^{12}\text{C}$ scattering at $E_{\text{lab}} = 300$ MeV.³⁹⁾ In the present CC calculation, we do not include the mutual-excitation channels for simplicity, but take into account the symmetry of the scattering amplitude due to the identity of the projectile and the target nuclei. The value of N_R is kept fixed at 1.0, while that of N_I is changed so that

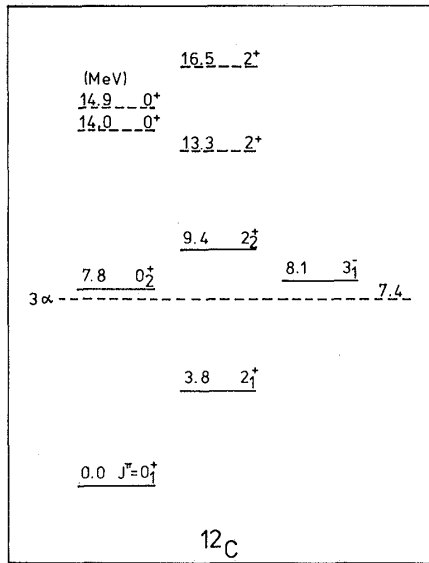


Fig. 36. The bound states and the 3α -breakup states of ^{12}C for the CC calculation. The excitation energy of each state is given. The dashed lines stand for the discretized non-resonant-breakup states.

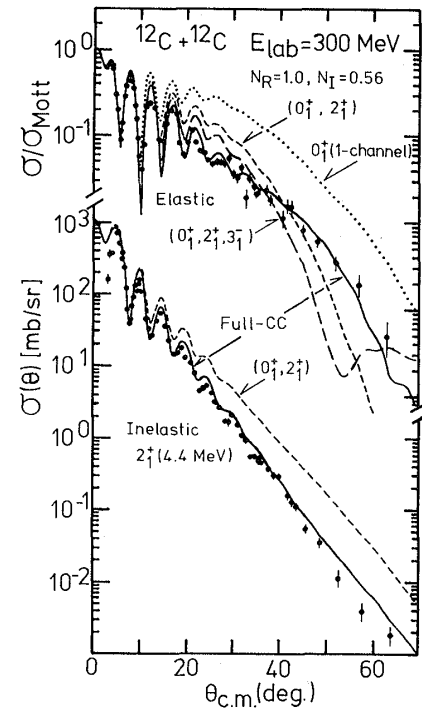


Fig. 37. The elastic scattering of $^{12}\text{C} + ^{12}\text{C}$ at $E_{\text{lab}} = 300$ MeV and the inelastic scattering leading to the 2_1^+ excited state. The experimental data are taken from Ref. 39). The solid curves represent the result of the full-CC calculation³⁵⁾ including all the states in Fig. 36. The result of the single-channel calculation is shown by the dotted curve. The short-dashed and the long-dashed curves show the result of the two-state CC calculation including the 0_1^+ and 2_1^+ states and that of the three-state one including the 0_1^+ , 2_1^+ and 3_1^- states, respectively. In all the calculation, the values of $N_R = 1.0$ and $N_I = 0.56$ have been used.

the full-CC calculation (including all the channels in Fig. 36) reproduces well the observed elastic cross section.

The solid curves in Fig. 37 show the result of the full-CC calculation.^(35),92) The optimum value of N_I is 0.56. The observed elastic cross section is reproduced well up to backward angles by the calculation with $N_R=1.0$. It should be stressed that any reduction of N_R is not necessary for fitting the data. The inelastic cross section for the 2_1^+ state is reproduced well. To see the role of the coupling of the individual channels, we make two other kinds of CC calculations, one of which includes 2_1^+ channel only (short-dashed curve in Fig. 37) and the other includes 2_1^+ and 3_1^- channels (long-dashed curve). The single-channel result is also shown by the dotted curve in the figure. The same values of N_R and N_I as for the full-CC calculation are used here. The large difference between the solid and the dotted curves shows a strong CC effect of the whole excited channels. Though the coupling of the 2_1^+ channel seems to have the largest effect, the 0_2^+ , the 2_2^+ and the non-resonant breakup channels also play important roles, especially in reproducing the enhancement of the observed cross section at large angles.

A similar successful result^(35),45),92) is obtained for the $^{12}\text{C}+^{12}\text{C}$ scattering at $E_{\text{lab}}(^{12}\text{C})=148\sim 289$ MeV (Fig. 38), for the $^{16}\text{O}+^{12}\text{C}$ scattering at $E_{\text{lab}}(^{16}\text{O})=168\sim 311$ MeV (Fig. 39), and for the $^{13}\text{C}+^{12}\text{C}$ scattering at $E_{\text{lab}}(^{13}\text{C})=260$ MeV (Fig. 40). In all these cases, the observed data are reproduced well by the full-CC calculation (solid curves in those figures) without any reduction of N_R from 1.0. The optimum value of N_I is given in each figure, and

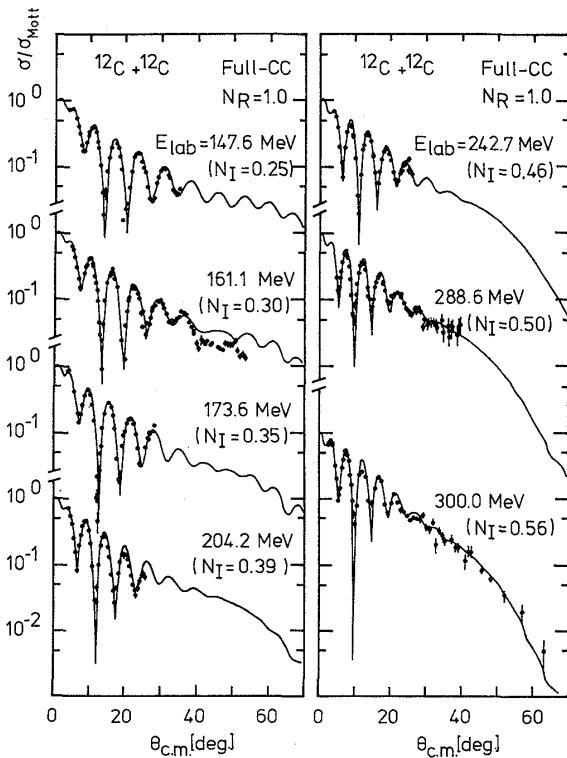


Fig. 38. The elastic scattering of $^{12}\text{C}+^{12}\text{C}$ at $E_{\text{lab}}=147.6\sim 300$ MeV. The experimental data are taken from Refs. 39)~41). The solid curves represent the result of the full-CC calculation⁽³⁵⁾ with $N_R=1.0$ with including all the states in Fig. 36. The optimum value of N_I for the calculation is given for each curve.

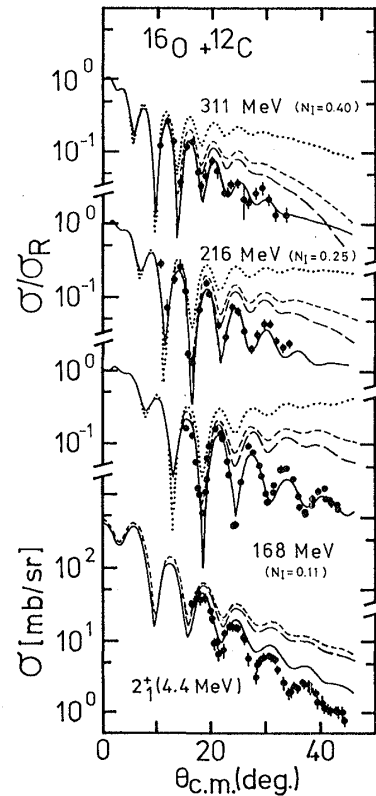


Fig. 39. The elastic and the inelastic scattering of ^{16}O from ^{12}C at $E_{\text{lab}}=168, 216$ and 311 MeV. The experimental data are taken from Refs. 43) and 44). The meaning of the curves is the same as in Fig. 37. This figure is taken from Ref. 45).

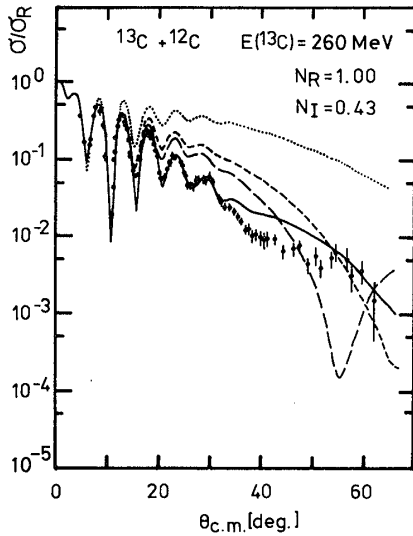


Fig. 40. The elastic scattering of ^{13}C from ^{12}C at $E_{\text{lab}} = 260$ MeV. The experimental data are taken from Ref. 42). The meaning of the curves is the same as in Fig. 37.

folding model still works well. Significant deviation of N_R from 1.0, however, is seen in the incident-energy region of $E_{\text{lab}}/A_p \gtrsim 10$ MeV. In this region, the CC effect is considered to induce a sizable real potential of repulsive nature. Therefore, the double-folding model with $N_R = 1.0$ does not work any more. Namely, the dynamical polarization potential due to the ^{12}C -excitation channels must have a sizable imaginary part and a small real part for $E_{\text{lab}}/A_p \lesssim 10$ MeV, while the real part increases and the imaginary part decreases as the incident energy increases.⁴⁵⁾ Explicit form of the potential will be calculated in the next section and the qualitative feature of the potential will be confirmed.

The incident-energy dependence of the ^{12}C -breakup effect seems much different from that of the ^6Li -breakup discussed in § 3.5. In the ^6Li case, the anomalous value of $N_R \approx 0.6$ in the double-folding model occurs almost independent of the incident energy and is explained as the repulsive effect of the ^6Li -breakup channels (Figs. 14~19). The optimum value of N_I for the CC calculation is also independent of the energy (Table III). In the present case, however, the value of N_R in the double-folding model begins to decrease rapidly at $E_{\text{lab}}/A_p \approx 10$ MeV (Fig. 35(a)), which can be explained by the CC calculation including the ^{12}C -breakup channels (Figs. 37~40). The optimum value of N_I for the CC calculation increases with the incident energy (Fig. 35(b)). Therefore, the energy dependence of the CC effect seems to correlate with the energy dependence of the optimum value of N_I . In fact, we have seen in § 3.4 that the imaginary part of the coupling potentials is very important in considering the CC effect. In § 7, we shall show that the imaginary part of the coupling largely affects the qualitative feature of the dynamical polarization potential induced by the coupling.

6.3. Inelastic scattering leading to the 2_1^+ , 3_1^- and 0_2^+ states of ^{12}C

In this section, we analyze the inelastic scattering of ^6Li , ^7Li , ^{12}C and ^{16}O projectiles leading to the 2_1^+ , 3_1^- and 0_2^+ states of the ^{12}C target, and discuss the effect of $^{12}\text{C} \rightarrow 3\alpha$ breakup channels on the inelastic cross sections. In the CC calculation throughout the

also illustrated in Fig. 35(b) in which N_I is plotted by open symbols versus the incident energy per nucleon, E_{lab}/A_p . It should be noted that the value of N_I for the CC calculation is much smaller than that for the double-folding model (solid symbols in Fig. 35(b)) in the lower incident-energy region, while the difference decreases as the incident energy increases. On the other hand, the optimum value of N_R for the double-folding model is close to 1.0 in the lower incident-energy region, while it decreases as the energy increases (Fig. 35(a)).

These results suggest the following. In the lower-energy region of $E_{\text{lab}}/A_p \lesssim 10$ MeV, the CC effect of the ^{12}C -excited channels can be taken into account by a renormalization of the imaginary potential in the single-channel calculation, and thus the double-

present subsection, we do *not* explicitly include any projectile-excited channels but include the 2_1^+ and the breakup channels of the ^{12}C target. The ground-state nucleon densities of the projectile nuclei are taken to be the same as those used before.

For ^{12}C and ^{16}O projectiles, N_R is kept fixed at 1.0 and the only adjustable parameter N_I is determined so that the CC calculation fit the *elastic* cross section. There is, therefore, no adjustable parameter left to fit the inelastic cross sections. In the case of ^6Li and ^7Li projectiles, however, we already know that, as long as we ignore the projectile breakup channels, the elastic cross section cannot be reproduced by the calculation with $N_R=1.0$. In order to take the projectile-breakup effects into account, we therefore allow N_R to change reasonably in fitting the *elastic* cross section by the CC calculation including the ^{12}C excited states in order to take the breakup effect approximately into account.

6.3.1. Inelastic scattering of $^{12}\text{C}+^{12}\text{C}$ and $^{16}\text{O}+^{12}\text{C}$

As already mentioned in the previous subsection, the full-CC calculation reproduces well the inelastic cross section for the 2_1^+ state in $^{12}\text{C}+^{12}\text{C}$ scattering at 300 MeV (solid curve in Fig. 37). Furthermore, the importance of the multi-step transition to the 2_1^+ state via the $^{12}\text{C} \rightarrow 3\alpha$ breakup channels is obvious from the difference between the solid and the

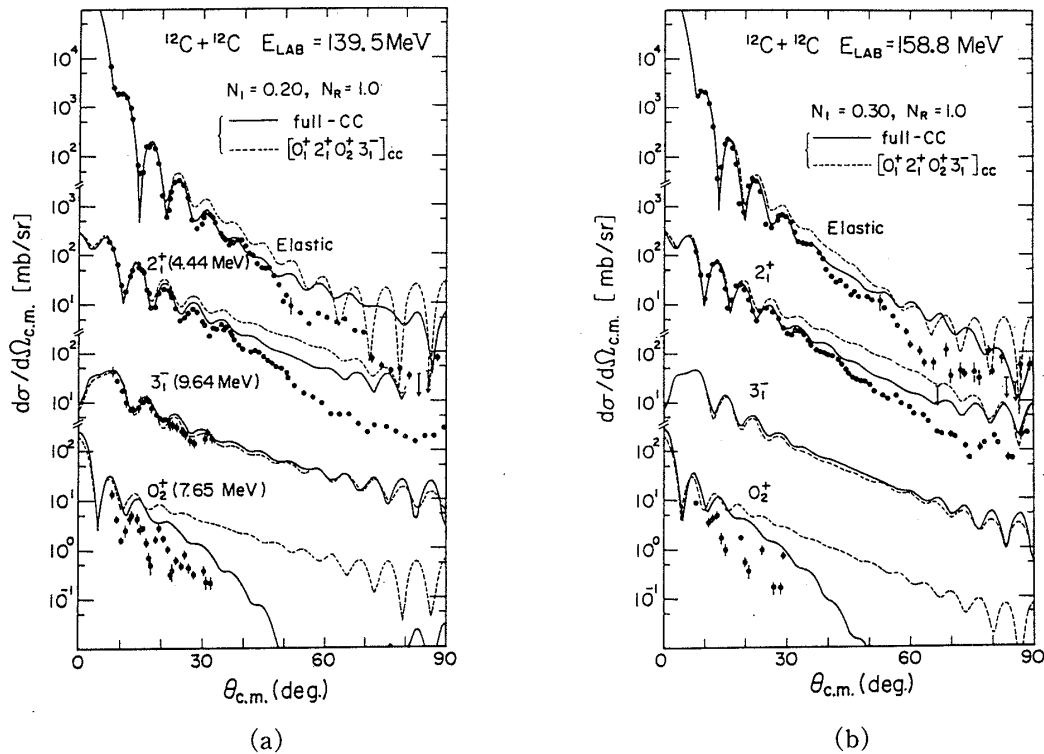


Fig. 41. (a) The elastic scattering of $^{12}\text{C}+^{12}\text{C}$ at $E_{\text{lab}}=139.5$ MeV and the inelastic one leading to the 2_1^+ , 3_1^- and 0_2^+ states of ^{12}C . The experimental data are taken from Ref. 94). The solid curves are the full-CC calculation with $N_I=0.20$ and $N_R=1.0$. The dashed curves are the small-space CC calculations which include only the 0_1^+ , 2_1^+ , 3_1^- and 0_2^+ states with the same renormalization as for the full-CC calculation. This figure is taken from Ref. 94).

(b) The same as Fig. 41(a) for $E_{\text{lab}}=158.8$ MeV. The experimental data are taken from Ref. 94). The solid curves are the full-CC calculation with $N_I=0.30$ and $N_R=1.0$. The dashed curves are the small-space CC calculations which include only the 0_1^+ , 2_1^+ , 3_1^- and 0_2^+ states with the same renormalization as for the full-CC calculation. This figure is taken from Ref. 94).

short dashed curves in the figure.

A similar result⁴⁵⁾ is obtained in the case of $^{16}\text{O} + ^{12}\text{C}$ scattering at $E_{\text{lab}}(^{16}\text{O})=168$ MeV (the lowest part of Fig. 39), though the calculation somewhat overestimates the data at large angles.

In Figs. 41(a) and (b), the full-CC calculation⁹⁴⁾ (solid curves) is compared with the experimental data⁹⁴⁾ for the $^{12}\text{C} + ^{12}\text{C}$ scattering at $E_{\text{lab}}=139.5$ and 158.8 MeV. The data are reasonably well reproduced by this calculation with the single adjustable parameter, except for the scattering at large angles. The optimum values obtained for N_i are 0.20 and 0.30 for the scattering at 139.5 and 158.8 MeV, respectively. (In the calculation, the calculated transition density⁴⁷⁾ between the ground state and the 3_1^- one has been reduced temporarily by 25 % so as to give better fit to the electron-scattering form factor.) In those figures, a small-space CC calculation including only the 0_1^+ , 2_1^+ , 3_1^- and 0_2^+ states is also shown with the dashed curves, in which the same values of N_R and N_i for the solid lines have been used. Coupling effects of the breakup channels excluded in this calculation are prominent at intermediate and large angles in the 0_1^+ , 2_1^+ and 0_2^+ states. Particularly, the calculated curve for the 0_2^+ state deviates considerably at large angles.

A same quality fit as the solid lines for the shell-model like states (0_1^+ , 2_1^+ and 3_1^-) can be obtained even in the small-space calculation by renormalizing the imaginary strength, N_i . The solid lines in Figs. 42(a) and (b) were obtained⁹⁴⁾ by the calculation with $N_i=0.33$ for 139.5 MeV and $N_i=0.43$ for 158.8 MeV. However, the fit is still very poor for the 7.65 MeV 0_2^+ state. This is because other 3α -breakup channels have a very strong coupling to

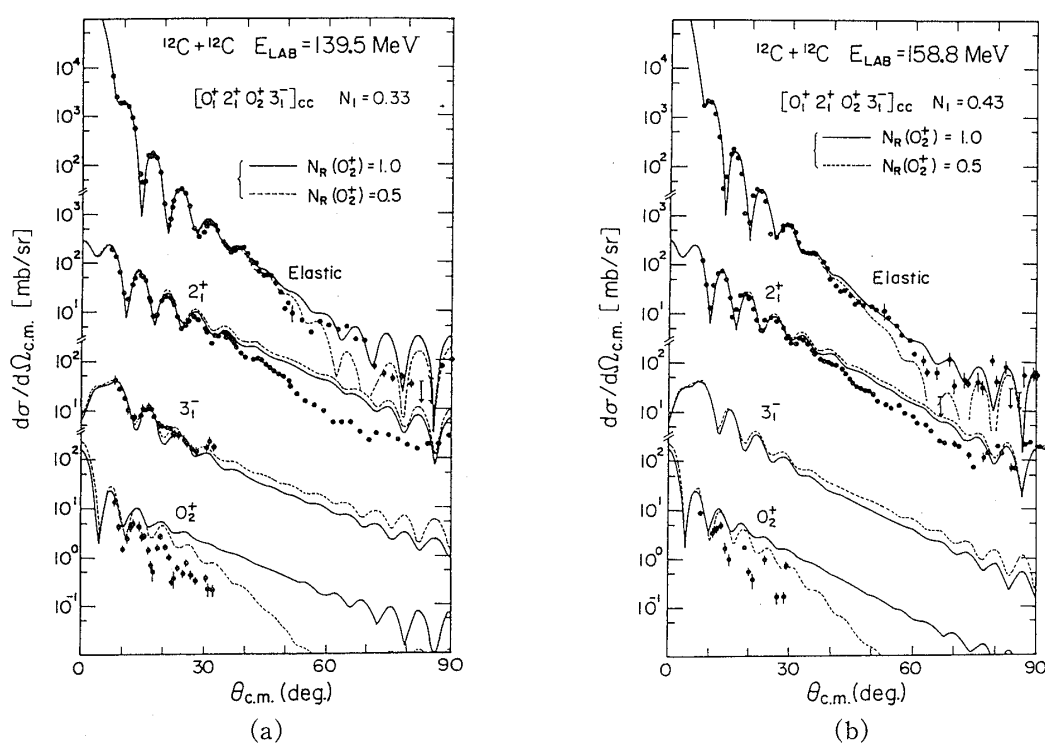


Fig. 42. (a) The same as Fig. 41(a). The solid curves are the result of the small-space CC calculation with $N_i=0.33$ and $N_R=1.0$, and the dashed curves are the same calculation except that $N_R(0_2^+)=0.50$ was used only in the 0_2^+ channel. This figure is taken from Ref. 94).
 (b) The same as Fig. 41(b). The solid curves are the result of the small-space CC calculation with $N_i=0.43$ and $N_R=1.0$, and the dashed curves are the same calculation except that $N_R(0_2^+)=0.50$ was used only in the 0_2^+ channel.

the 0_2^+ state, and thus the CC calculation requires explicit inclusion of those breakup channels with proper coupling potentials. This fact indicates that the angular distribution of the 0_2^+ state is a stringent test to verify the important role of the 3α -breakup channels.

Inelastic scattering of ^{12}C from ^{208}Pb was previously studied by Shyam et al.⁹⁵⁾ They pointed out that they needed to reduce artificially by a factor of two the real strength for the exit channel diagonal potential for the 7.65 MeV 0_2^+ state in DWBA calculations, although the origin for this phenomenon was not identified. For the present scattering of $^{12}\text{C}+^{12}\text{C}$, a treatment same as the one used by them is also confirmed to be effective to get a good fit to the 0_2^+ state phenomenologically. The dashed curves in Figs. 42(a) and (b) were obtained⁹⁴⁾ by reducing by a factor of two the real strength of the diagonal potential in the 0_2^+ channel alone, which we call $N_R(0_2^+)$, while all other parameters were set equal to those for the solid lines in the figures. This calculation roughly reproduces the full-CC calculation, especially for the 0_2^+ state. Thus, this result gives a good explanation for the unknown reduction factor⁹⁵⁾ for the state. This is clearly due to the fact that the 3α -clustering 0_2^+ state has a very strong coupling with other 3α -breakup channels, and the coupling induces a *repulsive* renormalization in the *exit-channel* distorting potential.

6.3.2. Inelastic scattering of $^6\text{Li}+^{12}\text{C}$ and $^7\text{Li}+^{12}\text{C}$

Next, we analyze recent data by Katori et al.,⁷⁴⁾ for the inelastic scattering of $^6\text{Li}+^{12}\text{C}$ at $E_{\text{lab}}=124$ and 169 MeV and of $^7\text{Li}+^{12}\text{C}$ at 132 MeV, leading to the 2_1^+ , 3_1^- and 0_2^+ states of ^{12}C . In § 3.5, we analyzed the *elastic* scattering of the same experiment with the CC calculation including the projectile-breakup channels (but excluding the ^{12}C excited channels), and shown that the breakup effect can be taken into account approximately by $N_R \approx 0.7$ in the single-channel calculation. Thus, we assume that, in the present CC analysis of the inelastic scattering, the projectile-breakup effect is approximately taken into account by the renormalization of $N_R \approx 0.7$. On the other hand, we have shown in § 6.2 that the coupling of the ^{12}C -excitation channels also induces a sizable real potential as well as the imaginary one at $E_{\text{lab}}/A_p \gtrsim 10$ MeV just as in the present case. Since, here we include the ^{12}C inelastic channels explicitly in the CC calculation, the optimum values of N_R and N_I for the present CC calculation must be somewhat different from those for the single-channel best-fit one. Therefore, we search an optimum values of them so that the CC calculation reproduces the elastic cross section; N_R and N_I are taken to be common for all the diagonal and the coupling potentials.

The solid curves in Figs. 43(a) and (b) represent the result of the CC calculation for the $^6\text{Li}+^{12}\text{C}$ scattering at $E_{\text{lab}}=169$ MeV and 124 MeV, respectively. The optimum value of N_R and N_I is determined by fitting the elastic data for $\theta_{\text{cm}} \lesssim 30^\circ$. The fit to the data at the larger angles is not satisfactory. It may be due to the crude assumption of the imaginary-potential shape (the same as real-potential shape), and/or the approximate treatment of the projectile-breakup effect by the reduction of N_R . It is encouraging that, once the elastic data is fitted, the absolute magnitude and the angular distribution of the observed inelastic cross sections are automatically reproduced quite well by the CC calculation without using any additional parameters.

A similar result is obtained for the $^7\text{Li}+^{12}\text{C}$ scattering at $E_{\text{lab}}=132$ MeV,⁷⁴⁾ as shown in Fig. 44. The peak-to-valley ratio of the calculated cross sections seems somewhat larger than that of the observed, which may be due to neglecting the re-orientation

coupling in the ground state of ${}^7\text{Li}$.

Finally, we briefly discuss the projectile-breakup effect on the *inelastic scattering* leading to the excited states of the target nucleus. In the present analysis, we have used the value of $N_R \approx 0.8$ for all the diagonal and the coupling potentials. There are two kinds of routes to reach the ${}^{12}\text{C}$ -excited channels, i.e., the direct transition from the initial channel and the multi-step transition via the projectile-breakup channels. They are illustrated by the dashed and the dotted lines, respectively, in Fig. 45. If the contributions of the latter process are negligible compared with the former, $N_R = 1.0$ should be used in the *coupling potential* for the ${}^{12}\text{C}$ excitation. In order to clarify the point, we perform a *modified-CC* calculation in which the value of N_R is taken to be 1.0 for the *coupling potentials* and ≈ 0.8 for the *diagonal potentials*, while the value of N_I is taken to be common to all the potentials. N_I is re-adjusted slightly so that the modified-CC calculation gives the same elastic cross section as the previous full-CC one. The dashed curves in Figs. 43(a) and (b) represent the result of the modified-CC calculation. N_I is taken to be 0.43 and 0.48 for 169 MeV and 124 MeV cases, respectively. The calculation overestimates the data in both cases. It is necessary to take $N_R \approx 0.8$ for *all the potentials* to reproduce simultaneously the elastic and the inelastic cross sections. This result shows the importance of the multi-step transition via the projectile-breakup channels in the excitation of the target nuclei by the weakly-bound heavy ions like ${}^{6,7}\text{Li}$.

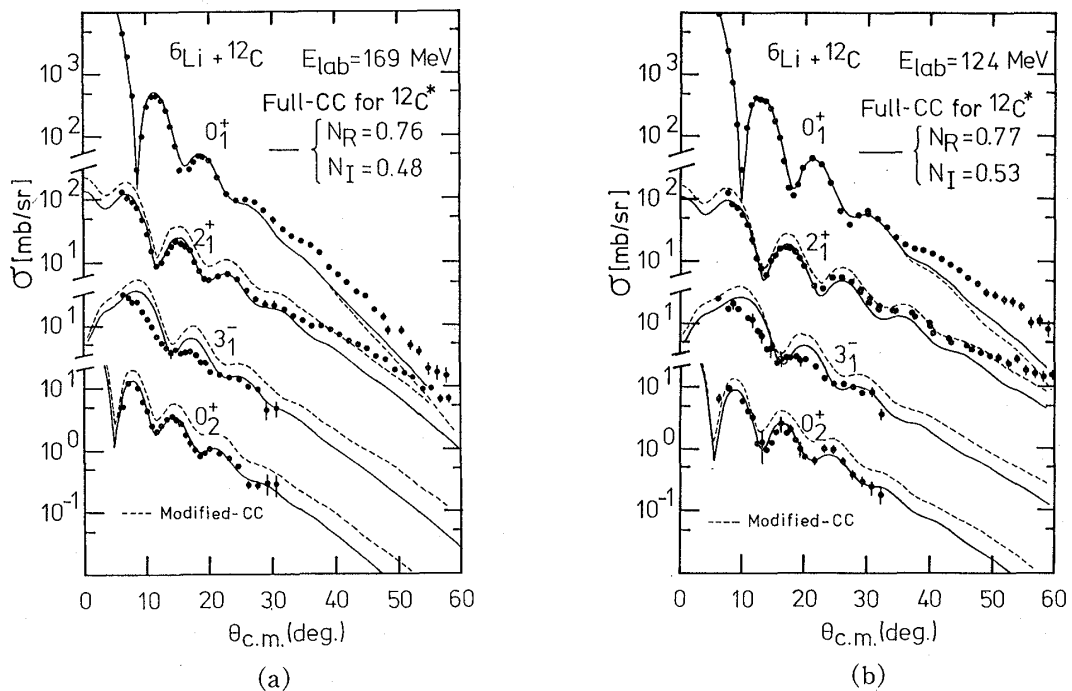


Fig. 43. (a) The elastic scattering of ${}^6\text{Li} + {}^{12}\text{C}$ at $E_{\text{lab}} = 169$ MeV and the inelastic scattering leading to the 2_1^+ , 3_1^- and 0_2^+ states of ${}^{12}\text{C}$. The experimental data are taken from Ref. 74). The solid curves represent the result of the CC calculation^{35),74)} with $N_R = 0.76$ and $N_I = 0.48$ with including all the states of ${}^{12}\text{C}$ shown in Fig. 36. The ${}^6\text{Li}$ -breakup channels are not included in the calculation. The dashed curves show the result of the “modified-CC” calculation³⁵⁾ in which $N_R = 1.0$ is used for all the coupling potentials and $N_R = 0.76$ for all the diagonal potentials, while N_I being 0.43 for all the diagonal and coupling potentials.

(b) The same as Fig. 43(a) for $E_{\text{lab}} = 124$ MeV. In the modified-CC calculation, N_R is taken to be 1.0 for all the coupling potentials and 0.77 for all the diagonal potentials, while $N_I = 0.48$ is used for all the diagonal and coupling potentials. The experimental data are taken from Ref. 74).

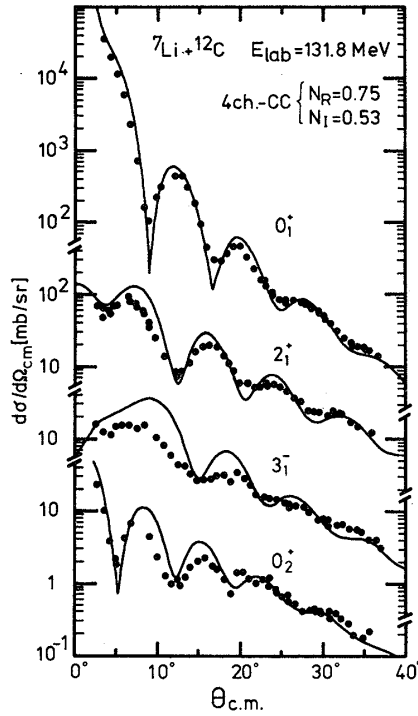


Fig. 44. The elastic scattering of ${}^7\text{Li} + {}^{12}\text{C}$ at $E_{\text{lab}} = 132$ MeV and the inelastic scattering leading to the 2_1^+ , 3_1^- and 0_2^+ states of ${}^{12}\text{C}$. The experimental data are taken from Ref. 74). The solid curves are the results of the four-state CC calculation including the 0_1^+ , 2_1^+ , 3_1^- and the 0_2^+ states of ${}^{12}\text{C}$. The values of N_R and N_I are 0.75 and 0.53, respectively.

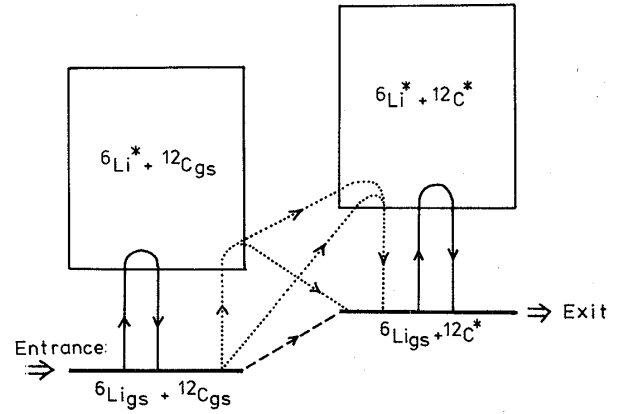


Fig. 45. An illustration of the single and the mutual excited channels of the ${}^6\text{Li} + {}^{12}\text{C}$ system and the possible routes of the direct and the multi-step transition to the ${}^{12}\text{C}$ -excited channel.

§ 7. New types of dynamical polarization potential

In the previous sections, we made microscopic CC analyses of the scattering and breakup of the ${}^6\text{Li}$, ${}^7\text{Li}$ and ${}^{12}\text{C}$ projectiles. The coupling of the excited and the breakup channels of the nuclei was found to be very strong and play important roles in reproducing the observed data. The anomalous reduction ($N_R = 0.5 \sim 0.6$) of the real potential strength in the double-folding model was explained well as the coupling effect of the channels on the elastic scattering. The effect was discussed in relation to the properties of the *dynamical polarization* (DP) *potential* induced by the coupling, though the potential was not explicitly calculated in the previous sections. The qualitative features of the potential suggested by the analyses are summarized as follows:

- (i) The DP potential due to the ${}^6\text{Li}$ -breakup channels must have a strongly repulsive real part and a very small imaginary part, at least in the peripheral region to which the elastic scattering is sensitive. The strength of the real part must amount to about 40~50 % of the double-folded potential of the incident channel. The incident-energy dependence of the DP potential seems to be very small.
- (ii) The property of the DP potential due to the ${}^7\text{Li}$ -breakup channels may be similar to that for the ${}^6\text{Li}$ case, though the re-orientation coupling in the ground state makes the interpretation of the potential somewhat complicated.

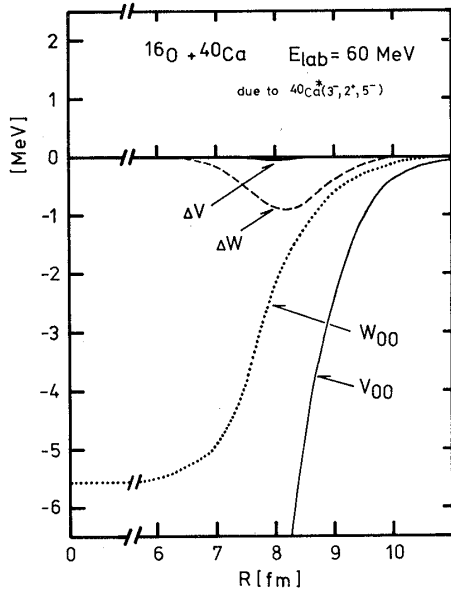


Fig. 46. The calculated dynamical polarization (DP) potential due to the coupling of the collective-phonon-like states of ^{40}Ca for the scattering of ^{16}O from ^{40}Ca at $E_{\text{lab}}=60$ MeV. ΔV and ΔW are the real and the imaginary part, respectively, of the DP potential to the second order of the coupling potential.²⁹⁾ V_{00} (W_{00}) is the real (imaginary) part of the optical potential.

- (iii) The excitation and the virtual breakup of ^{12}C must induce both an absorptive imaginary potential and a repulsive real one. However, the property of the DP potential seems to depend so much on the incident energy. In the lower-energy region of $E_{\text{lab}}/A < 10$ MeV, the imaginary part of the potential must dominate over the real part, while the latter part must be comparable with the former in the region of $E_{\text{lab}}/A \gtrsim 10$ MeV.

These properties of the DP potentials are quite new and have not been known for DP potentials in any other type of reactions. It was believed that collective excitations of nuclei usually induce mainly an absorptive imaginary potential. In fact, the coupling of some collective phonon states is known to induce a sizable imaginary potential and a very small real potential.^{25)~30)} We show in Fig. 46 an example of the DP potential²⁹⁾ of such a well-known type induced by the coupling of the 2^+ , 3^- and 5^- phonon states of ^{40}Ca in the scattering of $^{16}\text{O} + ^{40}\text{Ca}$ at $E_{\text{lab}}=60$ MeV.

In view of the situations described above, it is interesting to calculate *explicitly* the DP potentials for the present cases. Since the coupling between the channels is strong as shown before, we do not rely on perturbation theory, although it might be helpful for qualitative discussions of the DP potential. Thus, we calculate the DP potential in the form of the *wave-function-equivalent local potential* using the wave functions obtained as the solutions of the CC equations.^{34),45)} For the ^7Li case, however, we do not calculate the DP potential, since the non-zero spin in the ground state of ^7Li makes the derivation of the potential much complicated. In the following sections, therefore, we calculate the DP potential in the case of the ^6Li scattering where $V_{\text{eff}}^{(SO)}$ between α and d is switched off, and in the case of ^{12}C scattering.

7.1. Wave-function-equivalent local potential

Let us take up the equation for $i=0$ (for the incident channel) from Eq. (2.39) and rewrite it as follows:

$$\left[-\frac{\hbar^2}{2\mu} \nabla^2 + \frac{\hbar^2 L(L+1)}{2\mu R^2} + V_{00}(R) + iW_{00}(R) + V_{\text{coul}}(R) - (E - \varepsilon_0) \right] \hat{\chi}_{00}^{(J)}(R) \\ = \left\{ -\sum_{i \neq 0} \sum_l \sum_{|J-l| \leq L \leq J+l} [\Delta V_{iL}^{(J)}(R) + i\Delta W_{iL}^{(J)}(R)] \right\} \hat{\chi}_{00}^{(J)}(R). \quad (7.1)$$

Here, $V_{00} + iW_{00}$ stands for the nuclear part of the diagonal potential (J -independent) in the incident channel, $U_{00}^{(J),00J}$. Namely, from Eq. (2.37),

$$V_{00}(R) = N_R \cdot \hat{V}_{00J,00J}^{(J)}(R), \quad (7.2a)$$

$$W_{00}(R) = N_I \cdot \hat{V}_{00J,00J}^{(J)}(R). \quad (7.2b)$$

$\Delta V_{iL}^{(J)}$ and $\Delta W_{iL}^{(J)}$ in Eq. (7.1) represent the real and the imaginary part, respectively, of the wave-function-equivalent local potential³⁴⁾ which are defined as

$$\begin{aligned} \Delta V_{iL}^{(J)}(R) &\equiv \text{Re}\{U_{00J,iL}^{(J)}(R) \hat{\chi}_{iL}^{(J)}(R) / \hat{\chi}_{00J}^{(J)}(R)\} \\ &= \text{Re}\{(N_R + iN_I) \cdot \hat{V}_{00J,iL}^{(J)}(R) \hat{\chi}_{iL}^{(J)}(R) / \hat{\chi}_{00J}^{(J)}(R)\}, \end{aligned} \quad (7.3a)$$

$$\begin{aligned} \Delta W_{iL}^{(J)}(R) &\equiv \text{Im}\{U_{00J,iL}^{(J)}(R) \hat{\chi}_{iL}^{(J)}(R) / \hat{\chi}_{00J}^{(J)}(R)\} \\ &= \text{Im}\{(N_R + iN_I) \cdot \hat{V}_{00J,iL}^{(J)}(R) \hat{\chi}_{iL}^{(J)}(R) / \hat{\chi}_{00J}^{(J)}(R)\}, \end{aligned} \quad (7.3b)$$

where all the χ 's are the solutions of the CC equations of (2.39). By definition, the solution $\hat{\chi}_{00J}^{(J)}(R)$ of Eq. (7.1) is exactly the same in the whole region of R as that of Eq. (2.39). In the present work, we call the potential $\Delta V_{iL}^{(J)}(R) + i\Delta W_{iL}^{(J)}(R)$ the dynamical polarization (DP) potential for a subchannel specified by (iL) and J .

Since the number of the channels concerned is so large as shown in Figs. 8, 21 and 36, it is rather complicated to illustrate the DP potentials for all the channels. We therefore sum up the potentials over all the channels as

$$\Delta V^{(J)}(R) = \sum_l \sum_i \sum_{|J-l| \leq L \leq J+l} \Delta V_{iL}^{(J)}(R), \quad (7.4a)$$

$$\Delta W^{(J)}(R) = \sum_l \sum_i \sum_{|J-l| \leq L \leq J+l} \Delta W_{iL}^{(J)}(R). \quad (7.4b)$$

Note that, in principle, the total DP potential, $\Delta V^{(J)} + i\Delta W^{(J)}$, depends on J and the incident energy, E . If we want to see the contribution of the individual channels, the summation over l and i in Eqs. (7.4a) and (7.4b) is to be restricted to the corresponding values.

7.2. DP potential induced by the coupling with ${}^6\text{Li}$ -breakup channels

In this section, we calculate the DP potential induced by the coupling of the ${}^6\text{Li}$ -breakup channels. In all the calculation of the DP potential presented here, N_R is kept fixed at 1.0 and N_I is taken to be the same as the optimum value for the full-CC calculation of the corresponding scattering presented in § 3.

Before the discussion about the property of the DP potential, one should note that, in the heavy-ion scattering like the present case, the elastic scattering is almost insensitive to the potential in the nuclear interior, except few cases mentioned before. The detail of the potential in the internal region is completely masked by the strong absorptive potential. As a measure useful for defining the *insensitive internal region*, we adopt the change of the modulus of the elastic S-matrix element by less than 1 % against the switching-off of the DP potential in the region.^{34),35)} In such an insensitive region, the modulus of the incident-channel wave function, $|\hat{\chi}_{00J}^{(J)}(R)|$, has been found empirically to be less than about 10 % of its maximum value in the asymptotic region. Thus, we use the value of R satisfying the relation $|\hat{\chi}_{00J}^{(J)}(R)| \simeq 0.1 |\hat{\chi}_{00J}^{(J)}(\infty)|$ as the measure for defining the insensitive region.

First, the angular momentum dependence of the DP potential has been examined.³⁶⁾ Figure 47 shows the real and the imaginary part of the DP potential, $\Delta V^{(J)}$ and $\Delta W^{(J)}$ respectively, at nuclear surface in the case $J=0, 10, 21, 25$ and 28 of ${}^6\text{Li} + {}^{28}\text{Si}$ scattering at

99 MeV. Although the position of the boundary between the internal (insensitive) and the peripheral region depends on J , the “insensitive region” illustrated in the figure is for $J = 21$; the position is about 4 fm for $J=0$ and about 6 fm for $J=28$. Since the grazing angular momentum, J_{gr} , is 23 in the present case, the DP potentials for $J=0$ and $J=10$ do not take part in the scattering at all due to the strong absorption, although they are given in the figure for comparison. The figure shows that the DP potential is almost independent of J around J_{gr} in the peripheral region. For such J , the potential really has a repulsive real part ($\Delta V^{(J)} > 0$) whose strength amount up to about 40 % of the double-folded real potential, $V_{00}(R)$, of the incident channel. In other words, $V_{00}(R) + \Delta V^{(J)}(R)$ may almost J -independently be approximated well by $\approx 0.6 \cdot V_{00}(R)$. This is just the origin of the reduction of N_R in the DF model for ${}^6\text{Li}$ scattering. The figure also shows that the imaginary part, $\Delta W^{(J)}$, is negligibly small in the peripheral region for J around J_{gr} . This property, $\Delta V > 0$ and $\Delta W \approx 0$, of the DP potential due to the breakup channels is completely different from that of the *familiar type* of the potential due to the excitation of phonon-like states of nuclei which is usually dominated by an absorptive imaginary part^{25)~30)} ($\Delta V \approx 0$, $\Delta W < 0$) as shown in Fig. 46.

Secondly, the incident-energy dependence of the DP potential has been examined.³⁶⁾ Since the potential is almost independent of J in the peripheral region, it has been calculated for a specific J around J_{gr} for each incident energy. Figure 48 shows the real and the imaginary part of the DP potential for the scattering by ${}^{28}\text{Si}$ at 46, 99 and 135 MeV and for the scattering by ${}^{208}\text{Pb}$ at 50.6, 99 and 156 MeV. For both targets, the potential

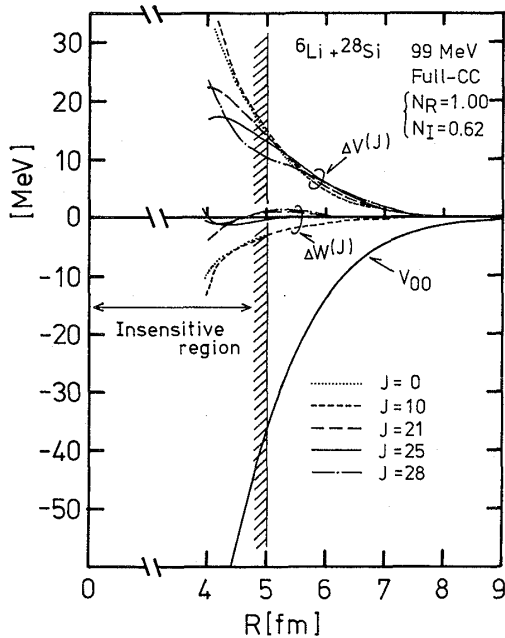


Fig. 47. The angular-momentum (J) dependence of the DP potential induced by the coupling to the breakup channels for ${}^6\text{Li} + {}^{28}\text{Si}$ at 99 MeV. $\Delta V^{(J)}$ ($\Delta W^{(J)}$) is the real (imaginary) part of the potential for $J=0$ (dotted), 10 (short-dashed), 21 (long-dashed), 25 (solid) and 28 (dot-dashed). The folding real potential for the elastic channel with $N_R=1$ (V_{00}) is shown for comparison. The insensitive region shown is for $J=21$.

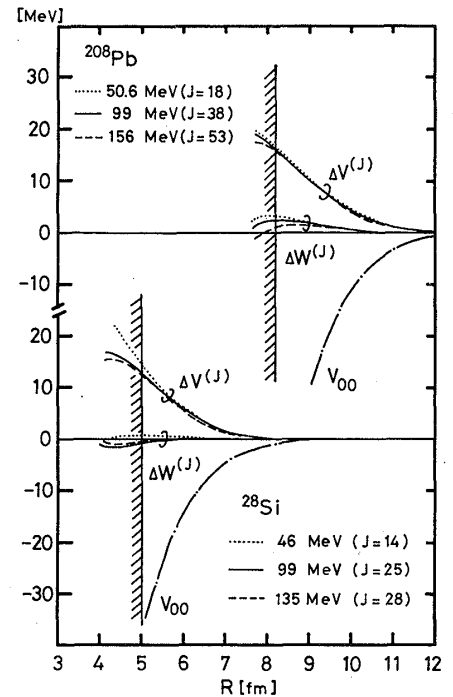


Fig. 48. The incident-energy dependence of the DP potential,³⁶⁾ the real and the imaginary part of which are shown in the case of ${}^{208}\text{Pb}$ target (upper) at 50.6, 99 and 156 MeV and ${}^{28}\text{Si}$ target (lower) at 46, 99 and 135 MeV. The folding real potential for the elastic channel with $N_R=1$ (V_{00}) is shown for comparison.

changes little as the incident energy changes, which explains well the fact that the renormalization factor of $N_R=0.5\sim 0.6$ occurs almost independently of energy in the DF-model studies of ${}^6\text{Li}$ scattering.

Next, the dependence of the DP potential on target mass number has been examined.³⁶⁾ To this end, calculated is the potential for ${}^{12}\text{C}$, ${}^{28}\text{Si}$, ${}^{40}\text{Ca}$, ${}^{58}\text{Ni}$ and ${}^{208}\text{Pb}$ at an incident energy of $E_{\text{lab}}=99\text{ MeV}$, which corresponds to the CC calculation shown in Fig. 17 of § 3. J has been chosen again to be the one around J_{gr} for each target; note that J_{gr} is 15, 23, 26, 30 and 38 for ${}^{12}\text{C}$, ${}^{28}\text{Si}$, ${}^{40}\text{Ca}$, ${}^{58}\text{Ni}$ and ${}^{208}\text{Pb}$, respectively. The solid and the dashed lines in Fig. 49 show the real and the imaginary part, respectively, of the DP potential for those targets. For comparison, the double-folded real potentials in the elastic channel are also shown by the dotted lines. For the cases of ${}^{28}\text{Si}$, ${}^{40}\text{Ca}$, ${}^{58}\text{Ni}$ and ${}^{208}\text{Pb}$ targets, the potentials are similar to each other, which satisfy approximately the relationships of $\Delta V^{(J)}(R) \simeq -0.4 V_{00}(R)$ and $\Delta W^{(J)}(R) \simeq 0$ in the peripheral region. These results are consistent with the target-independence of the reduction of $N_R (=0.5\sim 0.6)$ required in the DF-model studies. For the ${}^{12}\text{C}$ target, however, the ratio of $-\Delta V^{(J)}(R)/V_{00}(R)$ is about 0.2 $\sim$ 0.3 which is somewhat smaller than that for other targets, whereas a weak but non-negligible imaginary part of absorptive nature, $\Delta W^{(J)}(R) < 0$, is induced. The exceptional nature of the potential for ${}^{12}\text{C}$ may have some relation to the neglect of the strong coupling to the target (${}^{12}\text{C}$) excitation in the present CC calculation, and seems to correlate with the fact that, for ${}^{12}\text{C}$, the optimum value of N_i for the CC calculation is somewhat smaller ($\lesssim 0.5$) than that for the other targets ($\gtrsim 0.6$) as seen in Table III. In this respect, we refer the work of Ref. 96) in which the similar DP potential due to the ${}^6\text{Li}$ breakup was derived for ${}^6\text{Li}+{}^{12}\text{C}$ scattering at 156 MeV on the basis of the potential-inversion method⁹⁶⁾ with using the result of the adiabatic calculation of Ref. 31); the DP potential has strongly repulsive real part and an absorptive imaginary part with considerable magnitude, which is consistent with the present result.

These results suggest that the imaginary part of the coupling potentials is closely related to the qualitative feature of the DP potential. In fact, we have shown in § 3.4 that the imaginary part of the coupling potentials is very important in considering the CC effects of the ${}^6\text{Li}$ -breakup processes. This is indeed confirmed by Fig. 50 which shows the DP potentials corresponding to the two kinds of CC calculations given in Fig. 13 of § 3. The switching-off of the imaginary part of the coupling potentials causes an essential difference in the qualitative feature of the DP potential; the strength of the repulsive real part ($\Delta V > 0$) has been much reduced while a considerable absorptive imaginary part (ΔW

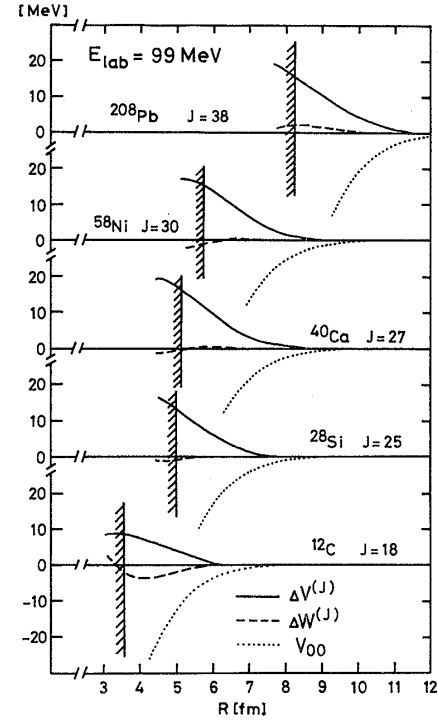


Fig. 49. The target-nucleus dependence of the DP potential.³⁶⁾ The solid (dashed) curves show the real (imaginary) part of the potential in the cases of ${}^{208}\text{Pb}$, ${}^{58}\text{Ni}$, ${}^{40}\text{Ca}$, ${}^{28}\text{Si}$ and ${}^{12}\text{C}$ from top to bottom. The folding potentials in the elastic channel are also shown by the dotted curves.

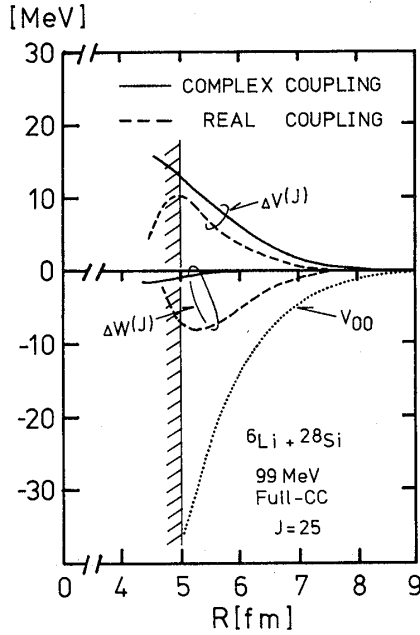


Fig. 50. The real part and the imaginary part of the DP potential for $J=25$ of the ${}^6\text{Li} + {}^{28}\text{Si}$ scattering at 99 MeV. The solid curve and the dashed one are given by the full-CC calculation with the complex coupling potentials and by the calculation with the real coupling potentials, respectively. The real folding potential in the elastic channel is also shown by the dotted curve.

<0) has grown up, which can hardly account for the empirical $\approx 40\%$ renormalization to the real folding potential. Thus, for a proper estimation of the ${}^6\text{Li}$ breakup effect, it is quite important to use complex coupling potentials with considerable imaginary part, since the essential feature of the DP potential due to the coupling is governed by the mixture of the imaginary part in the complex potential. In the present CC framework, it can be said that we have determined the mixture, namely, the ratio of N_I/N_R with N_R kept fixed at unity, phenomenologically in such a way that the CC calculation reproduces the observed elastic scattering. Through the CC analyses in § 3, the experimental data tell us that N_I must be $0.5 \sim 0.6$ almost independently of the incident energy and target nucleus (cf. Table III). This, therefore, implies that the DP potential due to the ${}^6\text{Li}$ breakup depends little on the energy and target, having a repulsive real part ($\Delta V \approx -0.4 V_{00}$) and a negligible imaginary part ($\Delta W \approx 0$). This result of the present CC study is absolutely consistent with the

fact that the ${}^6\text{Li}$ optical potential is systematically shallower than the M3Y double-folded potential by about $40 \sim 50\%$ irrespectively of the incident energy and target nucleus.^{13)~22)}

For the sake of a later discussion, we show in Fig. 51 the contribution of the individual breakup channels, S_{bu} , P_{bu} , $D_{bu}^{(res)}$ and $D_{bu}^{(off)}$, to the total DP potential for the ${}^6\text{Li} + {}^{28}\text{Si}$ scattering at $E_{lab} = 99$ MeV.³⁴⁾ It is seen that a dominant part of the total DP potential comes from the D -wave resonance channel, $D_{bu}^{(res)}$. The contribution of other channels seems rather small. However, this result does not necessarily mean that the effect of the other channels is negligible. Since the DP potential due to the $D_{bu}^{(res)}$ channel in Fig. 51 is derived by the full-CC calculation, it represents the contribution of the $D_{bu}^{(res)}$ channel in the *presence* of other breakup channels. It is different from the DP potential due to the $D_{bu}^{(res)}$ channel in the *absence* of all other channels, as shown by the difference between the solid and the dashed curves in Fig. 52. Furthermore, the importance of the re-orientation coupling in the resonance channel is clear from the difference between the dashed and the dotted curves; the re-orientation coupling is switched off in the latter curve.

Thus, we see the importance of the multi-step processes as well as that of the imaginary part of the coupling potential in considering the qualitative feature of the DP potential due to the ${}^6\text{Li}$ -breakup channels.

7.3. DP potential induced by the coupling with ${}^{12}\text{C}$ -excited states

In this section, we calculate the DP potential induced by the coupling of the ${}^{12}\text{C}$ -excited channels. All the DP potentials presented here are given by the full-CC

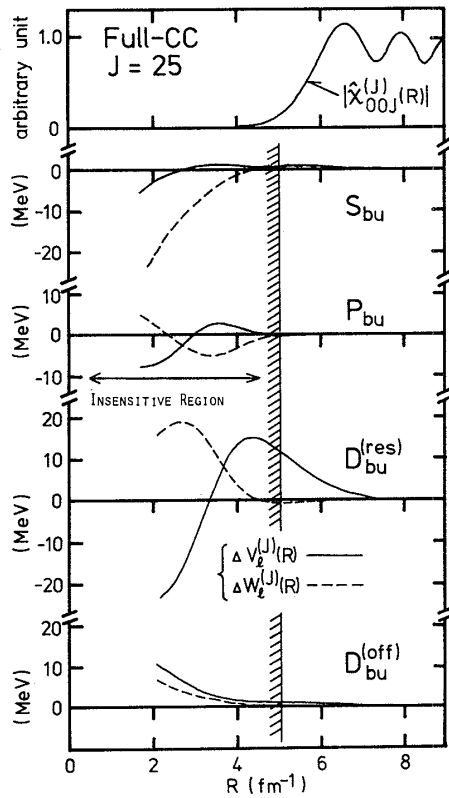


Fig. 51. The DP potentials due to the individual breakup channels for the ${}^6\text{Li} + {}^{28}\text{Si}$ scattering at 99 MeV³⁴⁾ given by the full-CDCC calculation. This figure is taken from Ref. 34).

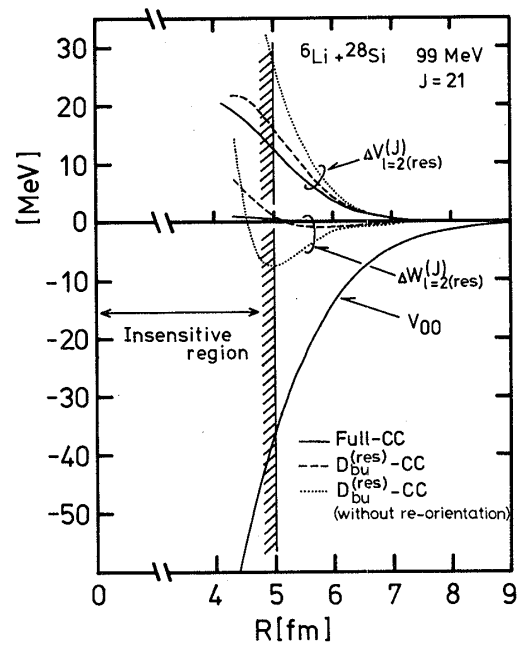


Fig. 52. The DP potential due to the $D_{bu}^{(res)}$ channel.³⁵⁾ The solid curves for $\Delta V^{(J)}$ and $\Delta W^{(J)}$ show the potential given by the full-CC calculation, while the dashed curves represent the potential given by the CC calculation with the ground-state channel coupled with the $D_{bu}^{(res)}$ channel only. In the dotted curves, the re-orientation coupling in the $D_{bu}^{(res)}$ channel is switched off.

calculation including all the channels shown in Fig. 36. In the calculation, N_R is kept fixed at 1.0 and N_I is taken to be the same as the optimum value for the full-CC calculation of the corresponding scattering presented in § 6. The DP potential only for $J = J_{gr}$ is presented here, since the potential is found to be almost independent of J in the peripheral region just as in the case of the ${}^6\text{Li}$ scattering.

We first investigate the incident-energy dependence of the potential. In Fig. 53, the energy dependence of the total DP potential is shown for the ${}^{12}\text{C} + {}^{12}\text{C}$ scattering at $E_{lab} = 148 \sim 300$ MeV. A similar calculation^{35),45)} is shown in Fig. 54 for the ${}^{16}\text{O} + {}^{12}\text{C}$ scattering at $E_{lab}({}^{16}\text{O}) = 168 \sim 311$ MeV which corresponds to the scattering discussed in the previous section. The value of N_I used here is shown in the figures. As is expected from the discussion in § 6.2, the DP potential has both a repulsive real part and an absorptive imaginary part. Furthermore, Figs. 53 and 54 clearly show that the DP potential depends so much on the incident energy just in the same manner as suggested before; i.e., the magnitude of the imaginary part is larger than that of the real part in the lower incident-energy region, while the former tends to decrease and the latter increases as the incident energy goes up. Therefore, these results well explain the reason why the single-channel (DF-model) calculation with $N_R = 1$ breaks down when the incident energy increases (cf. Fig. 35(a)); i.e., the strongly repulsive renormalization to the real potential causes the break down for the high energy scattering. They also explain the energy-dependence of the difference between the optimum value of N_I for the fit to the data by the CC calculation and that by the single-channel calculation (Fig. 35).

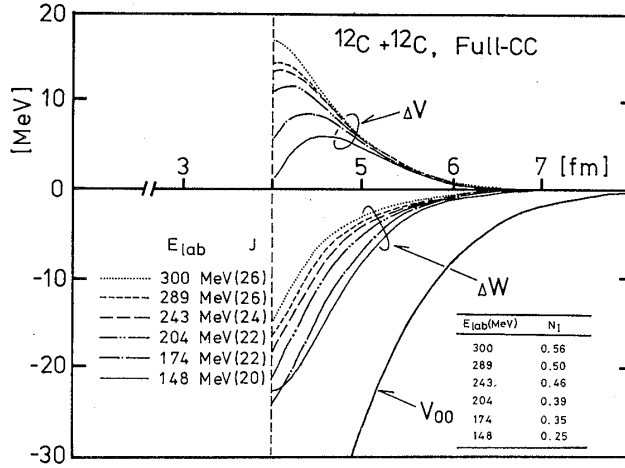


Fig. 53. The incident-energy dependence of the DP potential induced by the virtual excitation of ^{12}C nucleus in the $^{12}\text{C}+^{12}\text{C}$ scattering.³⁵⁾ In the calculation, all the states of ^{12}C shown in Fig. 36 are taken into account.

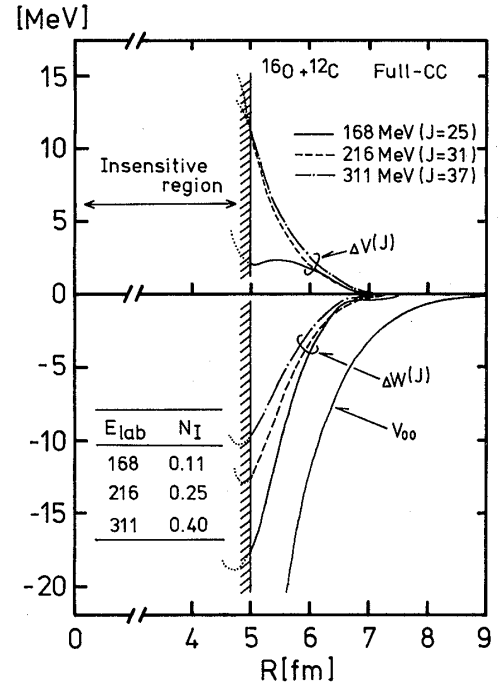


Fig. 54. The same as Fig. 53 for the $^{16}\text{O}+^{12}\text{C}$ scattering.³⁵⁾

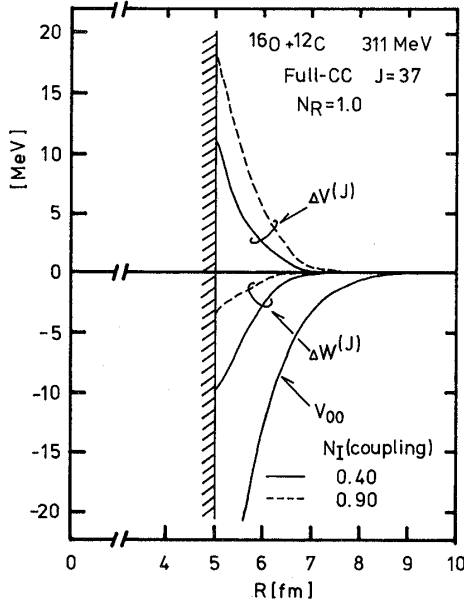


Fig. 55. The dependence of the DP potential on the imaginary-coupling strength in the case of $^{16}\text{O}+^{12}\text{C}$ scattering at 311 MeV.³⁵⁾ The solid curve is the same as the dashed one except that $N_I=0.9$ is used for the coupling potentials in the latter.

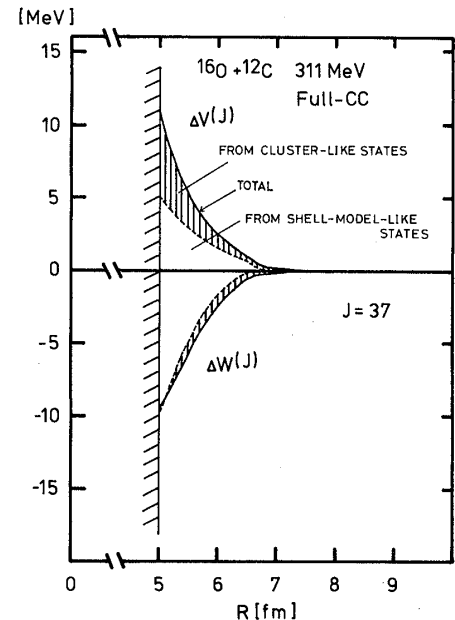


Fig. 56. The DP potential due to the "shell-model-like" states (2_1^+ and 3_1^-) and due to the "cluster-like" states (0_2^+ , 2_2^+ and the non-resonant-breakup channels) given by the full-CC calculation⁴⁵⁾ of the $^{16}\text{O}+^{12}\text{C}$ scattering at 311 MeV. The striped part represents the latter while the rest of the figure shows the former. The value of N_R and N_I used is the same as that in Fig. 54.

Now, it should be emphasized that the change of the qualitative feature of the DP potential with the increase of the incident energy seems to correlate with the increase of the optimum value of N_I for the CC calculation. Namely, for a small value of N_I , the coupling with the ^{12}C excited states induces a sizable imaginary part and relatively small real part, while both parts become comparable for a large value of N_I . This suggests that the qualitative change of the DP potential with the incident energy may be attributed to the change of N_I with the energy, not to the sensitivity of the potential to the incident energy itself. Therefore, one could expect by extrapolating the tendency seen in Figs. 53 and 54 that, when we use a value of N_I much larger than those adopted in the figures, the coupling may induce just the same type of DP potential as in the case of ^6Li ; $|\Delta W^{(J)}| \ll |\Delta V^{(J)}|$. In fact, such a potential can be realized if we increase N_I for the coupling potential artificially from 0.4 to 0.9 in the case of 311 MeV $^{16}\text{O} + ^{12}\text{C}$ scattering, as is shown by the dashed curve in Fig. 55. Thus, we see again the importance of the imaginary coupling potential in considering the qualitative feature of the DP potential.

Finally, we investigate the detail of the DP potential⁴⁵⁾ by taking the case of 311 MeV $^{16}\text{O} + ^{12}\text{C}$ scattering as an example. We decompose the DP potential given by Eqs. (7.4) into two components; one consists of the contribution from the 2_1^+ and the 3_1^- states which have the “shell-model-like” character and the other from the 0_2^+ , 2_2^+ and the non-resonant breakup states which are of the “cluster-like” character. In Fig. 56, the striped part represents the contribution from the “cluster-like” states, while the rest shows that from the “shell-model-like” states. It is interesting that the cluster-like, resonant-breakup states and the non-resonant-breakup states contribute mainly to the real part. The rest half of the real part and most of the imaginary part come from the shell-model-like states.

§ 8. A unified understanding of dynamical polarization potential

In the previous section, we calculated the dynamical polarization (DP) potential due to the ^6Li -breakup channels and that due to the ^{12}C -excited channels in the form of a wave-function-equivalent local potential. The calculation showed clearly that the DP potential has features unexpected from the conventional theories of the DP potential for collective excitations.^{25)~30)} Namely, (i) the potential due to the ^6Li -breakup channels has a repulsive real part ($\Delta V^{(J)} > 0$) and a very small imaginary part ($\Delta W^{(J)} \simeq 0$); this property depends little on the incident energy or on the target nucleus (Figs. 47~49). But, (ii) the coupling of the ^{12}C -excited channels induces both an absorptive imaginary potential ($\Delta W^{(J)} < 0$) and a repulsive real one ($\Delta V^{(J)} > 0$) which depend much on the incident energy (Figs. 53 and 54). These types of the DP potential are new and quite different from the well-known type of the potential such as that due to the excitation of weak phonon modes.^{25)~30)} The latter is known to

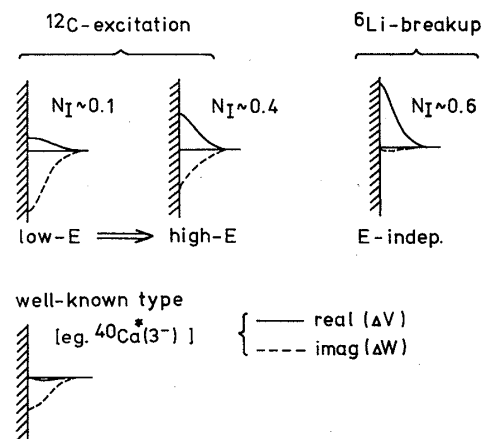


Fig. 57. Comparison of the typical features of the three types of DP potentials, the ones due to the ^6Li -breakup, due to the ^{12}C excitation and breakup, and due to weak-phonon states such as 2^+ and 3^- states of ^{40}Ca , respectively.

have a sizable, absorptive imaginary part ($\Delta W < 0$) and a very small real part ($\Delta V \approx 0$) as is shown in Fig. 46. We illustrate in Fig. 57 the typical features of the three types. It should be recalled that in the cases of the ${}^6\text{Li}$ scattering and the ${}^{12}\text{C}$ one the imaginary part of the coupling potential much affects the ratio of $\Delta V^{(J)}$ to $\Delta W^{(J)}$, as shown in Figs. 50 and 55.

The first part of this section is devoted to an investigation of the role of the real and the imaginary part of the *coupling* potential in the determination of the qualitative feature of the DP potential. For this purpose, we present a schematic model calculation in which the values of N_R and N_I for the coupling potential are artificially changed. We shall see that the model calculation simulates well the systematic change among the three types of the DP potential shown in Fig. 57. In the rest part of this section we present a quite simple but essential explanation for the systematic change. Although the idea is based on perturbation theory, we further discuss the higher-order effects through the multi-step processes between the channels.

8.1. A schematic model calculation of DP potential

In order to clarify the relation between the qualitative feature of the DP potential and the real and the imaginary part of the coupling potential, we present here a schematic-model calculation^(10),35),36) of the DP potential and show that the calculation simulates simultaneously well all the types of the potentials shown in Fig. 57.

Let us consider the ${}^6\text{Li} + {}^{28}\text{Si}$ scattering at 99 MeV and make a two-state CC calculation in which the incident channel is coupled only to the $D_{bu}^{(\text{res})}$ channel (Fig. 58). The re-orientation coupling in the $D_{bu}^{(\text{res})}$ channel is also included. In the calculation, we artificially change the strength of the coupling potentials by changing the values of N_R and N_I from 0.0 through 1.0 and calculate the DP potential for each set of N_R and N_I . The value of N_R and N_I for the diagonal potentials is, however, kept fixed at the “realistic” value of $N_R=1.0$ and $N_I=0.6$, since we have found that the strength of the diagonal potential does not affect so much the qualitative feature of the DP potential in the peripheral region (Fig.

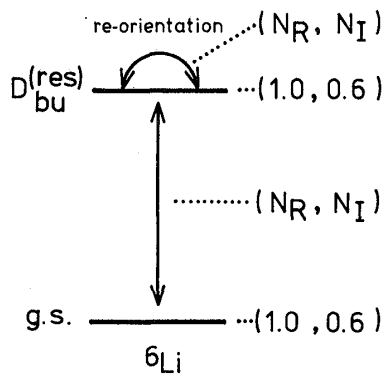


Fig. 58. The model space and the choice of the potential strength for the schematic-model calculation.

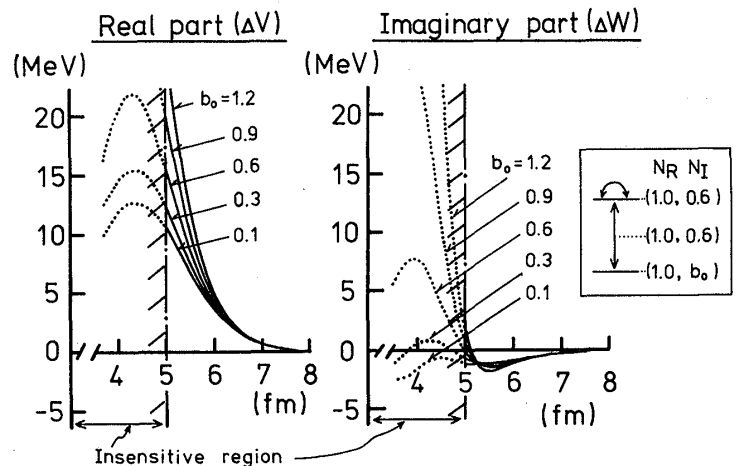


Fig. 59. The sensitivity of the DP potential to the strength of the imaginary potential (b_0) in the incident channel in the two-state-model calculation.³⁵⁾

59). The calculation is performed for $J=21$.

We show the result of the calculation^{10),35)} in a “(N_R , N_I)-map” (Fig. 60). In the map, the position of the origin of each sub-figure indicates the value of N_R and N_I used there for the coupling potentials. The systematic change of the DP potential with respect to N_R and N_I is apparent in the figure. When N_I is increased with N_R kept fixed, the strength of $\Delta W^{(J)}$ decreases rapidly and that of $\Delta V^{(J)}$ increases. $\Delta W^{(J)}$ comes to change its sign from negative to positive when the value of N_I becomes comparable with that of N_R , say $N_I/N_R=0.6\sim 1.0$. The absolute value of the strength of $\Delta W^{(J)}$, therefore, becomes very small there. Note that the value of $N_R=1.0$ and $N_I\approx 0.6$ which satisfies the condition for $\Delta W^{(J)}\approx 0$ (labelled by “A” in Fig.

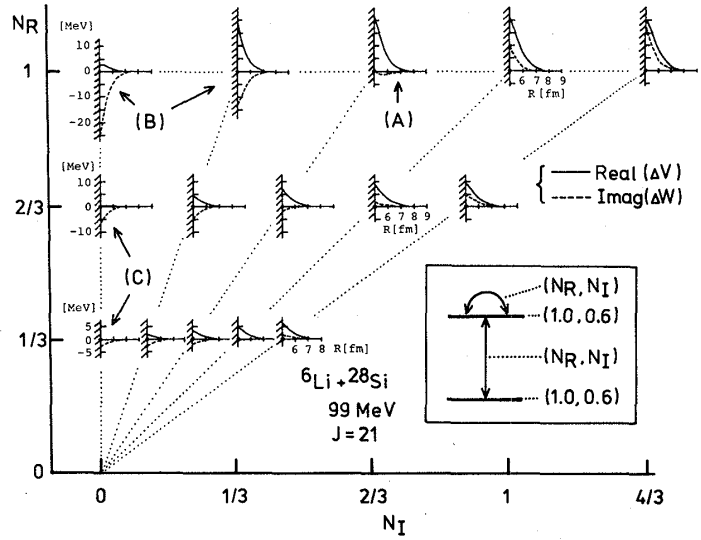


Fig. 60. The “(N_R , N_I)-map” of the DP potential given by the model calculation.^{10),35)} Namely, the systematic change of the potential with respect to the strength of the complex coupling potential, N_R and N_I . The position of the origin of each sub-figure indicates the value of N_R and N_I used there. The map simulates quite well the essential features seen in Fig. 57.

60) just happened to be the value for the realistic case of the ${}^6\text{Li}$ scattering. Thus, the DP potential for the ${}^6\text{Li}$ case is almost purely real, and repulsive. The values of $N_R=1.0$ and $N_I=0.1\sim 0.4$ correspond to the values for the realistic case of the ${}^{16}\text{O}+{}^{12}\text{C}$ scattering at $E_{\text{lab}}=168\sim 311$ MeV (cf. Figs. 39 and 54). Surprisingly, the potentials generated by those (N_R , N_I) (labelled by “B” in Fig. 60) are quite similar to those given by the realistic CC calculation for the scattering (cf. Figs. 54 and 57). Furthermore, the sets labelled by “C” in Fig. 60 correspond to the cases of weak coupling such as the excitation of weak-phonon modes. One indeed sees that the well-known type^{33)~38)} of the potential ($\Delta V^{(J)}\approx 0$, $\Delta W^{(J)}<0$) is generated there. For the sake of later discussions, it should also be noted that a set of DP potentials with a similar type lines up nearly along the dotted lines in Fig. 60. For a large value of N_R and N_I , however, the systematics deviate from the dotted lines a little upward. It will be shown in the next subsection that the deviation is explained qualitatively as effects of the multi-step processes among the channels.

In this subsection, we have seen that the schematic-model calculation explains simultaneously well all the types of DP potential shown in Fig. 57. In other words, the newly-discovered types as well as the familiar type can be classified unifiedly in terms of the strength of the *complex coupling potentials*. We see that the well-known type ($\Delta V\approx 0$, $\Delta W<0$) occurs only in the weak-coupling-limit case among the wide range of possibilities of the DP potential shown in Figs. 57 and 60.

8.2. A unified understanding of DP potential

8.2.1. A basic structure of the (N_R , N_I)-dependence

A basic structure of the (N_R, N_I) -dependence of the DP potential may easily be understood by the following consideration^{(35), (36)} based on perturbation theory. We start with the CC equation (2·2),

$$(T_i + U_{ii}(\mathbf{R}) - E_i)\chi_i(\mathbf{R}) = -\sum_{j \neq i} U_{ij}(\mathbf{R})\chi_j(\mathbf{R}), \quad (8.1)$$

where T_i stands for $-\hbar^2 \nabla^2 / 2\mu$ and E_i is $E - \varepsilon_i$. $U_{ij}(\mathbf{R})$ denotes the diagonal ($i=j$) and the coupling ($i \neq j$) potentials, which are complex and defined by

$$U_{ij}(\mathbf{R}) = (N_R + iN_I) \cdot V_{ij}(\mathbf{R}). \quad (8.2)$$

Here, $V_{ij}(\mathbf{R})$ is real since it is given by doubly-folding the real nucleon-nucleon potential into the nucleon densities of projectile and target nuclei. The explicit form of $V_{ij}(\mathbf{R})$ is given in Eq. (2·32).

If the coupling among the excited channels is negligibly small, the CC equation (8.1) can be reduced simply to the following non-local one-body equation in the incident channel which we denote by $i=0$;

$$(T_0 + U_{00} - E_0)\chi_0 = -\sum_{j \neq 0} U_{0j} G_j^{(+)} U_{j0} \chi_0, \quad (8.3)$$

where $G_j^{(+)}$ is the Green function, $(E_j - T_j - U_{jj} + i\varepsilon)^{-1}$, describing the propagation in the j -th channel. We define the DP potential $\Delta U(\mathbf{R})$ in the form of a wave-function-equivalent local potential as

$$\begin{aligned} \Delta U(\mathbf{R}) &= \sum_{j \neq 0} U_{0j}(\mathbf{R}) \int G_j^{(+)}(\mathbf{R}, \mathbf{R}') U_{j0}(\mathbf{R}') \chi_0(\mathbf{R}') d\mathbf{R}' / \chi_0(\mathbf{R}) \\ &= (N_R + iN_I)^2 \left\{ \sum_{j \neq 0} V_{0j}(\mathbf{R}) \int G_j^{(+)}(\mathbf{R}, \mathbf{R}') V_{j0}(\mathbf{R}') \chi_0(\mathbf{R}') d\mathbf{R}' / \chi_0(\mathbf{R}) \right\}, \end{aligned} \quad (8.4)$$

where the relation of Eq. (8.2) has been used. The DP potential, $\Delta U(\mathbf{R})$, defined here is different from that defined previously in § 7.1 in that the latter is defined in the partial-wave-expansion form, while the former is not expanded into partial waves. The basic structure of (N_R, N_I) -dependence which will be shown below is, however, the same. Thus, we call $\Delta U(\mathbf{R})$ defined by Eq. (8.4) the DP potential for a while. The form of Eq. (8.4) is convenient for comparison with previous perturbation theories of DP potential.^{(26), (29)}

In Eq. (8.4), the square, *not* the squared modulus, of the strength factor $(N_R + iN_I)$ is factorized out, because we assumed the factor to be common to all the coupling potentials in Eq. (8.2). Thus, the part in the bracket, $\{ \}$, on right-hand side of Eq. (8.4) is regarded as the DP potential induced by the *real* coupling potentials, V_{0j} and V_{j0} . Since the part is still complex in general, we define the real and the imaginary part by

$$\Delta v + i\Delta w = \sum_{j \neq 0} V_{0j}(\mathbf{R}) \int G_j^{(+)}(\mathbf{R}, \mathbf{R}') V_{j0}(\mathbf{R}') \chi_0(\mathbf{R}') d\mathbf{R}' / \chi_0(\mathbf{R}), \quad (8.5)$$

where both Δv and Δw are real. With the notations of Eqs. (8.4) and (8.5), the real and the imaginary part of ΔU , say ΔV and ΔW respectively, are written as

$$\Delta V = \{(N_R^2 - N_I^2)\Delta v - 2N_R N_I \Delta w\} \quad (8.6a)$$

and

$$\Delta W = \{(N_R^2 - N_I^2)\Delta w + 2N_R N_I \Delta v\}, \quad (8.6b)$$

respectively.

Let us consider the property of Δv and Δw along the work of Love, Terasawa and Satchler²⁶⁾ and of Kubo and Hodgson.²⁹⁾ In order to obtain a closed form for $\Delta v + i\Delta w$, they employed the following assumption: (i) The wave function of relative motion in the incident channel χ_0 and those in the intermediate channels in Eq. (8.5) can be replaced by the lowest-order plane wave, (ii) only the components for small momentum transfer are effective in the calculation of the DP potential. In the present discussion, excitations by nuclear interaction are of interest, although Kubo and Hodgson discussed both nuclear and Coulomb excitations.²⁹⁾ Under the assumption, the nuclear-excitation part of Eq. (8.5) is written²⁹⁾ as

$$\Delta v + i\Delta w = \frac{-ik}{8\pi E_{\text{cm}}} \sum_{j \neq 0} \sum_{\lambda} (2\lambda + 1) f_{j0}^{(\lambda)*}(R) \int_0^\infty dq q j_\lambda(qR) \int_0^\infty dx x^2 f_{j0}^{(\lambda)}(x) j_\lambda(qx), \quad (8.7)$$

where E_{cm} and k are the c.m. energy and the wave number of the relative motion respectively, λ is the multipolarity for the excitation of state j , and $f_{j0}^{(\lambda)}(R)$ is the radial part of the coupling potential $V_{j0}(\mathbf{R})$ defined as

$$V_{j0}(\mathbf{R}) = \sum_{\lambda\mu} f_{j0}^{(\lambda)}(R) Y_{\lambda\mu}^*(\hat{\mathbf{R}}). \quad (8.8)$$

It is clear that the right-hand side of Eq. (8.7) is pure-imaginary when $f_{j0}^{(\lambda)}$ is real. Furthermore, the imaginary part was found to be negative for realistic cases.²⁹⁾ Thus, the present assumptions lead to the DP potential of the well-known type,

$$\begin{cases} \Delta v \simeq 0, \\ \Delta w < 0. \end{cases} \quad (8.9)$$

This is just the case with $N_R = 1.0$ and $N_I = 0$ in Eqs. (8.6). Namely,

$$\begin{cases} \Delta V \simeq 0, \\ \Delta W < 0. \end{cases} \quad (8.10)$$

Let us apply the same assumption to the case of complex coupling potential, $N_I \neq 0$ but still the real V_{0j} . By definition, the property of Δv and Δw given in Eq. (8.7) is not affected at all by the change of N_R and N_I in the present case. The change only brings about a difference in ΔV and ΔW through Eqs. (8.6). Since $\Delta v \simeq 0$ under the present assumption,

$$\begin{cases} \Delta V \simeq -2N_R N_I \Delta w, \\ \Delta W \simeq (N_R^2 - N_I^2) \Delta w. \end{cases} \quad (8.11)$$

Thus, we can classify the qualitative feature of ΔV and ΔW by the ratio of N_I/N_R as follows:

- (i) In the case of a real coupling, namely $N_I = 0$, the DP potential of the conventional type is induced as

$$\begin{cases} \Delta V \simeq 0, \\ \Delta W \simeq N_R^2 \Delta w < 0. \end{cases} \quad (8 \cdot 12)$$

(ii) In the case of a strong-imaginary coupling such as $N_I \simeq N_R$,

$$\begin{cases} \Delta V \simeq -2N_R^2 \Delta w > 0, \\ \Delta W \simeq 0, \end{cases} \quad (8 \cdot 13)$$

which is just the type seen in the case of ${}^6\text{Li}$ -breakup.

(iii) In the intermediate case between (i) and (ii) such as $N_I \simeq N_R/2$,

$$\begin{cases} \Delta V \simeq -N_R^2 \Delta w > 0, \\ \Delta W \simeq 3/4 \cdot N_R^2 \Delta w < 0, \end{cases} \quad (8 \cdot 14)$$

which simulates well the case of ${}^{12}\text{C}$.

The meaning of these results becomes clear if one rewrite Eq. (8·11) in the form as

$$\Delta U = |\Delta U| \cdot e^{i\varphi}, \quad (8 \cdot 15)$$

where

$$|\Delta U| = N_R^2 (1 + \alpha^2) \cdot |\Delta w|, \quad (8 \cdot 16a)$$

$$\tan \varphi = \frac{(\alpha^2 - 1)}{2\alpha}, \quad (8 \cdot 16b)$$

$$\alpha = N_I / N_R. \quad (8 \cdot 16c)$$

The ratio, α , of the imaginary coupling to the real one essentially determines the *phase*, φ , of the complex DP potential. The ratio of $\Delta V / \Delta W$ increases as α increase from 0. It is this property of the DP potential that we have discovered empirically in the CC analyses shown in §§ 3~8.

8.2.2. Effects of multi-step processes

The weak-coupling assumptions which we have employed in the above discussion in order to obtain a closed form for Δv and Δw seem to be too simplified to be extended to the strong-coupling cases. As seen in §§ 3~7, multi-step processes between excited channels are as important (cf. Fig. 52) as the strength of the imaginary coupling potential in considering the property of the DP potential. Hence, the Green function which appears in Eqs. (8·4) and (8·5) has to be replaced by a complicated one with off-diagonal matrix elements depending on N_R and N_I themselves. The distortion effects on χ_0 as well as χ_i 's in the intermediate states due to the distorting potentials may also come to appear.

In the case of strong coupling, Δv in Eq. (8·5) will be expected to have a non-negligible strength even for a real-coupling potential. In fact, the schematic-model calculation (Fig. 60) shows that a small but non-negligible real part comes to appear along the line with $N_I = 0$ as N_R increases. Note that the DP potential for $N_I = 0$ in Fig. 60 is nothing but $\Delta v + i\Delta w$ itself. When Δv is not negligible, the structure of the (N_R, N_I) -dependence becomes somewhat more complicated than for the case of $\Delta v \simeq 0$. Since the ratio of $\Delta v / \Delta w$ itself

seems to depend on N_R , it seems better to discuss separately on a *weak-coupling* region where $\Delta v \approx 0$ and on a *strong-coupling* region where Δv is not negligible compared with Δw . In the weak coupling case, say in the region defined by $N_R \approx N_I \lesssim 0.5$ in Fig. 60, the behaviour of the DP potential can be simulated well by the relation in Eq. (8.11). ΔW changes its sign around the line of $N_R = N_I$, as predicted by the equation. On the other hand, in the strong coupling region which may correspond to the rest of the figure, the simple relation in Eq. (8.11) does not hold any more since Δv is not negligible there. Thus, we must go back to the Eqs. (8.6a) and (8.6b). According to Eq. (8.6b), ΔW changes its sign at the position on a line given by

$$N_I = N_R \sqrt{1 + 2(\Delta v / \Delta w)(N_I / N_R)}. \quad (8.17)$$

If we assume that the ratio $\Delta v / \Delta w$ changes with N_R as $\Delta v / \Delta w = c_0 N_R$, the position where ΔW changes the sign moves along a line of

$$N_I = N_R \sqrt{1 + 2c_0 N_I}. \quad (8.18)$$

Thus, the line deviates from a straight line of $N_R = N_I$ upward if Δv is positive (namely, $c_0 < 0$), while downward if Δv is negative (namely, $c_0 > 0$); in Fig. 61 three lines corresponding to $c_0 = 0$ and ± 0.5 are shown as examples. The negative value of c_0 is consistent with the tendency of Fig. 60. As the result, the DP potential induced at the position with $N_R = 1.0$ and $N_I = 0.6$ has a very small imaginary part.

Thus, one may understand that the drastic change in the behaviour of the DP potential with respect to the change in N_R and N_I seen in Fig. 60 is explained well by the *explicit* relationships of Eqs. (8.6a) and (8.6b) between the DP potential and the strength of the coupling potentials.

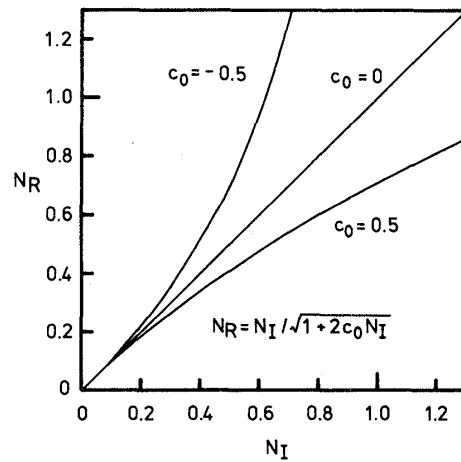


Fig. 61. A behaviour of the position where $\Delta W^{(j)}$ changes its sign, in the (N_R, N_I) -space when $\Delta v / \Delta w$ is assumed to change with N_R as $\Delta v / \Delta w = c_0 N_R$.³⁵⁾

§ 9. Summary

Microscopic method of CDCC

In order to study the scattering and breakup of light heavy-ions which have rather small breakup-threshold energy, we have employed a microscopic version³⁴⁾ of the method of coupled discretized continuum channels (CDCC).^{2)~5)} This method was originally developed^{1)~10)} for the study of deuteron breakup processes in deuteron involved reactions on the basis of the three-body model; the total wave function is expanded in terms of the eigenfunctions of the p-n system and the interaction between the projectile and the target is derived by folding the p-target and n-target optical potentials into the p-n relative wave functions of the ground state and the discretized unbound continuum states. In the present

study of the light heavy-ion scattering, the total wave function was expanded in terms of the microscopic cluster-model wave functions for the internal states of the projectile nucleus in the resonant and non-resonant breakup states as well as in the bound states. The projectile-breakup continuum was discretized into a finite number of momentum bins and the internal wave functions for the continuum states were averaged in each bin (Figs. 8, 21, 31 and 36). As for the interactions between the projectile and the target, the real parts of the diagonal and coupling potentials were constructed by doubly folding the effective nucleon-nucleon interaction of the M3Y type into the diagonal and transition densities of the projectile nucleus and the ground-state density of the target nucleus. The real potentials were multiplied by a factor N_R which is, however, fixed at unity in the present CDCC calculations; but, when we performed the single-channel (double-folding-model¹³⁾) calculation with no excited channels, N_R is varied as an adjustable parameter. The imaginary parts of the above potentials were simply assumed to have the same geometry of the corresponding real parts (for a reason, see Fig. 5 for the comparison of the form factor of the double-folding potential with that of the cluster-folding potential in the surface region) and its strength N_I is the only adjustable parameter in the present CDCC framework.

We then solve a set of coupled equations for the wave functions which describe the relative motion between the projectile and the target. The single parameter N_I was determined so that the observed differential cross section for the elastic scattering was satisfactorily fitted by the CDCC calculation; the fit was generally very good for most of the experimental data analyzed in the present paper. Since the cross sections for the inelastic scattering and the projectile-breakup reactions are automatically obtained in the same CDCC calculation without any additional parameter, it is a stringent test of the theory to examine those predicted cross sections by the observed ones.

Elastic scattering of ${}^6\text{Li}$

The microscopic CDCC method with the ${}^6\text{Li}$ states of Fig. 8(a) has been applied to the elastic scattering of ${}^6\text{Li}$ from ${}^{12}\text{C}$, ${}^{28}\text{Si}$, ${}^{40,48}\text{Ca}$, ${}^{58}\text{Ni}$ and ${}^{208}\text{Pb}$ at $E_{\text{lab}}=36\sim 169$ MeV. The observed differential cross sections were beautifully reproduced by the calculation except at very backward angles for several cases (Figs. 9~19). The effect of the virtual breakup of the ${}^6\text{Li}$ projectile is quite large and essentially important to obtain good fits to the data in a wide range of targets and bombarding energies. Since the ground state and the D -wave resonance state of ${}^6\text{Li}$ have much the same α -d cluster structure with $l=0$ and 2, respectively, for the α -d relative motion, the coupling of the elastic channel with the D -wave resonance channel is the strongest among those with the breakup channels. As the result, the resonance channel gives the most important effect on the elastic scattering; but it is also necessary to take into account the non-resonant breakup channels in order to obtain good agreement with the observed angular distributions (Fig. 12).

If one omits all the breakup channels in the calculation, one cannot obtain at all any reasonable fit to the observed angular distribution as long as $N_R \simeq 1.0$ is persisted; this is also the case when the Woods-Saxon shape is used for the imaginary potential in place of the same shape as that of the folding real potential with the strength parameter of N_I . That is, the reduction of the real potential strength N_R from 1.0 to $0.5\sim 0.6$ is inevitable to reproduce the experimental data by the single-channel calculation. This is the well-known

anomaly of N_R in the double-folding model.^{13)~22)} It was found that the single-channel calculation with $N_R=0.5\sim0.6$ but with the *same* value of N_I of the CDCC calculation well reproduced the elastic-scattering cross sections (Figs. 9 and 10) and the S -matrix elements (Fig. 11) given by the CDCC calculation. This means that, at least in the surface region which is most important for the scattering, the effect of the projectile breakup on the elastic scattering is strongly repulsive so as to account for the strong reduction of N_R , and furthermore it induces an unexpectedly small imaginary potential.

This property of the breakup effect was found to be almost independent of the bombarding energy and the target mass; i.e., almost irrespectively of the energy and target, the repulsive effect of the breakup channels can be simulated well simply by reducing N_R from 1.0 to $0.5\sim0.6$ in the single-channel calculation. However, when the effects are compared in the angular distributions, it seems as if the effect would strongly depend on the energy and target mass; i.e., the differential cross section (i) decreases due to the breakup effect for some cases but (ii) increases or (iii) becomes less oscillatory for other cases (for example, see Figs. 14~19). According to the nearside/farside decomposition of the scattering amplitude,^{76),77)} a *reduction* of the real-potential strength causes a decrease of the farside amplitude and an increase of the nearside one. Therefore, all the characteristics (i)~(iii) are understood consistently from the viewpoint of the repulsive effect of the virtual breakup of ${}^6\text{Li}$ (i) as a decrease of the farside-dominant cross section, (ii) as an increase of the nearside-dominant one and (iii) as a less interference between the nearside amplitude and the farside one, respectively (cf. Fig. 20).

In order to verify the repulsive property of the dynamical polarization (DP) potential induced by the virtual breakup of ${}^6\text{Li}$, we calculated the potential explicitly in the form of the wave-function-equivalent local potential (Eq. (7·3)). In a wide range of the bombarding energy and the target mass, it was clearly shown that, in the surface region which is most important for the scattering, the real part of the DP potential $\Delta V(R)$ is indeed strongly repulsive and the imaginary part ΔW is very small (Figs. 47~49). It is to be emphasized that ΔV nearly accounts for the reduction of N_R to $0.5\sim0.6$ in the double-folding model; namely, $N_R V_{00}(R) \simeq V_{00}(R) + \Delta V(R)$ in the surface region, V_{00} being the double-folding potential.

Elastic and projectile-inelastic scattering of ${}^7\text{Li}$

The microscopic CDCC method has been applied to the elastic scattering of ${}^7\text{Li}$ from ${}^{12}\text{C}$, ${}^{48}\text{Ca}$, ${}^{58}\text{Ni}$ and ${}^{208}\text{Pb}$ at $E_{\text{lab}}=63\sim148$ MeV with the ${}^7\text{Li}$ states of Fig. 21 taken into account. All the observed cross sections were well reproduced by the CDCC calculation with the use of the single adjustable parameter N_I , N_R being fixed at unity (Figs. 23 and 24). From the analysis of individual roles of the ${}^7\text{Li}$ excited states in the elastic scattering, the coupling with the $7/2^-$ resonance state ($l=3$) was found to be the strongest among the ${}^7\text{Li}$ states (Figs. 23 and 24). The effect of the re-orientation coupling of the $3/2^-$ ground state ($l=1$) and that of the coupling with the $1/2^-$ excited state ($l=1$) at $E_x=0.48$ MeV are comparatively weak for the present bombarding energies. Therefore, it may be concluded that the main origin of the strong reduction of N_R to $0.5\sim0.7$ in the double-folding-model calculations of the ${}^7\text{Li}$ scattering (e.g., Figs. 22(a)~(d)) is the coupling with the breakup channels, particularly with the $7/2^-$ resonance channel.

In the CDCC calculation, the cross section for the inelastic scattering leading to the

$1/2^-$ excited state of the ${}^7\text{Li}$ projectile was simultaneously predicted together with the elastic-scattering cross section with no additional parameter. For the scattering from ${}^{48}\text{Ca}$ at 89 MeV and from ${}^{12}\text{C}$ at 63 and 79 MeV, the predicted inelastic-scattering cross sections reproduced beautifully the measured ones (Figs. 25(a) and (b)) on account of a large effect of the breakup channels. This success gives a support for the usefulness of the present CDCC framework in the ${}^7\text{Li}$ scattering.

Breakup reactions of ${}^{6,7}\text{Li}$ projectiles

In the CDCC framework, cross sections for the breakup of ${}^6\text{Li}$ (${}^7\text{Li}$) projectile into $\alpha+d$ ($\alpha+t$) leaving the target nucleus in the ground state (i.e., the elastic-breakup cross sections) are simultaneously calculated together with the elastic-scattering cross section. Therefore, the comparison of them with the observed (${}^6\text{Li}$, αd) and (${}^7\text{Li}$, αt) elastic-breakup cross sections will provide a severe test of the present CDCC framework. The cross sections for ${}^{12}\text{C}$ (${}^6\text{Li}$, αd) and ${}^{208}\text{Pb}$ (${}^6\text{Li}$, αd) elastic-breakup reactions were measured⁸⁶⁾ at $E_{\text{lab}}=178$ MeV for the breakup from the α -d continuum at $0 < \varepsilon < 1.0$ MeV (region I) and that from the continuum at $1.0 < \varepsilon < 4.0$ MeV (region II) with the integration over ε in each region, ε being the kinetic energy of the α -d relative motion. For both targets, the agreement between the calculated cross section and the observed one was satisfactory for each of regions I and II both in the magnitudes and in the angular distributions (Figs. 26(a) and (b)). In region I the contribution of the breakup from the 3^+ resonance state is much larger (by about one order of magnitude) than that from the non-resonant states, while in region II the contribution of the non-resonant states is comparable with that of the 2^+ resonance state in the case of ${}^{12}\text{C}$ target and even exceeds the latter in the case of ${}^{208}\text{Pb}$ target. Furthermore, it was found that the couplings between the resonance states and the non-resonant ones are very strong and work to reduce the cross sections for the breakup from the resonance states by a factor of two to three (Fig. 27). As for the ${}^7\text{Li}$ projectile, the calculated results were compared at $E_{\text{lab}}=70$ MeV with the measured cross sections^{87),88)} for the breakup from the $7/2^-$ resonance state at $E_x=4.63$ MeV ($\varepsilon=2.16$ with respect to the α -t breakup threshold) and that from the non-resonant continuum at $0.29 < \varepsilon < 3.0$ MeV. In the case of ${}^{120}\text{Sn}$ target, the observed cross section for the $7/2^-$ resonance state and that for the non-resonant continuum are comparable with each other in magnitude, and both of them were reproduced satisfactorily by the CDCC calculation (Fig. 32(a)). For ${}^{12}\text{C}$ target, the observed cross section for the $7/2^-$ resonance state was fairly well reproduced by the calculation, whereas the cross section for the non-resonant continuum is negligibly small in the experiment but is only slightly smaller than that for the $7/2^-$ resonance state in the calculation (Fig. 32(b)). This discrepancy is to be investigated in the future. Also, in the case of low bombarding energies on heavy targets, the breakup due to the Coulomb interaction will be a future subject of the CDCC method; only a preliminary investigation of the Coulomb-breakup effects was done in the present paper (Figs. 33(a) and (b)).

Scattering and breakup of ${}^{12}\text{C}$

It has been known that the double-folding model with $N_R=1.0$ is successful for the elastic scattering of ${}^{12}\text{C}$ at $E_{\text{lab}}/12 \lesssim 10$ MeV.¹³⁾ As the bombarding energy increases, however, a significant reduction of N_R to $0.9 \sim 0.6$ becomes necessary to fit the experimental

data (Figs. 34 and 35(a)), which will mean that the effect of the ^{12}C virtual excitations becomes more repulsive with the increase of the bombarding energy. In order to clarify this mechanism and the roles of individual excited states of ^{12}C , we analyzed the elastic and the inelastic scattering of $^{12}\text{C}+^{12}\text{C}$ at $E_{\text{lab}}(^{12}\text{C})=140\sim 300$ MeV, $^{13}\text{C}+^{12}\text{C}$ at $E_{\text{lab}}(^{13}\text{C})=260$ MeV and $^{16}\text{O}+^{12}\text{C}$ at $E_{\text{lab}}(^{16}\text{O})=168\sim 311$ MeV with the use of the microscopic CC method in which the ^{12}C states in Fig. 36 (including the $^{12}\text{C}\rightarrow 3\alpha$ breakup states) were taken into account explicitly. The CC calculation with the single adjustable parameter N_l (N_r being fixed at 1.0) well reproduced the elastic-scattering cross sections and *simultaneously* the inelastic-scattering ones for the 2_1^+ , 3_1^- and 0_2^+ states of ^{12}C (Figs. 39~42). The effect of the ^{12}C virtual excitations is striking; although the coupling of the 2_1^+ state seems to give the largest contribution, the 0_2^+ and 2_2^+ resonant breakup states and the non-resonant breakup states also play important roles, especially in reproducing the characteristic behaviours of the observed cross sections at large angles (Figs. 37~40). It was also found that the coupling between the 0_2^+ state and other 3α -breakup states was very strong and the coupling effect much improved the fit to the observed inelastic-scattering cross section for the state. The breakup effect on the 0_2^+ state can be simulated well by reducing the real-potential strength of the state by a factor of two,⁹⁵⁾ which means that a strongly repulsive DP potential is induced in the 0_2^+ -channel by the coupling with the other 3α -breakup channels (Figs. 42(a) and (b)).⁹⁴⁾

The optimum value of N_l in the DF-model calculation is significantly larger than that in the CC calculation (Fig. 35(b)). This means that, contrary to the case of ^6Li scattering, the ^{12}C excited channels induce an absorptive potential with a considerable strength together with the repulsive real potential. The explicit calculation of the DP potential, $\Delta V + i\Delta W$, with the use of the CC wave functions clearly verified the property of the ^{12}C virtual-excitation effects mentioned above (Figs. 53~55). It was also found that the shell-model-like states (namely, 2_1^+ and 3_1^- states) generate most of ΔW and the half of ΔV , while the cluster-like states (0_2^+ and 2_2^+ and the non-resonant states) induce the rest half of ΔV and their contribution to ΔW is very small (Fig. 56); the DP potential induced by the cluster-like states is like that induced by the excitation of ^6Li projectile into the cluster-like resonant and non-resonant continuum states as mentioned before. It is to be noted that both ΔV and ΔW depend much on the bombarding energy; as the energy increases, ΔV increases but $|\Delta W|$ decreases (Figs. 53 and 54), which explains well the reason for the breakdown of the double-folding model with $N_r=1.0$ for ^{12}C scattering at $E_{\text{lab}}/12 \gtrsim 10$ MeV (cf. Fig. 35(a)).

Unified understanding of various types of DP potentials

The DP potential induced by the breakup of ^6Li and that induced by the excitation of ^{12}C are quite different from each other and from the well-known type of DP potential for heavy-ion scattering which is induced by the excitation of collective phonon modes of nucleus^{25)~30)} and has usually a sizable imaginary part ($\Delta W > 0$) and a very small real part ($\Delta V \simeq 0$) (see, for example, Fig. 46). Characteristics of ΔV and ΔW of these different types of DP potentials are illustrated in Fig. 57; the behaviour of the relative magnitude of ΔV and ΔW is of interest. In order to understand the mechanism for the appearance of such different types, we performed schematic CC-model calculation of DP potentials in which we varied widely the strengths of the real and imaginary parts of the diagonal and

coupling potentials. It was found that, in the surface region which is most important for the scattering, the relative magnitude of ΔV to ΔW is governed mainly by the *relative* strength of the imaginary part to the real one of the *coupling* potentials; when the relative strength is increased from zero, the real potential ΔV rapidly grows up from zero to a positive value with a considerable magnitude, whereas the imaginary potential ΔW starts from a negative value with a considerable magnitude, quickly attenuating to $\Delta W \simeq 0$ (Fig. 60). Here, beautifully simulated are the above-mentioned prototypes of the DP potentials (Fig. 57) by the CC-model calculation with the relative strength varied; i.e., the DP potentials induced (i) by the collective modes, (ii) by the breakup of ${}^6\text{Li}$ projectile and (iii) by the excitation of ${}^{12}\text{C}$ are classified respectively at (i) small, (ii) large and (iii) intermediate magnitudes of the relative strength of the imaginary part to the real one of the coupling potentials (Fig. 60). The reason why the characteristics of the DP potentials is governed by the relative strength was presented in § 8.2 by extending the perturbation theories^{26),29)} of the DP potentials to the cases of the excitation of strong-coupling states. We derived an explicit relationship between the DP potential and the strength of the real and the imaginary part of the coupling potential, which explains well the general property of the DP potential mentioned above. Thus, we can say that we have revealed the wide possibility for appearance of various types of the DP potentials in heavy-ion scattering and the conventional type of the DP potentials are realized only in the weak-coupling limit which occupies only a small part of the possibility.

Projectile breakup effects on the non-central interactions

The virtual excitations of ${}^{6,7}\text{Li}$ projectiles induce not only central potentials but also non-central potentials such as of spin-orbit and tensor types. In the CC framework with the use of the cluster-folding interaction and the double-folding one, the virtual-excitation effects on the non-central interactions have extensively been studied by Hosei-Kyushu group including the present authors⁹⁷⁾ and by the Surrey-Oxford group;⁹⁸⁾ the vector and tensor analyzing powers observed in the scattering of polarized ${}^{6,7}\text{Li}$ ions at $E_{\text{lab}} = 14 \sim 44$ MeV have been successfully been explained in terms of the effective non-central interactions induced by the coupling of the excited states of ${}^{6,7}\text{Li}$ projectiles as well as by the folding tensor potentials in the ground state. Though not reviewed in the present paper, the works of Refs. 97) and 98) are quite interesting and important from the same viewpoint of the present paper on the virtual-excitation effects of light heavy-ions projectiles as well as that of the target nucleus.

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